SEVEN OPEN UNIT EQUILATERAL TRIANGLES DO NOT COVER A REGULAR UNIT HEXAGON

AUTHOR

Abstract. We prove that the regular hexagon of side length one cannot be covered by seven open equilateral triangles of side length one. Equivalently, no regular hexagon of side length greater than one is covered by seven closed unit equilateral triangles. Seven distinguished points force one center role and six vertex roles. An exhaustive classification of those roles reduces the proof to four mechanisms: direct boundary, radial, diagonal, and conditional-skeleton length sums; exact propagation of boundary and radial demands; normalized area loss; and a center-independent residual-core obstruction based on exact AB-unions. The machine-assisted steps are stated as finite, independently checkable interval certificates with directed rounding or exact rational arithmetic.

1. Introduction and main result

Let H be the regular hexagon of side length one. The problem is to decide whether seven open equilateral triangles of the same side length can cover H. The use of open triangles is essential at boundary handoffs: when two open traces meet at a common endpoint, that endpoint is still uncovered.

Theorem 1.1 (Main theorem)**.** *The regular side-one hexagon H cannot be covered by seven open unit equilateral triangles.*

The compactness and scaling lemma proved in Proposition 2.1 gives the following equivalent form.

Corollary 1.2 (Expanded closed formulation)**.** *For every $L > 1$, the regular hexagon H_L of side length L cannot be covered by seven closed unit equilateral triangles.*

The proof is organized by mechanism rather than by the eventual case tree. Section 2 fixes the geometric model, assigns the seven roles, and proves exhaustive center- and vertex-role classifications. It also transfers the supercritical-row count from actual maximal boundary reaches to strictly selected handoff demands.

Section 3 establishes direct one-dimensional trace bounds on the boundary, the six radial arms, the three main diagonals, and the full skeleton when the additional hypotheses needed for a skeleton estimate are present. These estimates dispose of the coarse branches and several exceptional positive-support configurations.

Section 4 develops exact local admissibility and capped demand maps. Exact center-trace and radial-exit formulas turn a center gap or a missing midpoint into

2020 *Mathematics Subject Classification.* 52C15, 52A40, 51M04.

Key words and phrases. equilateral triangles, geometric covering, regular hexagon, boundary propagation, area loss, caliper certificate.

boundary and radial demands. Iteration of the local maps returns a demand that is incompatible with the remaining trace capacity. This section contains the delicate CE1/CE2 branches.

Section 5 normalizes the area of a unit equilateral triangle to one. It proves unconditional local square-loss estimates and a separate loss estimate for the recurrent 걸거치는 (straddling) T3-like configuration. Their cyclic sum closes the remaining CE0 branches with one or at least two supercritical rows.

Section 6 treats the all-Vd0, exactly-one-supercritical, zero-gap configuration without any center-class assumption. Monotonicity of exact corner-clipped AB-unions forces a compact residual witness consisting of a disk and three further points. A caliper inequality shows that no open unit equilateral triangle can contain that witness.

Finally, Section 7 gives the exhaustive routing table and proves Theorem 1.1. Long formulas are collected in Appendix A. Appendix B records the finite certificates used in the proof, including their domains, pruning rules, exact or outward-rounded arithmetic, and expected outputs. No plot, floating-point optimization, or unproved structural conjecture is used as a logical dependency.

2. Geometric setup and structural reductions

Throughout, indices are taken in $\mathbb{Z}/6\mathbb{Z}$. Put

$$O = (0,0), \qquad V_i = \left(\cos\frac{i\pi}{3}, \sin\frac{i\pi}{3}\right), \qquad H = \operatorname{conv}\{V_0, \dots, V_5\}.$$

Thus H is the regular hexagon of side length one. We use the notation

$$e_{i,i+1} = [V_i, V_{i+1}], \qquad r_i = [O, V_i], \qquad M_i = \frac{1}{2}V_i,$$

and, when needed,

$$D = \bigcup_{i=0}^{5} r_i, \qquad S = \partial H \cup D, \qquad S_{1/2} = \partial H \cup \{O, M_0, \dots, M_5\}.$$

The arms r_i meet only at O, up to sets of one-dimensional measure zero, and consequently

$$\mathcal{H}^1(\partial H) = 6, \qquad \mathcal{H}^1(D) = 6, \qquad \mathcal{H}^1(S) = 12. \tag{1}$$

For $L > 0$, $H_L := LH$ denotes the concentric regular hexagon of side length L. All symmetries used below belong to the dihedral group D_6; applying a symmetry always transports the indices, orientations, and incoming and outgoing labels together.

2.1. Open, closed, and scaled formulations.

Proposition 2.1 (Compactness and scaling equivalence)**.** *The following assertions are equivalent.*

(1) *The set H is covered by seven open equilateral triangles of side length* 1.
(2) *For some $\varepsilon > 0$, the set H is covered by seven closed equilateral triangles of side length $1 - \varepsilon$.*
(3) *For some $L > 1$, the set H_L is covered by seven closed equilateral triangles of side length* 1.

Proof. Suppose that open unit triangles $U_1, \dots, U_7$ cover H. Define

$$m_j(p) = \operatorname{dist}(p, \mathbb{R}^2 \setminus U_j), \qquad m(p) = \max_{1 \le j \le 7} m_j(p).$$

The continuous function m is positive on the compact set H, so $\eta := \min_{p \in H} m(p) > 0$. Choose

$$0 < \rho < \min\left\{\eta, \frac{\sqrt{3}}{6}\right\}.$$

Moving each side of U_j inward through distance ρ gives a closed equilateral triangle K_j of side $1 - 2\sqrt{3}\rho$. Every $p \in H$ has $m_j(p) \ge \eta > \rho$ for some j, and hence belongs to K_j. This proves (1)⇒(2), with $\varepsilon = 2\sqrt{3}\rho$.

Conversely, enlarging a closed triangle of side $1 - \varepsilon$ about its incenter to an open triangle of side 1 preserves containment, proving (2)⇒(1). Finally, dilation by $(1 - \varepsilon)^{-1}$ identifies (2) with (3), with

$$L = (1 - \varepsilon)^{-1} > 1.$$

□

We conduct the geometric argument in the open side-one model. Denote the original open triangles by U_C and U_i for $0 \le i \le 5$, and put

$$T_C = \overline{U_C}, \qquad T_i = \overline{U_i}.$$

The distinction between U_i and T_i will be maintained: closure may add endpoints or tangencies, although it does not change area or one-dimensional trace measure.

Lemma 2.2 (Distinct roles)**.** *The seven triangles can be labelled so that*

$$O \in U_C, \qquad V_i \in U_i \quad (0 \le i \le 5),$$

and these seven roles are distinct. Consequently

$$O \in \operatorname{int}(T_C), \qquad V_i \in \operatorname{int}(T_i).$$

Proof. The seven distinguished points $O, V_0, \dots, V_5$ are pairwise at distance at least 1. Any two points of an open unit equilateral triangle are at distance strictly less than its diameter 1. Thus one open triangle contains at most one distinguished point. Selecting a covering triangle for each of the seven points gives an injection from seven points to seven triangles, hence a bijection and the asserted labelling. □

2.2. **The center classification.** For a closed center role define

$$N_E(T_C) = |\{i : T_C \cap e_{i,i+1} \text{ contains a positive-length interval}\}|.$$

Point contacts, endpoint contacts, and tangencies do not contribute. We say that T_C has type CE0, CE1, or CE2 according as $N_E(T_C)$ is 0, 1, or 2.

Proposition 2.3 (Exhaustive center types)**.** *Every center role has exactly one of the types CE0, CE1, and CE2. Equivalently,*

$$N_E(T_C) \le 2.$$

Proof. Among any three edges of the six-cycle ∂H, two have no common endpoint. It is therefore enough to show that relative-interior points of two nonadjacent edges are more than unit distance apart. Up to symmetry there are two cases. For $0 < t, s < 1$, put

$$x = (1 - t)V_0 + tV_1.$$

If $y = (1-s)V_2 + sV_3$, direct calculation gives
$$\|x-y\|^2 - 1 = (1-t)(2-t) + s(s+t) > 0.$$
If instead $y = (1-s)V_3 + sV_4$ and $u = t+s$, then
$$\|x-y\|^2 - 1 = (u-1)^2 + 2 > 0.$$
A unit equilateral triangle has diameter 1. It therefore cannot contain relative-interior points on two nonadjacent hexagon edges. Three positive-length edge traces would supply such a pair, a contradiction. □

2.3. **Vertex types and the T3-like normalization.** For an original vertex role T_i, let
$$o(T_i) = |\{\text{vertices of } T_i \text{ outside } H\}|$$
and let $n(T_i)$ be the number of adjacent arms r_{i-1}, r_{i+1} having positive-length intersection with T_i. We use the names

type	(o, n)
Vd0	$(1,0), (2,0), (3,0)$
Vd1	$(1,1)$
Vd2	$(1,2)$
T3-like	$(2,1)$.

Lemma 2.4 (Local wedge)**.** *Every original vertex role has $1 \le o \le 3$. If it has positive-length intersection with one adjacent arm, at least one of its vertices belongs to H; if it has positive-length intersection with both adjacent arms, at least two of its vertices belong to H.*

Proof. Normalize to $i = 0$ and write
$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0).$$
Then $V_0 = (0,0)$,
$$\|(x,y)\|^2 = x^2 + y^2 - xy,$$
and, within the closed unit ball about V_0, membership in H is equivalent to membership in the wedge
$$W = \{(x,y) : x \ge 0, y \ge 0\}.$$
Indeed,
$$x^2 + y^2 - xy = \left(y - \frac{x}{2}\right)^2 + \frac{3x^2}{4} = (x-y)^2 + xy$$
shows that $x, y \ge 0$ and norm at most one imply the remaining hexagon inequalities. The adjacent arms are
$$r_1 = \{(1,y) : 0 \le y \le 1\}, \qquad r_5 = \{(x,1) : 0 \le x \le 1\}.$$

Suppose $T_0 \cap r_1$ has positive length, and let m be the largest x-coordinate of a vertex of T_0. If $m > 1$, a maximizing vertex (m,y) lies in the unit ball and hence satisfies $0 < y < 1$; it belongs to H. If $m = 1$, positive-length contact with the supporting line $x = 1$ forces a side of T_0 to lie there, and both endpoints satisfy $1 + y^2 - y \le 1$, hence belong to H. The same argument applies to r_5. If both arms are met but only one triangle vertex belonged to H, that vertex would have to have both $x > 1$ and $y > 1$. This is impossible because $(x-y)^2 + xy > 1$.

Finally, if all three triangle vertices belonged to the convex set H, then $T_0 \subset H$. This is incompatible with the extreme point V_0 lying in the interior of T_0. Hence $o \ge 1$. □

Proposition 2.5 (Exhaustive vertex types)**.** *Every original vertex role has exactly one of the four types Vd0, Vd1, Vd2, and T3-like.*

Proof. We have $n \in \{0, 1, 2\}$. Lemma 2.4 gives $o \le 2$ when $n \ge 1$, and $o \le 1$ when $n = 2$. Together with $o \ge 1$, the possibilities are

n	possible o
0	$1, 2, 3$
1	$1, 2$
2	$1,$

which are precisely the four named classes. □

The next normalization is useful for trace estimates, but its final warning is essential.

Proposition 2.6 (T3-like closed-trace domination)**.** *Let T be the closure of an original T3-like V_i-role. There is a translate $\widehat{T}$ of T, with the same orientation and side length, such that*

$$V_i \in \operatorname{relint}(\textit{a side of } \widehat{T}), \qquad T \cap H \subsetneq \widehat{T} \cap H,$$

and $\widehat{T}$ remains of type $(o, n) = (2, 1)$. Every old open interior point in $H \setminus \{V_i\}$ remains an interior point of $\widehat{T}$. The point V_i itself becomes a boundary point, so $\widehat{T}$ is a closed-trace majorant and is not a replacement open role.

Proof. Use the wedge coordinates of Lemma 2.4. Set

$$D_t = 1 - t + t^2, \qquad z = \sqrt{D_t}, \qquad 0 \le t \le 1.$$

After relabelling triangle vertices, every orientation belongs to one of the following two families:

	first side vector	second side vector
I	$z^{-1}(1, t)$	$z^{-1}(1 - t, 1)$
II	$z^{-1}(t, -(1 - t))$	$z^{-1}(1, t)$.

For Type I write

$$U = \alpha + y - tx, \qquad V = \beta + x - (1 - t)y, \qquad T = \{U \ge 0, V \ge 0, U + V \le z\}.$$

Here $\alpha, \beta > 0$ and $\alpha + \beta < z$. Direct inversion shows that if exactly two triangle vertices are outside W, the sole wedge vertex has both coordinates strictly below 1: in the two possible cases its coordinates are bounded respectively by $(1 - t)/z$ and t/z, with the endpoint strictness supplied by $\alpha > 0$ or $\beta > 0$. The support argument in Lemma 2.4 then excludes positive-length contact with either $x = 1$ or $y = 1$. Thus Type I cannot be T3-like.

For Type II write

$$U = \alpha + (1 - t)x + ty, \qquad V = \beta + tx - y, \qquad T = \{U \ge 0, V \ge 0, U + V \le z\}.$$

Reflecting in $x = y$ if necessary, assume the supported adjacent arm is r_1. The two outside vertices have respectively a negative x- and a negative y-coordinate, while the unique wedge vertex is

$$Q = \left(\frac{z - \alpha - t\beta}{D_t}, \frac{t(z - \alpha) + (1 - t)\beta}{D_t} \right).$$

The endpoint cases $t = 0, 1$ give no positive adjacent interval, so $0 < t < 1$. The exact axis reaches are

$$A = \beta, \qquad B = z - \alpha - \beta,$$

and T3-like support is equivalent to

$$Q_x > 1, \qquad Q_y \leq 1.$$

Define $\widehat{T}$ by replacing α with 0 and keeping t, β fixed. Since the linear map $(x, y) \mapsto (U, V)$ is nonsingular, this changes only the translation. At the origin, $\widehat{U} = 0$ and $0 < \widehat{V} = \beta < z$, so V_0 is in the relative interior of a side. For $(x, y) \in W$,

$$\widehat{U} = (1-t)x + ty \geq 0, \qquad \widehat{V} = V, \qquad \widehat{U} + \widehat{V} = U + V - \alpha.$$

Thus $T \cap W \subset \widehat{T} \cap W$. The old x-axis reach is B and the new one is $B + \alpha$; because $B < 1$ and $\alpha > 0$, the inclusion is strict inside H.

The two outside vertices remain outside, whereas V_0 and the new wedge vertex lie in H, so $o = 2$. Moreover $\widehat{Q}_x > Q_x > 1$. From $Q_x > 1$ one gets $t\beta < z - \alpha - D_t$, and hence

$$tD_t(1 - \widehat{Q}_y) > D_t(1 - z) + (1 - t)\alpha > 0.$$

Thus $\widehat{Q}_y < 1$ and $n = 1$. Finally, for every nonzero point of W, $\widehat{U} > 0$; the displayed slack relations show that every old interior point other than the origin remains interior. □

2.4. **Reach coordinates, supercritical rows, and gaps.** For the original vertex role U_i, define its actual maximal incoming, outgoing, and radial reaches by

$$\begin{aligned} A_i &= \sup\{t \in [0,1] : V_i + t(V_{i-1} - V_i) \in U_i\}, \\ B_i &= \sup\{t \in [0,1] : V_i + t(V_{i+1} - V_i) \in U_i\}, \\ C_i &= \sup\{t \in [0,1] : V_i + t(O - V_i) \in U_i\}. \end{aligned}$$

Thus the closure T_i contains the corresponding endpoints. We reserve lowercase (a_i, b_i, c_i) for selected or prescribed lower-bound demands whenever actual and selected values occur together. A realizing role then satisfies

$$A_i \geq a_i, \qquad B_i \geq b_i, \qquad C_i \geq c_i.$$

An actual row is supercritical if $A_i + B_i > 1$, and

$$N_+ = |\{i : A_i + B_i > 1\}| . \tag{2}$$

This definition never counts a smaller demand pair in place of the actual row.

Parametrize $e_{i,i+1}$ by

$$X_i(x) = V_i + x(V_{i+1} - V_i), \qquad 0 \leq x \leq 1.$$

The two incident open traces are

$$U_i \cap e_{i,i+1} = [0, B_i), \qquad U_{i+1} \cap e_{i,i+1} = (1 - A_{i+1}, 1]$$

in this coordinate. A *boundary gap* on this edge is the nonempty closed complement between these two traces,

$$[B_i, 1 - A_{i+1}] \qquad \text{when } B_i \leq 1 - A_{i+1}.$$

In particular, equality gives a singleton gap. This convention records the actual open traces; a point missing from both open roles is not erased merely because both closures contain it.

Lemma 2.7 (T3-like rows are nonsupercritical)**.** *Every original T3-like row satisfies*

$$A_i + B_i \le 1. \tag{3}$$

In particular, it is not counted by N_+.

Proof. The proof of Proposition 2.6 showed that a T3-like role has Type II orientation with

$$A_i = \beta, \qquad B_i = z - \alpha - \beta, \qquad z = \sqrt{1 - t + t^2} \le 1.$$

Thus $A_i + B_i = z - \alpha \le 1$. The same conclusion holds after reflection. □

We record the precise transfer from actual rows to strict handoff demands. Its hypothesis that the six vertex roles themselves cover the boundary is part of the statement.

Proposition 2.8 (Strict boundary handoffs)**.** *Assume that $U_0, \dots, U_5$ cover ∂H. On each edge choose*

$$x_i \in (1 - A_{i+1}, B_i)$$

and define

$$a_i = 1 - x_{i-1}, \qquad b_i = x_i.$$

Then $0 < a_i < A_i$, $0 < b_i < B_i$, the two selected anchors lie in U_i, and

$$a_i^2 + a_i b_i + b_i^2 < 1.$$

Moreover:

1. *if exactly one actual row, indexed by p, is supercritical, every strict selection has exactly one selected supercritical row, also indexed by p;*
2. *if at least two actual rows are supercritical, some simultaneous strict selection has at least two selected supercritical rows.*

Proof. Because V_i is interior to a diameter-one closed triangle,

$$0 < A_i < 1, \qquad 0 < B_i < 1.$$

No nonincident vertex role contains a relative-interior point of $e_{i,i+1}$, since such a point is more than unit distance from its distinguished vertex. The two displayed incident traces therefore cover the edge and, being relatively open and nonempty, overlap. Hence each interval $(1 - A_{i+1}, B_i)$ is nonempty. The anchor and distance assertions follow from the choice and from the fact that two points of U_i are at distance strictly less than one.

Put $\ell_i = 1 - A_{i+1}$ and $u_i = B_i$. Then

$$A_i + B_i > 1 \iff \ell_{i-1} < u_i, \qquad a_i + b_i > 1 \iff x_{i-1} < x_i. \tag{4}$$

If actual row i is nonsupercritical, then

$$x_i < u_i \le \ell_{i-1} < x_{i-1},$$

so every selected version of it is strictly subcritical. Also

$$\sum_{i=0}^{5} (a_i + b_i) = \sum_{i=0}^{5} (1 - x_{i-1} + x_i) = 6. \tag{5}$$

If p is the sole actual supercritical index, the other five selected sums are strictly below one; equation (5) forces the pth sum above one. This proves (1).

For (2), first take two nonadjacent supercritical indices p, q. For each $r \in \{p, q\}$ choose

$$\ell_{r-1} < z_r < u_r,$$

then choose

$$x_{r-1} \in (\ell_{r-1}, \min\{u_{r-1}, z_r\}), \quad x_r \in (\max\{\ell_r, z_r\}, u_r).$$

The two pairs of variables are disjoint and give ascents at p, q. Any set of at least three indices on a six-cycle contains two nonadjacent indices. It remains to consider exactly two adjacent supercritical indices $p, p+1$. For the other four nonsupercritical rows, successive inequalities $\ell_i < u_i \leq \ell_{i-1}$ give

$$\ell_{p-1} < \ell_{p+1}.$$

Together with supercriticality at $p, p+1$, this implies

$$\max\{\ell_{p-1}, \ell_p\} < \min\{u_p, u_{p+1}\}.$$

Choose x_p between these bounds and then choose $x_{p-1} < x_p < x_{p+1}$ in their respective overlap intervals. By (4), both adjacent rows are selected supercritical. □

2.5. **Midpoint forcing.** We first need a local fact for vertex roles. Take unit vectors

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \qquad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).$$

Lemma 2.9 (Self-midpoint obstruction)**.** *Let a closed unit triangle contain*

$$0, \qquad au, \qquad bv, \qquad \frac{u+v}{2}.$$

Then $a + b \leq 1$. Consequently, for every vertex role,

$$M_i \in T_i \quad \Longrightarrow \quad A_i + B_i \leq 1.$$

If the two demanded boundary points and the midpoint are contained with a positive margin, the conclusion is strict.

Proof. Suppose $a + b > 1$ and set

$$K = \operatorname{conv}\left\{0, au, \frac{u+v}{2}, bv\right\}.$$

For outward unit normals n_0, n_1, n_2 separated by 120°, the least equilateral triangle with those normals and containing K has side length

$$\frac{2}{\sqrt{3}} \sum_{j=0}^{2} h_K(n_j).$$

On an angular interval where the three support vertices are fixed, the support sum S satisfies $S'' = -S < 0$. Hence its angular minimum occurs at a breakpoint, namely at an outward normal to one of the four edges of K.

For the edge from 0 to au, direct projection gives

$$S_{0A} = \max\left\{\frac{\sqrt{3}}{4}, \frac{\sqrt{3}a}{2}\right\} + \frac{\sqrt{3}b}{2} > \frac{\sqrt{3}}{2};$$

if $a \geq 1/2$ this is $(\sqrt{3}/2)(a+b)$, and if $a < 1/2$ then $b > 1/2$. The edge through bv is symmetric. For the edge from au to $(u+v)/2$, put $D_a = \sqrt{4a^2 - 2a + 1}$. Projection gives

$$S_{AM} = \begin{cases} \dfrac{\sqrt{3}(a+b)}{2D_a}, & a \leq \frac{1}{2}, \\ \dfrac{\sqrt{3}(2a^2+b)}{2D_a}, & a > \frac{1}{2}. \end{cases}$$

The first value is strictly above $\sqrt{3}/2$ because $D_a \leq 1$. In the second case, $b > 1-a$ and

$$(2a^2 - a + 1)^2 - D_a^2 = a^2(2a-1)^2 \geq 0,$$

so again $S_{AM} > \sqrt{3}/2$. The fourth edge is symmetric. Every enclosing equilateral triangle therefore has side greater than one, a contradiction.

If equality $a + b = 1$ occurred while all relevant points had positive margin, both boundary demands could be increased slightly; the closed conclusion would then be violated. $\square$

The corresponding center fact is exact and uses the interior condition at O.

Proposition 2.10 (Unique center midpoint)**.** *Let T_C be a closed unit equilateral triangle with $O \in \operatorname{int}(T_C)$. If T_C has positive-length intersection with a boundary edge of H, then it contains exactly one of $M_0, \ldots, M_5$.*

Proof. Normalize a positive overlap to

$$T_C \cap e_{0,1} = [s,t], \qquad 0 \leq s < t \leq 1,$$

where $e_{0,1}$ is parametrized from V_0 to V_1. In affine coordinates

$$X = V_0 + b(V_1 - V_0) + a(V_5 - V_0),$$

reflect if necessary so that the ratio of the two cutting slopes satisfies $0 < \lambda < 1$, and put $\rho = \sqrt{1 - \lambda + \lambda^2}$. The triangle has the exact side-slack representation

$$\begin{aligned} F_1 &= \lambda b + (1-\lambda)a - \lambda s, \\ F_2 &= -b + \lambda a + t, \\ F_0 &= (1-\lambda)b - a + \rho + \lambda s - t, \end{aligned} \qquad T_C = \{F_0, F_1, F_2 \geq 0\},$$

with $F_0 + F_1 + F_2 = \rho$. The case $\lambda = 1$ would give $F_0(O) = s - t < 0$; the reflected labelling supplies $0 < \lambda < 1$.

Set $C_j = F_j(O)$. Since O is interior,

$$C_0, C_1, C_2 > 0, \qquad C_0 + C_1 + C_2 = \rho.$$

The positive overlap is equivalent to

$$C_0 + (1-\lambda)C_2 < P, \qquad P = \rho(1-\rho).$$

Using $1 - \rho^2 = \lambda(1-\lambda)$, we have

$$P = \frac{\lambda(1-\lambda)\rho}{1+\rho} < \frac{\lambda(1-\lambda)}{2}, \qquad P < \frac{1-\lambda}{2}.$$

Substitution of the six midpoints gives the exact criteria

$$\begin{array}{c|l} M_0 & C_1 \geq \frac{1}{2} \\ M_1 & C_1 \geq \frac{1-\lambda}{2},\ C_2 \geq \frac{\lambda}{2} \\ M_2 & C_2 \geq \frac{1}{2} \\ M_3 & C_0 \geq \frac{\lambda}{2},\ C_2 \geq \frac{1-\lambda}{2} \\ M_4 & C_0 \geq \frac{1}{2} \\ M_5 & C_0 \geq \frac{1-\lambda}{2},\ C_1 \geq \frac{\lambda}{2}. \end{array}$$

Now

$$C_0 + C_2 < \frac{P}{1-\lambda},$$

and therefore

$$C_1 > \rho - \frac{P}{1-\lambda} = \frac{\rho(\rho-\lambda)}{1-\lambda} > \frac{1}{2};$$

the last inequality follows from

$$2\rho(\rho-\lambda) - (1-\lambda) = \frac{(1-\lambda)(\rho-\lambda)}{\rho+\lambda} > 0.$$

Thus $M_0 \in T_C$. On the other hand,

$$C_2 < \frac{P}{1-\lambda} < \frac{\lambda}{2}, \qquad C_0 < P < \min\left\{\frac{\lambda}{2}, \frac{1-\lambda}{2}, \frac{1}{2}\right\}.$$

The table excludes $M_1, \ldots, M_5$. Undoing the reflection and rotation proves the assertion. □

Remark 2.11. The hypothesis $O \in \operatorname{int}(T_C)$ cannot be weakened to $O \in T_C$: the triangle $\operatorname{conv}\{O, V_0, V_1\}$ has a boundary-edge overlap and contains both M_0 and M_1.

3. Strategy 1: direct length-sum obstructions

For a closed triangle T and a rectifiable target E, write

$$L_E(T) = \mathcal{H}^1(T \cap E).$$

The passage from an original open role U to its closure $T = \overline{U}$ can only enlarge the trace. Hence a cover of E by the open roles implies

$$\mathcal{H}^1(E) \leq L_E(T_C) + \sum_{i=0}^{5} L_E(T_i). \tag{6}$$

All estimates in this section concern full traces. They therefore remain valid for any assigned measurable subsets of those traces.

The strategy uses three one-dimensional targets: ∂H, D, and $S = \partial H \cup D$, whose measures were recorded in (1). The skeleton estimates below are conditional on specific center and row hypotheses; no unconditional noncoverage assertion for S is made.

3.1. **Exact neighboring-arm capacity.** We first record the exact neighboring-arm capacity. Normalize at V_0 and put

$$A_a = V_0 + a(V_5 - V_0), \quad B_b = V_0 + b(V_1 - V_0), \quad X_c = (1-c)V_1.$$

Let $C_+(a,b)$ be the largest $c \in [0,1]$ for which a closed unit equilateral triangle contains V_0, A_a, B_b, X_c; call it undefined when no such c exists. Reflection gives $C_-(a,b) = C_+(b,a)$ for the other neighboring arm.

Theorem 3.1 (Exact neighboring-arm capacity)**.** *For $0 \le a \le 1/2$, let $p(a)$ be the unique zero in $[a,1]$ of*

$$f_a(x) = x^3 - (a+2)x^2 + 2(a+1)x - 1,$$

and set

$$\sigma(a) = 1 - p(a), \qquad \tau(a) = 1 - a - (p(a)-a)(1-p(a)).$$

Then

$$C_+(a,b) = \begin{cases} \text{undefined}, & a+b > 1, \\ 1-b, & a+b \le 1,\ a > 1/2, \\ 1-b, & 0 \le a \le 1/2,\ 0 \le b \le \sigma(a), \\ p(a), & 0 \le a \le 1/2,\ \sigma(a) \le b \le \tau(a), \\ a + \dfrac{1}{2} - \sqrt{(a+b)^2 - \dfrac{3}{4}}, & 0 \le a \le 1/2,\ \tau(a) < b \le 1-a. \end{cases}$$

At $b = \tau(a)$ the plateau value $p(a)$ is used. In particular, if a vertex role has actual reaches $A_i + B_i > 1$, it has no positive-length trace on either adjacent radial arm; in fact it contains no point of either adjacent arm.

Proof. Use the cone basis

$$E_- = V_5 - V_0, \qquad E_+ = V_1 - V_0.$$

Then the four points have coordinates

$$(0,0), \quad (a,0), \quad (0,b), \quad (c,1),$$

and $\|uE_- + vE_+\|^2 = u^2 + v^2 - uv$. For $R^2 + S^2 - RS = 1$, define

$$U = (R-S)u + Sv, \quad V = Su - Rv, \quad W = Ru + (S-R)v.$$

A translate of the unit equilateral triangle with this orientation contains the four points exactly when

$$(*) \qquad \max W - \min U - \min V \le 1.$$

As in the support argument in Lemma 2.9, on every angular cell with fixed support vertices the left side of $(*)$ is a sinusoid with negative second derivative at an interior minimum candidate. Thus a minimizing orientation lies at a support breakpoint, where a side contains one of the hull edges. Checking OA, AX, XB, BO against the three possible support sides leaves, up to dominated reflected patterns, precisely the following three configurations:

active support	candidate c
OA and BX	$1-b$
$AX,\ B$ inactive	$p(a)$
$AX,\ B$ active	$a + \dfrac{1}{2} - \sqrt{(a+b)^2 - \dfrac{3}{4}}$.

We give the algebra selecting the cells, which also verifies that no squared branch has been retained incorrectly.

For the first pattern one may take $R = S = 1$. Its side length is $b + c$, and A is contained exactly when $a \le c$. Thus $c = 1 - b$ is feasible exactly when $a + b \le 1$.

For the second and third patterns, the side AX gives, with $d = c - a$, $R = Sd$. When B is inactive, the opposite support conditions give

$$(d(c-1)+1)^2 = d^2 - d + 1,$$

which factors as

$$(c-a)f_a(c) = 0.$$

For $0 \le a \le 1/2$, one has $f_a(a) = 2a - 1 \le 0$, $f_a(1) = a \ge 0$, and

$$f_a'(x) = 3x^2 - 2(a+2)x + 2(a+1) > 0$$

because its discriminant is $4(a^2 - 2a - 2) < 0$. Hence the required root $p(a)$ is unique. The remaining containment inequality is

$$b \le 1 - p(a)(1 + a - p(a)) = \tau(a),$$

and this branch dominates $1 - b$ exactly when $b \ge \sigma(a)$.

When B is active, the side-length equation gives $S = -1/(a+b)$ and

$$d^2 - d + 1 = (a+b)^2.$$

The relevant root $0 \le d \le 1/2$ yields the last expression in (3.1). Its nonactive inequalities reduce to

$$0 \le a \le \frac{1}{2}, \qquad \tau(a) \le b \le 1 - a.$$

The square root is real on this cell. Indeed, writing $p = p(a)$ gives

$$a = \frac{(1-p)(p^2-p+1)}{p(2-p)}, \qquad a + \tau(a) = \frac{p^2-p+1}{p(2-p)} \ge \frac{\sqrt{3}}{2}.$$

Finally, for $s = a + b \in [\sqrt{3}/2, 1]$,

$$s - \frac{1}{2} - \sqrt{s^2 - \frac{3}{4}} \ge 0,$$

so this branch dominates $1 - b$. The three selected cells are ordered because $\tau(a) - \sigma(a) = (p(a) - a)p(a) \ge 0$ and $\tau(a) \le 1 - a$. The support-edge enumeration shows that when $a + b > 1$ none is feasible, proving undefinedness. Reflection exchanges a, b and the two neighboring arms. □

3.2. **Conditional skeleton trace caps.** We now develop the skeleton estimate used only when the center is CE1 or CE2. The first lemma is the center contribution.

Lemma 3.2 (Center skeleton cap)**.** *If T_C is a CE1 or CE2 center role and contains exactly one radial midpoint, then*

$$L_S(T_C) < \frac{3}{2}.$$

Proof. Normalize the unique midpoint as M_0 and a positive boundary overlap to $e_{0,1}$. Use the side slacks from Proposition 2.10, with $0 < \lambda < 1$ and $\rho = \sqrt{1 - \lambda + \lambda^2}$. Put

$$X = F_0(O), \qquad Y = F_2(O), \qquad P = \rho(1-\rho).$$

The positive-length condition $s < t$ is equivalent to

$$X + (1-\lambda)Y < P. \tag{7}$$

The boundary trace on $e_{0,1}$ is

$$\ell_{01} = 1 - \lambda + Y - \frac{X + Y + 1 - \rho}{\lambda},$$

and the possible second trace on $e_{5,0}$ has length

$$\ell_{50} = \left[\frac{P - (\lambda X + Y)}{1-\lambda}\right]_+ .$$

Parametrizing the six radial arms by distance from O, direct substitution in the three side slacks gives the upper bounds

i	0	1	2	3	4	5
$\mathcal{H}^1(T_C \cap r_i)$	$\rho - X - Y$	Y/λ	Y	$\min\{X/\lambda, Y/(1-\lambda)\}$	X	$X/(1-\lambda)$.

Adding boundary and radial contributions yields

$$L_S(T_C) \le K_\lambda + \mathcal{E}(X,Y),$$

where

$$K_\lambda = 1 - \lambda + \rho - \frac{1-\rho}{\lambda}$$

and

$$\mathcal{E}(X,Y) = -\frac{X}{\lambda} + \frac{X}{1-\lambda} + Y + \min\left\{\frac{X}{\lambda}, \frac{Y}{1-\lambda}\right\} + \left[\frac{P - (\lambda X + Y)}{1-\lambda}\right]_+ .$$

Splitting according to which term realizes the minimum and whether the positive part vanishes gives, in all four cases,

$$\mathcal{E}(X,Y) \le \frac{P}{1-\lambda}. \tag{8}$$

For example, if $X/\lambda \le Y/(1-\lambda)$, the expression without the positive part is $X/(1-\lambda)+Y$; if the positive part is active, the excess over $P/(1-\lambda)$ is $X - \lambda Y/(1-\lambda) \le 0$. In the opposite order, the corresponding excess is $Y - (1-\lambda)X/\lambda \le 0$. If the positive part vanishes, (7) gives the same bound strictly.

It remains only to estimate a one-variable expression:

$$\frac{3}{2} - K_\lambda - \frac{P}{1-\lambda} = \frac{1 + A - 2\rho A}{2\lambda(1-\lambda)}, \qquad A = 1 + \lambda - \lambda^2.$$

Both sides in $1 + A > 2\rho A$ are positive, and

$$(1+A)^2 - 4A^2\rho^2 = \lambda^2(1-\lambda)^2(5 + 4\lambda - 4\lambda^2) > 0.$$

Thus $K_\lambda + P/(1-\lambda) < 3/2$, proving the lemma. ☐

Lemma 3.3 (Positive-support skeleton cap)**.** *Let T_i be the closure of an original open vertex role, so $V_i \in \mathrm{int}(T_i)$. If T_i has positive-length intersection with an adjacent radial arm, then*

$$L_S(T_i) < \frac{3}{2}.$$

Proof. Normalize to $i = 0$ and translate the hexagon by V_0:

$$\widetilde{H} = V_0 + H, \qquad \widetilde{O} = V_0, \qquad W_j = V_0 + V_j.$$

Then $W_3 = O$, $W_2 = V_1$, and $W_4 = V_5$. The five original skeleton components on which a unit triangle containing V_0 can have positive-length trace are

$$[V_0, V_1],\ [V_0, V_5],\ [O, V_1],\ [O, V_0],\ [O, V_5].$$

All five belong to the skeleton $\widetilde{S}$ of $\widetilde{H}$. Since $V_0 = \widetilde{O}$ is interior to T_0, the triangle is a legitimate center role for $\widetilde{H}$. Its positive adjacent-arm trace is a positive boundary-edge trace of $\widetilde{H}$, so it is CE1 or CE2 by Proposition 2.3. Proposition 2.10 and Lemma 3.2 therefore give

$$L_{\widetilde{S}}(T_0) < \frac{3}{2}.$$

A nonlocal component of S is more than unit distance from the interior point V_0, apart from zero-measure endpoints. Hence $L_S(T_0) \le L_{\widetilde{S}}(T_0)$. ☐

Lemma 3.4 (No-support skeleton cap)**.** *If a vertex role has $A_i + B_i \le 1$ and no positive-length trace on either adjacent radial arm, then*

$$L_S(T_i) \le 2.$$

Proof. Its boundary contribution is $A_i + B_i \le 1$ by (14). For $P = sV_j \in r_j \setminus \{O\}$,

$$\|V_i - P\|^2 = 1 + s^2 - 2s\cos\frac{(j-i)\pi}{3} > 1$$

when $j \notin \{i-1, i, i+1\}$. The two adjacent arms have zero-length trace by hypothesis, so only the own unit arm can contribute positive length. This adds at most one. ☐

For the remaining cap we use one specialized part of the exact local equilateral enclosure criterion. In local directions u, v meeting at $120°$, suppose a closed unit triangle contains

$$0, \quad au, \quad bv, \quad c(u+v),$$

with $0 \le a \le b \le 1$ and $a + b > 1$. The support-line calculation gives the selected supercritical cell

$$\begin{aligned} &a^2 + ab + b^2 \le 1, \qquad c \le \frac{1}{2}, \\ &(a^2-1)c^2 + (2ab^2 + b)c + (b^4 - b^2) \le 0, \end{aligned} \tag{9}$$

where c lies on the connected component containing $c = 0$. To see the selector directly, the quadratic is negative at $c = 0$, positive at $c = 1/2$, and opens downward. For the middle assertion, four times its value at $1/2$ is increasing in a and, at $a = 1 - b$, equals $b^2(2b-1)^2$; it is therefore positive when $a + b > 1$. Hence $c \le 1/2$ selects its smaller-root component. Formula (9) follows by fixing in the support sum the three active contacts $\{B\}$, $\{B, c(u+v)\}$, and $\{A\}$, and squaring the positive unit-side equation.

Lemma 3.5 (Supercritical skeleton cap)**.** *If $A_i + B_i > 1$, then*

$$L_S(T_i) < \frac{3}{2}.$$

Proof. Theorem 3.1 excludes both adjacent arms. Thus, after normalizing and writing $a = A_i$, $b = B_i$, the skeleton trace has length $a + b + c$, where c is the own-arm reach. Reflect so that $a \leq b$, and put

$$s = a + b, \qquad \delta = b - a, \qquad c_0 = \frac{3}{2} - s.$$

The diameter constraint gives

$$1 < s \leq \frac{2}{\sqrt{3}}, \qquad 0 \leq \delta \leq \sqrt{4 - 3s^2}.$$

Let $P(c)$ denote the quadratic in (9). Substitution and multiplication by 16 give

$$\begin{aligned} 16P(c_0) = Q(s, \delta) = \delta^4 &+ 8\delta^3 s - 6\delta^3 + 14\delta^2 s^2 - 18\delta^2 s + 5\delta^2 \\ &- 8\delta s^3 + 30\delta s^2 - 34\delta s + 12\delta \\ &+ s^4 - 6s^3 - 19s^2 + 60s - 36. \end{aligned}$$

Its derivative factors as

$$\frac{\partial Q}{\partial \delta} = 2(2\delta + 4s - 3)\big(\delta^2 + \delta(4s - 3) + (s - 1)(2 - s)\big) \geq 0.$$

Therefore

$$Q(s, \delta) \geq Q(s, 0) = (s - 1)(s^3 - 5s^2 - 24s + 36) > 0.$$

Indeed, the cubic is decreasing on $[1, 2/\sqrt{3}]$ and at its right endpoint equals $88/3 - 136\sqrt{3}/9 > 0$. Hence $P(c_0) > 0$. Since (9) selects the smaller-root component, $P(c) \leq 0$ forces $c < c_0$. Thus $a + b + c < 3/2$. □

Lemma 3.6 (A positive-support rescuer is forced)**.** *Suppose T_C is CE1 or CE2 and, after symmetry, contains exactly M_0. If at least two vertex rows are supercritical, then a vertex role distinct from a chosen supercritical role has positive-length adjacent-arm support.*

Proof. Choose a supercritical index $s \neq 0$. Lemma 2.9 gives $M_s \notin T_s$, and $M_s \notin T_C$ by the unique-midpoint property. In the original open cover, M_s is therefore contained by some other open vertex role. Diameter one restricts this role to U_{s-1} or U_{s+1}. Because it contains an interior point of the adjacent arm, its intersection with that arm has positive length. □

Theorem 3.7 (Conditional skeleton obstruction)**.** *If T_C is CE1 or CE2, $O \in \operatorname{int}(T_C)$, and $N_+ \geq 2$, the seven roles cannot cover S.*

Proof. Proposition 2.10 supplies the exact-one-midpoint hypothesis. Two supercritical roles, the forced positive-support rescuer, and the center each contribute strictly less than $3/2$, by Lemmas 3.5, 3.6, 3.3, and 3.2. The remaining three rows contribute at most 2 each; a further positive-support or supercritical row only improves this estimate. Therefore

$$L_S(T_C) + \sum_{i=0}^{5} L_S(T_i) < \frac{3}{2} + 2\left(\frac{3}{2}\right) + \frac{3}{2} + 3 \cdot 2 = 12,$$

contradicting (6) and $\mathcal{H}^1(S) = 12$. □

Proposition 3.8 (CE2 mixed positive-support obstruction)**.** *Assume T_C is CE2, $N_+ = 1$, exactly one vertex role is Vd1 or Vd2, and $k \geq 1$ additional vertex roles have positive-length adjacent-arm support. Then the seven roles cannot cover S.*

Proof. The unique supercritical role has no adjacent support by Theorem 3.1; it is therefore distinct from the Vd1/Vd2 role and the k additional positive-support roles. The center, supercritical role, Vd1/Vd2 role, and the k additional roles all have skeleton trace strictly below 3/2. Each of the remaining $4-k$ roles is nonsupercritical and has no adjacent support, so its trace is at most 2. Consequently

$$L_S(T_C) + \sum_{i=0}^{5} L_S(T_i) < \frac{3}{2}(3+k) + 2(4-k) = \frac{25-k}{2} \le 12.$$

For $k = 1$ the numerical right side equals 12, but the preceding trace bounds are strict, so the total is still strictly below 12. ☐

3.3. **Diagonal trace caps.**

Theorem 3.9 (Diagonal trace table)**.** *For closures of original open roles,*

role	*hypothesis*	L_D
T_C	CE1/CE2, $O \in \operatorname{int}(T_C)$	$< \frac{3}{2}$
T_i	Vd0, $A_i + B_i \le 1$	≤ 1
T_i	Vd0, $A_i + B_i > 1$	$< \frac{1}{2}$
T_i	*T3-like*	$< \frac{1}{2}$.

Proof. A vertex role T_i can meet only r_{i-1}, r_i, r_{i+1} in positive length. Indeed, for $P = sV_j \in r_j \setminus \{O\}$,

$$\|V_i - P\|^2 = 1 + s^2 - 2s\cos\frac{(j-i)\pi}{3} > 1$$

when $j \notin \{i-1, i, i+1\}$. A Vd0 role therefore meets only its own unit arm, proving the first Vd0 cap. If it is supercritical, the self-midpoint obstruction gives $M_i \notin T_i$. Since $T_i \cap r_i$ is an interval containing V_i, its length is then strictly below 1/2.

For the center cap, use the notation $\lambda, \rho, C_0, C_1, C_2$ from Proposition 2.10. The positive boundary overlap gives

$$C_0 + (1-\lambda)C_2 < \rho(1-\rho).$$

Direct radial exit bounds are

i	0	1	2	3	4	5
$\mathcal{H}^1(T_C \cap r_i)$ is at most	C_1	C_2/λ	C_2	C_0/λ	C_0	$C_0/(1-\lambda)$.

Thus, using $C_0 + C_1 + C_2 = \rho$,

$$\begin{aligned} L_D(T_C) &\le \rho + \frac{C_0 + (1-\lambda)C_2}{\lambda(1-\lambda)} \\ &< \rho + \frac{\rho(1-\rho)}{1-\rho^2} = \rho + \frac{\rho}{1+\rho} < \frac{3}{2}. \end{aligned}$$

The last inequality follows from

$$\frac{3}{2} - \rho - \frac{\rho}{1+\rho} = \frac{(1-\rho)(2\rho+3)}{2(1+\rho)} > 0.$$

It remains to treat T3-like roles. Apply the closed-trace majorant $\widehat{T}$ from Proposition 2.6, reflected so its supported adjacent arm is r_1. In its Type II parameters,

$$0 < t < 1, \quad \Delta = 1 - t + t^2, \quad z = \sqrt{\Delta}, \quad 0 < \beta < z,$$

and positive support is equivalent to

$$\beta < \frac{z-\Delta}{t} = \frac{(1-t)z}{1+z}.$$

The own-arm length and adjacent-arm length are respectively

$$c = \frac{\beta}{1-t}, \qquad d = \frac{z-\Delta-t\beta}{1-t}.$$

No other arm contributes. Hence

$$L_D(T_i) \le L_D(\widehat{T}) = \beta + \frac{z-\Delta}{1-t} < \frac{z-\Delta}{t(1-t)} = \frac{z}{1+z} < \frac{1}{2}.$$

□

3.4. **Vd1/Vd2 corner estimates.** We next isolate the much stronger trace loss caused by Vd1 and Vd2 geometry. For the following two results use the temporary coordinates

$$X = V_0 + x(V_5 - V_0) + y(V_1 - V_0).$$

Thus $V_0 = (0,0)$, $V_5 = (1,0)$, $V_1 = (0,1)$, $O = (1,1)$, and $\|(x,y)\|^2 = x^2 + y^2 - xy$.

Proposition 3.10 (Vd1/Vd2 corner normal form)**.** *Let T be the closure of an original Vd1 or Vd2 role at V_0, and let a, b be its exact reaches toward V_5, V_1. There is a unique $t > 0$ such that, with $d = \sqrt{t^2+t+1}$,*

$$T = \left\{ (x,y) : \begin{array}{r} x-(t+1)y \le a, \\ ty-(t+1)x \le tb, \\ tx+y \le d-a-tb \end{array} \right\}.$$

The raw traces, in coordinates measured from the indicated outer vertex, are

$$\begin{array}{c|c} r_0 & 0 \le s \le \dfrac{d-a-tb}{t+1} \\ r_1 & \dfrac{t(1-b)}{t+1} \le s \le \dfrac{d-a-tb-1}{t} \\ r_5 & \dfrac{1-a}{t+1} \le s \le d-a-tb-t. \end{array}$$

Intersecting with $[0,1]$ gives the actual traces. Moreover,

$$a + b < \frac{1}{2}. \tag{10}$$

Proof. A Vd1/Vd2 triangle has one vertex W outside H and two vertices U, Z in H. Since V_0 is an interior convex combination of them, both coordinates of W are negative. The two axis endpoints $(a,0), (0,b)$ lie on distinct sides incident with W. Put $p = U - W$, $q = Z - W$. Both vectors have positive coordinates and satisfy

$$\|p\| = \|q\| = \|p - q\| = 1.$$

The order in which the two sides cross the positive axes selects the $60°$ rotation

$$q = (p_x - p_y, p_x), \qquad p_x > p_y > 0.$$

Writing $t = (p_x - p_y)/p_y$ gives uniquely

$$p = \frac{(t+1, 1)}{d}, \qquad q = \frac{(t, t+1)}{d}.$$

The two side lines through the axis endpoints and the parallel third side are exactly those in (3.10). Substitution of $(s,s),(s,1),(1,s)$ gives (3.10).

If the r_1 trace has positive length, its lower endpoint is smaller than its upper endpoint. Clearing denominators gives

$$(t+1)a+tb<(t+1)(d-1)-t^2.$$

Since the left side is at least $t(a+b)$,

$$a+b<\frac{(t+1)(d-1)-t^2}{t}<\frac{1}{2}.$$

For the last inequality, both sides are positive and

$$\left(t^2+\frac{3t}{2}+1\right)^2-(t+1)^2(t^2+t+1)=\frac{t^2}{4}>0.$$

Reflection proves the same conclusion when the supported arm is r_5. □

Lemma 3.11 (Vd2 neighboring-midpoint cap). *If an original Vd2 role contains either neighboring midpoint, then its exact boundary reaches satisfy*

$$a+b<\frac{1}{3}.$$

Proof. In Proposition 3.10, reflection sends (t,a,b,M_1,M_5) to $(1/t,b,a,M_5,M_1)$, so assume $t\geq 1$ and put $U=a+tb$.

If $M_5=(1,1/2)$ is contained, the third half-plane in (3.10) gives

$$a+b\leq U\leq d-t-\frac{1}{2}\leq\sqrt{3}-\frac{3}{2}<\frac{1}{3};$$

the middle expression decreases for $t\geq 1$.

If $M_1=(1/2,1)$ is contained, the second and third half-planes give

$$b\geq\frac{t-1}{2t},\qquad a+b\leq S(t):=d-t-\frac{1}{2t}.$$

Positive-length contact with the other adjacent arm r_5 gives, by (3.10),

$$a+b<R(t):=\frac{2(t+1)d-3t^2-t-2}{2t}.$$

Direct differentiation shows that S is increasing and R decreasing on $[1,\infty)$. For example, the sign calculation for S' reduces to

$$(2t+1)^2t^4-(t^2+t+1)(2t^2-1)^2=(t+1)(t^3+3t^2-1)>0,$$

while $R'<0$ follows from $d>t+1/2$. Splitting at $t=4/3$ yields

$$\begin{aligned}1\leq t\leq\frac{4}{3}:&\quad a+b\leq S(4/3)=\frac{8\sqrt{37}-41}{24}<\frac{1}{3},\\ t\geq\frac{4}{3}:&\quad a+b<R(4/3)=\frac{7\sqrt{37}-39}{12}<\frac{1}{3}.\end{aligned}$$

The final inequalities follow respectively from $8\sqrt{37}<49$ and $7\sqrt{37}<43$. □

3.5. **Exact center intervals and boundary trace caps.** The next two lemmas turn the unique-midpoint normalization from Proposition 2.10 into exact boundary data.

Lemma 3.12 (CE1 maximal interval)**.** *Normalize an exact-M_0 CE1 center triangle so that its only positive-length boundary trace is*

$$T_C \cap e_{0,1} = [s, t].$$

Then $0 < s < t < 1$ *and*

$$t \le f(s) := \sqrt{s^2 - s + 1} - (1 - s)^2. \tag{11}$$

In the closed relaxation $O \in T_C$, the maximal intervals under inclusion are exactly $[s, f(s)]$, $0 < s < 1$. They are mutually incomparable.

Proof. Use the side slacks in the proof of Proposition 2.10. Exact M_0 and positive overlap force $0 < \lambda < 1$. The center condition gives

$$t \le \rho - \lambda(1 - s), \qquad \rho = \sqrt{1 - \lambda + \lambda^2}.$$

Since CE1 has no positive-length trace on $e_{5,0}$, restricting the slacks to that edge gives

$$t \ge \rho - \frac{\lambda^2}{1 - \lambda} s.$$

Compatibility implies $\lambda \ge 1 - s$. Put $q = 1 - s$ and write $\lambda = q + d$, $d \ge 0$. Then

$$t \le \sqrt{1 - \lambda + \lambda^2} - q\lambda \le \sqrt{1 - q + q^2} - q^2 = f(s).$$

For completeness, the second inequality follows after squaring from

$$d(1 - q^2) \le 1 - 2q + 2q\sqrt{1 - q + q^2};$$

its left side is at most $(1 - q)(1 - q^2)$, and the difference is

$$q\sqrt{1 - q + q^2}\left(2 - \sqrt{1 - q + q^2}\right) > 0.$$

For $0 < s < 1$, choose $\lambda = 1 - s$ and $t = f(s)$. Both displayed center and no-$e_{5,0}$ inequalities are equalities. The target length is

$$t - s = \rho - \rho^2 = \rho(1 - \rho) > 0, \qquad \rho = \sqrt{s^2 - s + 1},$$

and direct substitution gives the strict exclusions of M_1 and M_5; the remaining midpoints are then excluded by the same side slacks. Thus the frontier is feasible in the closed relaxation. Finally,

$$f'(s) = 2 - 2s + \frac{2s - 1}{2\sqrt{s^2 - s + 1}} > 0,$$

so two different frontier intervals have both left and right endpoints strictly ordered and neither contains the other. □

Lemma 3.13 (CE2 interval-pair characterization)**.** *Normalize an exact-M_0 CE2 center triangle so that*

$$T_C \cap e_{5,0} = [x, u], \qquad T_C \cap e_{0,1} = [y, v].$$

Put $S_0 = x + y$ and $\Delta = \sqrt{x^2 + xy + y^2}$. The feasible interval pairs are exactly those satisfying

$$0 < x < u < 1, qquad 0 < y < v < 1,$$

$$(u+v)S_0 - xy = \Delta, \tag{12}$$

$$uS_0 \geq x, \quad vS_0 \geq y, \qquad uS_0 < x + \frac{y}{2}, \quad vS_0 < y + \frac{x}{2}. \tag{13}$$

Every such pair is maximal under product interval inclusion. This is the exact maximal closed *interval-pair domain. When the pair is the closure of an original open center role, the two center inequalities retain the strict form*

$$uS_0 > x, \qquad vS_0 > y.$$

Proof. The point V_0 cannot belong to T_C: otherwise O, V_0, at distance one, would be two vertices of the unit triangle, and its third vertex would be V_1 or V_5, allowing only one of the two stated boundary overlaps. The unique side violated at V_0 therefore cuts the two initial endpoints x, y. Its outward normal and the two rotations by $120°$ are

$$n_2 = \frac{1}{2\Delta}(\sqrt{3}(x+y), y-x), \qquad n_0 = R_{2\pi/3}n_2, \qquad n_1 = R_{-2\pi/3}n_2.$$

Writing the triangle as $n_j \cdot X \leq \alpha_j$, the four endpoints give

$$\alpha_2 = \frac{\sqrt{3}(S_0 - xy)}{2\Delta},$$
$$\alpha_0 = \frac{\sqrt{3}(vS_0 - y)}{2\Delta},$$
$$\alpha_1 = \frac{\sqrt{3}(uS_0 - x)}{2\Delta}.$$

The unit side condition $\alpha_0 + \alpha_1 + \alpha_2 = \sqrt{3}/2$ is precisely (12). The conditions $O \in T_C$ are the first two weak inequalities in (13). Substitution of M_1 and M_5 shows that their exclusion is exactly the last two strict inequalities; these also exclude M_2, M_3, M_4, while the weak inequalities imply $M_0 \in T_C$. Restricting the three side slacks to the two target edges gives exactly $[x, u]$ and $[y, v]$. On each remaining edge one of the following necessary slack sums is negative:

$$\Delta + xy - (2x + y), \qquad \Delta + xy - (x + 2y),$$

because

$$(2x + y - xy)^2 - \Delta^2 = x\big(x(1-y)(3-y) + y(3-2y)\big) > 0,$$
$$(x + 2y - xy)^2 - \Delta^2 = y\big(x(3-2x) + y(1-x)(3-x)\big) > 0.$$

Thus there are no further positive boundary traces.

For maximality define

$$F(x,y) = \frac{\Delta + xy}{x+y} = u + v.$$

The weak center inequalities imply the start domain $2x + 2y - xy \geq 1$. On it,

$$\frac{\partial F}{\partial x} = \frac{y(x - y + 2y\Delta)}{2\Delta(x+y)^2} > 0, \quad \frac{\partial F}{\partial y} = \frac{x(y - x + 2x\Delta)}{2\Delta(x+y)^2} > 0.$$

If another feasible pair product-dominated the given one, its initial coordinates would be no larger and its terminal coordinates no smaller. Monotonicity would make its terminal sum no larger, while domination makes it no smaller. Equality forces all four endpoints to agree. $\square$

Theorem 3.14 (Boundary trace table)**.** *For closures of original open roles, the following bounds hold:*

role	*hypothesis*	$L_{\partial H}$
T_C	CE0	0
T_C	CE1	$\leq \dfrac{\sqrt{3}}{2} - \dfrac{3}{4}$
T_C	CE2	$< \dfrac{1}{2}$
T_i	Vd0, $A_i + B_i \leq 1$	≤ 1
T_i	Vd0, $A_i + B_i > 1$	$\leq \dfrac{2}{\sqrt{3}}$
T_i	Vd1/Vd2	$< \dfrac{1}{2}$
T_i	*T3-like*	$< 1.$

Proof. For every vertex role,

$$L_{\partial H}(T_i) = A_i + B_i. \tag{14}$$

Indeed, the two incident traces are initial intervals of these lengths. A relative-interior point of any nonincident edge is more than unit distance from V_i, whereas V_i is interior to the diameter-one triangle T_i. This excludes any other positive-length boundary trace.

The CE0 row is the definition. In the CE1 row put

$$\rho = \sqrt{s^2 - s + 1} \in [\sqrt{3}/2, 1).$$

Lemma 3.12 gives

$$L_{\partial H}(T_C) = t - s \leq \rho(1 - \rho) \leq \frac{\sqrt{3}}{2} - \frac{3}{4}.$$

For CE2, Lemma 3.13 gives

$$\begin{aligned} L_{\partial H}(T_C) &= (u - x) + (v - y) \\ &= \frac{\Delta + xy}{S_0} - S_0 = \frac{\Delta(1 - \Delta)}{S_0}. \end{aligned}$$

The center inequalities imply $\Delta + xy \geq S_0$, whence $S_0 \geq (1 + xy)/2 > 1/2$ after squaring. Since the trace is positive, $0 < \Delta < 1$, and therefore

$$L_{\partial H}(T_C) \leq \frac{1}{4S_0} < \frac{1}{2}.$$

For a nonsupercritical Vd0 row, (14) is enough. For a supercritical row, its incident endpoints are at squared distance $A_i^2 + A_iB_i + B_i^2 \leq 1$, and hence

$$\frac{3}{4}(A_i + B_i)^2 \leq 1.$$

This proves the $2/\sqrt{3}$ cap. The Vd1/Vd2 row is (10). Finally, the Type II calculation in Proposition 2.6 gives for an original T3-like role

$$A_i + B_i = z - \alpha < z < 1$$

because $0 < t < 1$ and $\alpha > 0$. $\square$

3.6. **Terminal length contradictions.**

Proposition 3.15 (Boundary-length branches)**.** *The following branches are impossible:*

(1) *CE0 with $N_+ = 0$;*
(2) *CE0 with $N_+ = 1$ and at least one Vd1/Vd2 role;*
(3) *CE1 or CE2 with $N_+ = 0$ and at least one Vd1/Vd2 role;*
(4) *CE1 with $N_+ = 1$ and at least one Vd1/Vd2 role;*
(5) *CE2 with $N_+ = 1$ and at least two Vd1/Vd2 roles.*

Proof. In CE0, the center open role contains no relative-interior boundary point. No nonincident vertex role can contain such a point either. Hence on every edge the two incident open vertex traces form two nonempty relatively open sets covering a connected interval; they overlap in positive length. Thus

$$1 < B_i + A_{i+1}$$

on every edge. Summing gives $6 < \sum_i (A_i + B_i)$, contradicting $N_+ = 0$. This proves (1).

For the other cases apply Theorem 3.14. Put

$$\gamma_1 = \frac{\sqrt{3}}{2} - \frac{3}{4}.$$

Vd1, Vd2, and T3-like roles are nonsupercritical, so whenever $N_+ = 1$ the unique supercritical role in these cases is Vd0. The respective total trace bounds are

case	$L_{\partial H}(T_C) + \sum_i L_{\partial H}(T_i)$
CE0, $N_+ = 1$, ≥ 1 Vd1/Vd2	$< \frac{1}{2} + \frac{2}{\sqrt{3}} + 4 < 6$
CE1, $N_+ = 0$, ≥ 1 Vd1/Vd2	$< \gamma_1 + \frac{1}{2} + 5 < 6$
CE2, $N_+ = 0$, ≥ 1 Vd1/Vd2	$< \frac{1}{2} + \frac{1}{2} + 5 = 6$
CE1, $N_+ = 1$, ≥ 1 Vd1/Vd2	$< \gamma_1 + \frac{1}{2} + \frac{2}{\sqrt{3}} + 4 < 6$
CE2, $N_+ = 1$, ≥ 2 Vd1/Vd2	$< \frac{1}{2} + 2\left(\frac{1}{2}\right) + \frac{2}{\sqrt{3}} + 3 < 6.$

Each contradicts (6) for the length-six boundary. The numerical inequalities follow, for example, from $9/2 + 2/\sqrt{3} < 6$ and $15/4 + 7/(2\sqrt{3}) < 6$. □

Proposition 3.16 (Vd2 neighboring-midpoint branch)**.** *Assume a CE2, $N_+ = 1$ configuration has exactly one Vd1/Vd2 role, that role is Vd2 and contains a neighboring midpoint, and all other vertex roles are Vd0. Then the boundary cannot be covered.*

Proof. Lemma 3.11 and Theorem 3.14 give

$$L_{\partial H}(T_C) + \sum_i L_{\partial H}(T_i) < \frac{1}{2} + \frac{1}{3} + \frac{2}{\sqrt{3}} + 4 < 6.$$

The last inequality is equivalent to $12 < 7\sqrt{3}$, whose square is $144 < 147$. □

Proposition 3.17 (At least two T3-like roles)**.** *Assume the center is CE1 or CE2, $N_+ = 1$, no vertex role is Vd1 or Vd2, and at least two vertex roles are T3-like. Then the diagonal target D cannot be covered.*

Proof. Let $k \geq 2$ be the number of T3-like roles. By Lemma 2.7, the unique supercritical role is Vd0; $2 \leq k \leq 5$, and the remaining $5-k$ roles are nonsupercritical Vd0. Theorem 3.9 gives

$$L_D(T_C) + \sum_i L_D(T_i) < \frac{3}{2} + \frac{1}{2} + \frac{k}{2} + (5-k) = 7 - \frac{k}{2} \leq 6,$$

contradicting $\mathcal{H}^1(D) = 6$ and (6). □

Proposition 3.18 (Branches closed by Strategy 1)**.** *Direct length-sum obstructions close precisely the following rows of the final routing table:*

(1) *CE0, $N_+ = 0$;*
(2) *CE0, $N_+ = 1$, with a Vd1/Vd2 role;*
(3) *CE1/CE2, $N_+ = 0$, with a Vd1/Vd2 role;*
(4) *CE1, $N_+ = 1$, with a Vd1/Vd2 role;*
(5) *CE2, $N_+ = 1$, with at least two Vd1/Vd2 roles;*
(6) *CE1/CE2, $N_+ = 1$, with no Vd1/Vd2 role and at least two T3-like roles;*
(7) *CE1/CE2, $N_+ \geq 2$;*
(8) *within the CE2, $N_+ = 1$ package with exactly one Vd1/Vd2 role, the Vd2 neighboring-midpoint subcase and every subcase with an additional positive-support role.*

Proof. Items (1)–(5) are Proposition 3.15; item (6) is Proposition 3.17; item (7) is Theorem 3.7; and the two subfamilies in item (8) are Propositions 3.16 and 3.8. These are only the indicated subcases of the exceptional CE2 row; its remaining placements are handled by the exact-demand strategy. □

4. Strategy 2: exact boundary–radial demand propagation

This section treats the configurations in which a coarse length sum does not by itself record enough information. The mechanism is always the same. A center trace fixes boundary inputs and complementary radial demands; an exact local map propagates these data through nonsupercritical rows; and the returning demand exceeds the remaining boundary or radial capacity.

Actual maximal reaches continue to be denoted by (A_i, B_i, C_i), while lowercase letters denote prescribed demands. The only machine-assisted statement used in this section is the rational interval certificate Theorem B.2 in Appendix B. All other estimates in this section are proved analytically. In particular, no empirical branch catalogue and no formal root from a discarded algebraic component is used.

4.1. **The exact local feasible set.**

Put

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \qquad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).$$

For $0 \leq a, b, c \leq 1$ define

$$K(a,b,c) = \operatorname{conv}\{0, au, bv, c(u+v)\}.$$

Thus a, b are lower demands on the two incident boundary rays and c is a lower demand on the inward radial ray.

Definition 4.1. The local admissible set is

$$\mathcal{A} = \{(a,b,c) \in [0,1]^3 : K(a,b,c) \text{ is contained in a closed unit equilateral triangle}\}.$$

For $a, c \in [0,1]$ put

$$\begin{aligned} B_c(a) &= \max\{b : (a,b,c) \in \mathcal{A}\}, \\ F_c(a) &= \min\{B_c(a), 1-a\}, \\ G_c(a) &= 1 - F_c(a). \end{aligned}$$

The distinction between demands and actual reaches is important. If a role has actual reaches (A,B,C) and $A \geq a$, $C \geq c$, then (a,B,c) is a demand triple realized by the same role. If in addition $A + B \leq 1$, then

$$B \leq F_c(a). \tag{15}$$

No assertion about the type of a maximizing triangle is required.

Proposition 4.2 (Exact admissible set and radial envelope)**.** *Put*

$$s = a+b, \qquad d = a^2+ab+b^2, \qquad m = \min\{a,b\}, \qquad M = \max\{a,b\},$$

and

$$q = s^4 - s^2 + ab.$$

Define

$$\begin{aligned} F_L(a,b,c) &= c^4 - c^2 + mc - m^2, \\ F_T(a,b,c) &= (s^2-1)c^2 + Mc - M^2, \\ F_S(a,b,c) &= (m^2-1)c^2 + (2mM^2+M)c + M^4 - M^2. \end{aligned}$$

Then $\mathcal{A} = \mathcal{A}_L \cup \mathcal{A}_T \cup \mathcal{A}_S$*, where*

$$\begin{aligned} \mathcal{A}_L &= \{d \leq 1, s \leq 1, q \leq 0, F_L \leq 0\}, \\ \mathcal{A}_T &= \{d \leq 1, s \leq 1, q \geq 0, F_T \leq 0, c \leq 2M\}, \\ \mathcal{A}_S &= \{d \leq 1, s > 1, F_S \leq 0, c \leq \tfrac{1}{2}\}. \end{aligned}$$

All variables in these three sets are also constrained to $[0,1]$*.*

For $d \leq 1$*, the maximal radial demand is*

$$c_{\max}(a,b) = \begin{cases} 1, & a = b = 0, \\ C_L(m), & s \leq 1, q \leq 0, (a,b) \neq (0,0), \\ C_T(m,M), & s \leq 1, q > 0, \\ C_S(m,M), & s > 1, \end{cases} \tag{16}$$

where $C_L(m)$ *is the unique root in* $[\sqrt{3}/2, 1]$ *of*

$$z^4 - z^2 + mz - m^2 = 0,$$

and

$$C_T(m,M) = \frac{2M}{1+\sqrt{4s^2-3}}, \qquad C_S(m,M) = \frac{M(2mM+1) - M\sqrt{4d-3}}{2(1-m^2)}.$$

If $d > 1$*, the radial envelope is undefined. At* $q = 0$ *the two subcritical formulas agree and equal* s*.*

Proof. If $s > 0$ and $cs \le ab$, then $c/a + c/b \le 1$ (with the zero-coordinate cases obtained by continuity), so $c(u+v)$ lies in $\mathrm{conv}\{0, au, bv\}$. Its minimum enclosing equilateral side is the diameter

$$\sqrt{a^2+ab+b^2}.$$

Assume now that $cs > ab$. The four points occur cyclically as $0, bv, c(u+v), au$. For three outward unit normals separated by $120°$, the required side equals $2/\sqrt{3}$ times the sum of the three support values. On an angular cell with fixed support points this sum has second derivative equal to its negative. Hence an angular minimum occurs when a normal is perpendicular to one of the four hull edges. The four resulting side lengths are

$$L_{0A} = b + \max\{a, c\}, \qquad L_{B0} = a + \max\{b, c\},$$

$$L_{AC} = \begin{cases} (a^2+bc)/\sqrt{a^2-ac+c^2}, & a \ge c, \\ c(a+b)/\sqrt{a^2-ac+c^2}, & a < c \le a+b, \\ c^2/\sqrt{a^2-ac+c^2}, & c > a+b, \end{cases}$$

and L_{CB} is obtained by interchanging a, b. Thus the minimum side is the minimum of these four expressions.

Ordering a, b as $m \le M$, squaring the positive support equations gives $F_L = 0, F_T = 0, F_S = 0$ in the three respective contact regimes; the inactive support inequalities give $q \le 0, q \ge 0, s > 1$. Squaring creates one spurious component in each of the last two cells. In the T cell the two roots are

$$\frac{2M}{1+\sqrt{4s^2-3}}, \qquad \frac{2M}{1-\sqrt{4s^2-3}},$$

and $c \le 2M$ selects the first. In the S cell the smaller root lies below $1/2$ and the other above $1/2$; hence $c \le 1/2$ is exactly the geometric selector. The L fiber is the interval from zero to its unique selected root. Solving the selected equations gives (16). This also proves that $\mathcal{A}$ is coordinatewise down-closed. ☐

4.2. **The exactly-one-supercritical all-**Vd0 **branch.** For a center interval, call the nonempty closed set missed by its two open endpoint roles a *V-gap*. It may be a singleton. If the maximal closed center trace is $[s, t]$, open coverage gives the weak bounds $b_i \ge s$ and $a_{i+1} \ge 1 - t$; thus none of the arguments below discards a singleton gap.

Proposition 4.3 (CE1 one-gap five-map obstruction)**.** *Assume that every vertex role is* Vd0*,* $N_+ = 1$*, and the center role is CE1. If its boundary trace contains a V-gap, the roles do not cover* H*.*

Proof. By midpoint locality, rows $1, \dots, 5$ contain their own radial midpoints; hence they are nonsupercritical and row 0 is the unique supercritical row. Put

$$R = \lambda, \quad w = 1 - R, \quad E = \sqrt{1 - Rw}, \quad \eta = 1 - E, \quad P = \eta E,$$

and let $A = C_0$, $D = C_2$ in Proposition 4.13. Then, with $X = R - D$, $m = A/R$ and $H = (\eta + A + D)/(2R) = s/2$, the exact strict domain is

$$\begin{gathered} A > 0, \quad D > 0, \quad A + wD < P, \quad D + RA \ge P, \\ A < \frac{w}{2}, \quad D < \frac{R}{2}, \quad RD - wA > 0, \end{gathered} \tag{17}$$

and the six complementary demands are

$$c_0=\eta{+}A{+}D,\quad c_1=1{-}\frac{D}{R},\quad c_2=1{-}D,\quad c_3=1{-}m,\quad c_4=1{-}A,\quad c_5=1{-}\frac{A}{w}.$$

The gap gives $B_0\ge s$ and $A_1\ge X$. Boundary propagation through the five nonsupercritical rows therefore gives

$$A_0\ge Z:=(G_{c_5}\circ G_{c_4}\circ G_{c_3}\circ G_{c_2}\circ G_{c_1})(X),\qquad A_0\le B_{c_0}(s).$$

It remains to prove the reverse strict inequality.

Repeated use of (30) reduces it to

$$(G_{c_2}\circ G_{c_3}\circ G_{c_4})(H)>1-X. \tag{18}$$

The elementary low-root estimates give

$$e(A)<X,\qquad H>A.$$

At row 4, every label except T_+^{hi} already yields $G_{1-A}(H)>1-X$. On T_+^{hi} its selector forces $X>1/2$. Indeed, if $X\le 1/2$, the selector and the two center inequalities give

$$Q(A):=\eta+P+wA-2R(2A+5A^2)\le 0,\qquad 0<A<A_0:=P-w(R-\tfrac12).$$

The quadratic Q is strictly concave and $Q(0)=\eta+P=Rw>0$. Rationalize by

$$k=\frac{R}{1+E},\qquad R=\frac{k(2-k)}{1-k^2},\quad E=\frac{1-k+k^2}{1-k^2},\quad \eta=\frac{k(1-2k)}{1-k^2}.$$

When $R\le 1/2$, one has $k\le 2-\sqrt3<27/100$ and $A_0\ge P$, while

$$Q(P)=\frac{k(1-2k)(-f_8(k))}{(1-k^2)^5},$$

where

$$f_8(k)=16k^8-77k^7+175k^6-244k^5+199k^4-101k^3+13k^2+12k-3.$$

For $0\le k\le 27/100$, put

$$q(k)=\frac{2333}{5}k^3-303k^2+26k+12.$$

Then

$$f_8'(k)-q(k)=\frac{k^3}{5}\left(640k^4+k^2(5250-2695k)+1647-6100k\right)\ge 0.$$

Writing $t=27/100-k$, one also has

$$q(k)\ge\frac{30576939}{5000000}+\frac{1089743}{50000}t>0,$$

because the difference between the two sides is

$$-\frac{t(100t-27)(116650t+12759)}{25000}\ge 0.$$

Thus f_8 is increasing. At $k_0=2-\sqrt3$,

$$f_8(k_0)=77724-44874\sqrt3<0,\qquad 3\cdot 44874^2-77724^2=7452>0.$$

Hence $f_8(k)<0$ for $k\le k_0$, and so $Q(P)>0$. When $R>1/2$, $0<A_0<P$ and

$$Q(A_0)=\frac{(1-2k)(-g_5(k))}{2(1-k^2)^3},\qquad g_5(k)=32k^5-118k^4+150k^3-86k^2+18k-1.$$

At $k_0 = 2 - \sqrt{3}$, $g_5(k_0) = 3471 - 2004\sqrt{3} < 0$ because $3 \cdot 2004^2 - 3471^2 = 207$. If $y = 4k - 1 \in [0, 1]$, then

$$8g_5'(k) = 5y^4 - 39y^3 + 78y^2 - 51y - 29 \le y(-34y^2 + 78y - 51) - 29 < 0,$$

because the displayed quadratic has discriminant -852. Thus g_5 decreases and remains negative for $k > k_0 > 1/4$, so $Q(A_0) > 0$. In either range concavity makes $Q(A) > 0$ throughout the allowed interval, contradicting $Q(A) \le 0$ and proving $X > 1/2$.

Put $p_1 = G_{1-A}(H)$. If row 3 is not T_+^{hi}, then $p_1 \le A + e(A) < 3A + 5A^2 < 29/64 < 1/2$, and $2R(H - m) = \eta + D - A > \eta - P > 0$. Thus both Full branches are impossible; on L, T_-, or the low-c tie one has $F_{1-m}(p_1) \le p_1$, proving (18). We use the following elementary property of a selected T_+ arc. Put $d = 1 - c$, let $q = G_c(p)$, and write $u = q - p$. The selected equation is

$$f(q) := (1 - q)(q - d) = g(u) := (1 - d)^2 u(2 - u).$$

On the interior of the arc, $f', g' > 0$. If

$$A_0 = f'(q) = 1 + d - 2q, \qquad B_0 = g'(u) = 2(1 - d)^2(1 - u),$$

then

$$u' = \frac{A_0}{B_0}, \qquad u'' = \frac{2((1 - d)^2 A_0^2 - B_0^2)}{B_0^3}.$$

Moreover

$$B_0 - (1 - d)A_0 = (1 - d)(1 - 3d + 2p + 2du) > 0$$

and

$$B_0 - A_0 \ge (1 - d)(1 - 2d) > 0.$$

Thus $p = q - u$ is increasing and strictly convex as a function of q. Consequently $q = G_{1-d}(p)$ is increasing and strictly concave on each selected arc; a tie endpoint follows by continuity.

When $d < 1 - \sqrt{3}/2$, the arc endpoints are (d, d) and $(e(d), d + e(d))$. Its chord and $e(d) < 2d/(1 - 2d)$ give

$$q \ge p + \frac{d}{e(d) - d}(p - d) > p + \frac{1 - 2d}{1 + 2d}(p - d).$$

This is stronger than both the row-4 coefficient $1 - 4d$ and the row-3 coefficient $1 - 5d$. In the low-c row-3 range $1 - \sqrt{3}/2 \le d < 1/5$, the other endpoint is $(h(1 - d), 1 - h(1 - d))$, and the chord coefficient is

$$\frac{1 - 2h(1 - d)}{h(1 - d) - d} > \frac{1}{3} > 1 - 5d.$$

Indeed $h(1 - d) < (3 + d)/7$: substitution of $(3 + d)/7$ into its defining upward quadratic gives

$$\frac{-1 + 30d - 65d^2 + 16d^3 + 4d^4}{49} > 0.$$

Here $d \ge 1 - \sqrt{3}/2 > 2/15$, and $-1 + 30d - 65d^2$ is increasing and already equals $83/45$ at $2/15$. For $d \ge 1/5$, extensivity alone is stronger than the row-3 estimate.

Applying these chord bounds first with $d = A$ and then with $d = m$ proves, without an omitted selected branch,

$$p_1 > L_1 := (2 - 4A)H - (1 - 4A)A,$$
$$p_2 := G_{1-m}(p_1) > L_2 := (2 - 5m)L_1 - (1 - 5m)m.$$

Finally let

$$D_0 = P - RA, \qquad D_h = A(4R-1) + 10RA^2 - \eta, \qquad x = Rw.$$

Feasibility puts $D_0 \le D \le D_h$. We first prove that the actual feasible value satisfies $D < 1/10$. Since

$$P = \frac{xE}{1+E} < \frac{x}{2}, \qquad \eta = \frac{x}{1+E} > \frac{x}{2},$$

the contrary assumption $D \ge 1/10$ and $A + wD < P$ give

$$A < \overline{A} := \frac{w(5R-1)}{10}.$$

The function $D_h(A)$ is increasing, so

$$D \le D_h(A) < D_h(\overline{A}) < U(R) := \overline{A}(4R-1) + 10R\overline{A}^2 - \frac{x}{2}.$$

Direct expansion yields

$$\frac{1}{10} - U(R) = -\frac{R}{10}P_4(R), \qquad P_4(R) = 25R^4 - 60R^3 + 26R^2 + 22R - 14.$$

For $y = 2R - 1 \in [0,1]$,

$$16P_4(R) = 25y^4 - 20y^3 - 106y^2 + 124y - 39 \le -101y^2 + 124y - 39 < 0.$$

The last quadratic has discriminant -380. Hence $U(R) < 1/10$, a contradiction, and therefore $D < 1/10$.

It remains to compare L_2 with $2D + 3D^2$. Put

$$J(D) = L_2(D) - \frac{23}{10}D.$$

Since

$$L_2 - 2D - 3D^2 - J(D) = 3D\left(\frac{1}{10} - D\right) > 0,$$

it is enough to prove $J(D) > 0$. Exact differentiation gives

$$J'(D) = \frac{S(R,A)}{10R^2}, \qquad S(R,A) = 100A^2 - (40R+50)A + R(20-23R).$$

The inequality $D_0 \le D_h$ is equivalent to

$$x \le A(5R - 1 + 10RA).$$

Because $A < P < x/2 \le 1/8$,

$$A > L := \frac{4x}{25R-4}.$$

Also $\partial_A S = 200A - 40R - 50 < -45$, and direct substitution gives

$$S(R,L) = -\frac{Rq_3(R)}{(25R-4)^2},$$

where

$$q_3(R) = 8775R^3 - 14260R^2 + 7928R - 1120.$$

For $y = 2R - 1$,

$$8q_3(R) = 8775y^3 - 2195y^2 + 997y + 3007 \ge 812 > 0.$$

Thus $J'(D) < 0$ and $J(D) \ge J(D_h)$.

At $D = D_h$ one has $H = 2A + 5A^2$. Write $L_{2,h} = L_2(D_h)$. Expansion gives

$$L_{2,h} - A(1 + 4R) = \frac{A}{R^2}\mathcal{Q}(R, A),$$

where

$$\mathcal{Q}(R, A) = 100A^3R - 40A^2R^2 - 30A^2R + 12AR^2 - 15AR + 5A \\ - 4R^3 + 5R^2 - R.$$

On $0 \le A \le x/2$,

$$\partial_A^2\mathcal{Q} = 20R(30A - 4R - 3) < 0.$$

Its two endpoint values are

$$\mathcal{Q}(R, 0) = x(4R - 1) > 0, \qquad \mathcal{Q}\left(R, \frac{x}{2}\right) = \frac{x}{2}p(R),$$

where

$$p(R) = 25R^5 - 30R^4 + 20R^3 - 3R^2 - 7R + 3.$$

For $y = 2R - 1$,

$$32p(R) = 25y^5 + 65y^4 + 90y^3 + 106y^2 - 35y + 5 > 0,$$

because the terms of degree at least three are nonnegative and $106y^2 - 35y + 5$ has discriminant -895. Concavity therefore gives $L_{2,h} > A(1 + 4R)$.

On the other hand, $A < P = \eta E$ and $A < P < x/2$ imply

$$\frac{D_h}{A} = 4R - 1 + 10RA - \frac{\eta}{A} < C_0 := 4R - 2 + 5Rx - \frac{x}{2}.$$

Here we used

$$\frac{1}{E} > 1 + \frac{x}{2}, \qquad (1 - x)\left(1 + \frac{x}{2}\right)^2 = 1 - \frac{x^2(3 + x)}{4} < 1.$$

Finally,

$$1 + 4R - \frac{23}{10}C_0 = \frac{h_3(R)}{20}, \qquad h_3(R) = 230R^3 - 253R^2 - 81R + 112.$$

Since $h_3''(R) = 46(30R - 11) \ge 184$, strong convexity at $R_0 = 20/23$ gives

$$h_3(R) \ge \frac{788}{529} - \frac{(17/23)^2}{368} = \frac{289695}{194672} > 0.$$

Consequently

$$J(D_h) > A\left(1 + 4R - \frac{23}{10}C_0\right) > 0,$$

and hence

$$p_2 > L_2 > 2D + 3D^2.$$

The low root $e(D)$ is the smaller root of

$$H_D(z) = z^2 - (1 - D)z + (1 - D)^2 - (1 - D)^4,$$

and direct substitution gives $H_D(2D + 3D^2) = D^2(8D^2 + 19D - 2) < 0$. Hence $e(D) < 2D + 3D^2 < p_2$. Every row-2 label now yields $G_{1-D}(p_2) > 1 - X$, proving (18).

The first and fifth maps are extensive, and therefore

$$Z > 1 - H = 1 - \frac{s}{2}.$$

The diameter endpoint $\beta(s) = (-s+\sqrt{4-3s^2})/2$ satisfies $B_{c_0}(s) \le \beta(s) < 1-s/2$. Thus $Z > B_{c_0}(s)$, contradicting the two bounds on a_0. □

Proposition 4.4 (CE2 one-gap five-map obstruction)**.** *Assume that every vertex role is* Vd0*,* $N_+ = 1$*, and the center role is CE2. If exactly one of its two boundary traces contains a V-gap, the roles do not cover* H*.*

Proof. Again row 0 is uniquely supercritical. Put

$$R = \frac{x}{x+y}, \quad w = 1-R, \quad E = \sqrt{1-Rw}, \quad \eta = 1-E, \quad P = \eta E,$$

and

$$\alpha = u - R, \qquad \delta = v - w, \qquad k = \eta + \alpha + \delta.$$

The exact CE2 domain becomes

$$\alpha, \delta > 0, \qquad \delta + R\alpha < P, \qquad \alpha + w\delta < P, \tag{19}$$

with $x = k/w$, $y = k/R$, and demands

$$\begin{aligned} c_0 &= k, & c_1 &= 1-\delta/R, \\ c_2 &= 1-\delta, & c_3 &= 1-\min\{\delta/w, \alpha/R\}, \\ c_4 &= 1-\alpha, & c_5 &= 1-\alpha/w. \end{aligned}$$

Write $G_i = G_{c_i}$ for $1 \le i \le 5$. Suppose the gap is in $[y, v]$. Its endpoint bounds give the initial input $z_0 = 1 - v = R - \delta$ and target y. From (19), the concavity of

$$\pi(q) = (\eta+q)(1-2q) - 2qw, \qquad \pi(0) = \eta, \quad \pi(P) = \eta(\eta + 2ER^2),$$

and (31),

$$z_0 > \frac{\eta+\alpha}{w} > \frac{2\alpha}{1-2\alpha} > e(\alpha). \tag{20}$$

Put $Q = y/2 = k/(2R)$. We also have

$$\min\{e(\alpha), e(\delta)\} < Q. \tag{21}$$

For if both roots were at least Q, then $\alpha, \delta > Q/(2(1+Q))$; eliminating η with the first inequality in (19) would give $H(Q) > 0$, where

$$H(q) = \left(2R - \frac{1}{1+q}\right)(E+R) - 2R^2 - \frac{w}{2(1+q)}.$$

But $H'(q) > 0$, $Q < w/(2R)$, and

$$H\left(\frac{w}{2R}\right) = \frac{R(2RE - R - 1)}{R+1} < 0,$$

the last sign following from $(R+1)^2 - 4R^2E^2 = (1-R)(4R^3+3R+1) > 0$.

If $e(\alpha) < Q$, row 4 and (32) give $z_4 > 1-Q$; otherwise (21) gives $e(\delta) < Q$, and (20) makes row 2 give $z_2 > 1-Q$. Extensivity of the remaining maps yields

$$Z_R = (G_5 \circ G_4 \circ G_3 \circ G_2 \circ G_1)(1-v) > 1 - y/2.$$

Moreover the two inequalities in (19) imply $E(\alpha+\delta) < \eta$, hence $x^2 + xy + y^2 < 1$. The exact inner selector at the diameter endpoint is therefore active and

$$B_{c_0}(y) = \beta(y) < 1 - y/2.$$

Thus $Z_R > B_{c_0}(y)$, contrary to the supercritical row's admissibility. Reflection exchanges (x, u, R, α) with (y, v, w, δ) and reverses the five maps, proving the other orientation. □

Proposition 4.5 (Positive-gap all-Vd0, $N_+ = 1$)**.** *If the center role is CE1 or CE2, every vertex role is* Vd0*, and $N_+ = 1$, then no cover of H can have a boundary gap in the actual open vertex-role traces.*

Proof. The midpoint argument used above makes T_0 uniquely supercritical. In an assumed cover, every boundary gap in the vertex-role traces must lie in a center trace. CE1 therefore has the one-gap state excluded by Proposition 4.3. For CE2, the exactly-one-gap state is Proposition 4.4; if both center intervals contain gaps, the endpoint bounds and exact exits are those of Lemma 4.15, so the two far outgoing reaches sum to less than one. Rows $2, 3, 4$ would then have to cover more than three boundary units, contradicting their three nonsupercritical caps. This exhausts the one- and two-gap states, including singleton gaps. The zero-gap state is intentionally owned by the center-independent obstruction of Strategy 4. □

4.3. **The exactly-one-$T3$ branch.**

Proposition 4.6 (Exactly one T3-like row)**.** *Assume the center role is CE1 or CE2, $N_+ = 1$, exactly one vertex role is T3-like, and no role is Vd1 or Vd2. Then the roles do not cover H.*

Proof. Normalize the center midpoint to M_0. A T3-like row misses its own midpoint, a supercritical row misses all its local midpoints, and a Vd0 row cannot rescue a neighboring midpoint. Consequently, after reflection,

$$T_0 \text{ is T3-like}, \qquad M_1 \in T_0, \qquad T_1 \text{ is uniquely supercritical},$$

and $T_2, \ldots, T_5$ are nonsupercritical Vd0 rows.

In V_0 wedge coordinates put

$$1 \le D \le \frac{2}{\sqrt{3}}, \quad E = \sqrt{4 - 3D^2}, \quad R = \frac{D+E}{2},$$

and let a be the $e_{5,0}$ reach of T_0. Its interval on r_1 is

$$[c, u], \qquad c = \frac{D(1+a) - 1}{R}, \qquad u = 1 - \frac{R}{D} + a,$$

where $E/(2D) \le a \le (4 - 3D + E)/(4D)$. Its own-radial exit is $x_0 = aD/R$. For $B(c) = \mathcal{B}(c)$ and $A(c) = 1 - B(c)$, direct substitution gives

$$c - \frac{x_0(2 - x_0)}{1 + x_0} = \frac{2N(D, a)}{(D+E)(2Da + D + E)},$$

where

$$N = 4D^2a^2 + D^2a + D^2 - DEa + DE - 2Da - D - E.$$

This convex quadratic is nonnegative: on $1 \le D \le 8/7$ its minimum on the allowed interval is at $a = E/(2D)$, and squaring reduces the sign to $4(D-1)(7D^3 - 10D^2 - 8D + 12) \ge 0$; on $8/7 \le D \le 2/\sqrt{3}$ its critical value is positive because $(9D - 10)E + 9D^2 - 6D - 4 > 0$. Before using monotonicity, note explicitly that

$$x_0 = \frac{aD}{R} \le \frac{4 - 3D + E}{4R} \le \frac{1}{2};$$

the second inequality is equivalent, using $2R = D + E$, to $D \ge 1$. Also $c \le 1/2$ because $M_1 \in [c, u]$, and hence $0 \le A(c) = 1 - \mathcal{B}(c) \le 1/2$. Since $X(2-X)/(1+X)$ is increasing on $[0, 1/2]$, this proves

$$x_0 \le A(c). \tag{22}$$

Radial coverage forces the supercritical row's own demand $c_1 \geq c$. Lemma 4.12 gives $b_1 < B(c_1) \leq B(c)$. On $e_{5,0}$ the far row T_5 is forced to reach at least $h \geq B(c)$. Indeed this is $h \geq 1-a$ unless a CE2 interval hides the boundary gap. In the hiding case the side model gives $T \leq x_0$, and (22) again gives $h \geq 1-T \geq B(c)$. Therefore

$$\sum_{i=2}^{5}(a_i + b_i) \geq (1 - b_1) + 1 + 1 + 1 + h > 4,$$

contrary to the four row caps. The case $c = 1/2$ is already impossible because the strict supercritical feasible set is empty there. ☐

4.4. **The nonsupercritical CE1/CE2 boundary packages.**

Proposition 4.7 (All-Vd0 boundary loss)**.** *Assume that T_C is CE1 or CE2, every vertex role is Vd0, and*

$$A_i + B_i \leq 1 \qquad (0 \leq i \leq 5).$$

Then the seven roles do not cover H.

Proof. If the center boundary traces are already covered by the open vertex roles, then those six roles cover all of ∂H. Put $W_i = U_i \cap \partial H$. These are relatively open subsets of the connected circle ∂H, each of length at most one. At least two are nonempty, and a finite relatively open cover of a connected space cannot have its nonempty members pairwise disjoint. Thus two overlap on a positive-length relatively open set, whence

$$6 = \mathcal{H}^1(\partial H) < \sum_{i=0}^{5} \mathcal{H}^1(W_i) \leq \sum_{i=0}^{5}(A_i + B_i) \leq 6,$$

a contradiction.

Suppose next that T_C is CE1 and its trace $[s,t] \subset e_{0,1}$ contains a boundary gap. Put $u = 1-t$ and $w = t-s$. Boundary coverage gives $B_0 \geq s$ and $A_1 \geq u$. Since the center has no trace on $e_{5,0}$,

$$A_0 + B_5 \geq 1, \qquad A_0 + B_0 \leq 1,$$

and hence $B_5 \geq B_0 \geq s$. The exact CE1 formulas, after putting $R = \lambda$ in the orientation selected there, have the form

$$c_1 = u/R, \qquad c_5 = 1 - \left(u - \frac{R}{1 + \sqrt{1 - R + R^2}} - \frac{R}{1-R}w\right).$$

All hypotheses of Lemma 4.14 hold. Therefore the outgoing reaches of rows 1 and 5 satisfy $\widehat{b}_1 + \widehat{b}_5 < 1$, where

$$\widehat{b}_1 \leq F_{c_1}(u), \qquad \widehat{b}_5 \leq F_{c_5}(s).$$

Rows $2, 3, 4$ must then supply boundary length at least

$$(1 - \widehat{b}_1) + 1 + 1 + (1 - \widehat{b}_5) > 3,$$

contrary to their three row caps.

Now let T_C be CE2. If exactly one of $[x,u]$, $[y,v]$ contains a gap, reflect so that it is $[y,v]$. The interval $[x,u]$ is covered by its endpoint roles, so the same endpoint propagation gives inputs y and $1-v$. With

$$R = \frac{x}{x+y}, \qquad s = y, \qquad u' = 1 - v, \qquad w' = v - y,$$

the CE2 coupling and exit formulas are exactly those of Lemma 4.14. The preceding three-row contradiction applies unchanged. If both center intervals contain gaps, put

$$p = 1 - u, \quad q = 1 - v, \quad r = \frac{x}{x+y}, \quad w = \frac{y}{x+y}.$$

The far endpoint roles have inputs at least p, q and radial demands $p/w, q/r$. Lemma 4.15 again makes their two outgoing reaches sum to less than one, and the same middle three-row budget is impossible. These are the no-, one-, and two-active-interval states, and are exhaustive. □

We next treat the additional T3-like rows in the $N_+ = 0$ branch. The following local facts also reappear later.

Lemma 4.8 (T3-like midpoint and side tradeoff)**.** *Let T_i be T3-like. Then $M_i \notin T_i$. After applying the closed-trace domination of Proposition 2.6, suppose the selected adjacent support is on r_{i+1} and M_{i+1} belongs to the dominated trace. If b is its boundary reach toward V_{i+1}, c its exit on its own arm, and ℓ its entry on r_{i+1}, all measured from the relevant vertex, then*

$$b + c < 1, \qquad b + \ell > 1. \tag{23}$$

Consequently two adjacent T3-like traces that cover each other's midpoint, whose selected supports point toward one another, cannot cover both outer half-arms if no other role has positive support there.

Proof. Normalize at V_0 with wedge coordinates (x, y) and metric $x^2 + y^2 - xy$. Suppose first, toward the own-midpoint claim, that a unit triangle with two outside vertices A, B contains both $(0,0)$ and $(1/2, 1/2)$. Diameter one implies that an outside vertex has a negative coordinate and that each of its coordinates is strictly below one. Order A, B and write

$$B - A = (r, s), \qquad C - A = (s, s - r), \qquad r^2 + s^2 - rs = 1.$$

Since $(0,0)$ lies in the triangle, for some $\lambda, \mu \geq 0$ with $\lambda + \mu \leq 1$,

$$A = -\lambda(r, s) - \mu(s, s - r).$$

The midpoint condition is exactly

$$\lambda + \frac{r}{2} \geq 0, \qquad \mu + \frac{s-r}{2} \geq 0, \qquad \lambda + \mu + \frac{s}{2} \leq 1.$$

Put $t = s - r$. The four choices of negative coordinates of A, B give the required bounds on the third vertex as follows.

(1) If $A_x, B_x < 0$, then $C_x = A_x + s < 1$ when $s \leq 1$. When $s > 1$, the unit equation gives $r, t > 0$, and $B_x < 0$ gives $C_x < t < 1$. For the other coordinate, $C_y = B_y - r < 1$ when $r \geq 0$; when $r < 0$, put $q = -r + s$ to obtain

$$C_y \leq q - \frac{q^2 - 1}{2} \leq 1.$$

(2) If $A_y, B_y < 0$, the reflected argument gives the same two bounds.
(3) If $A_x < 0$ and $B_y < 0$, the alternative $s > 1$ would imply $\mu > 1 - \lambda$, which is impossible. Hence $C_x < 1$. Also, $r < -1$ would imply $s < 0$ and then $A_x \geq 0$; consequently $C_y = B_y - r \leq 1$.
(4) If $A_y < 0$ and $B_x < 0$, reflection of the preceding case applies.

In the $r<0$ entry, $q=-r+s$ and $(-r)^2+s^2-rs=1$ give $(-r)s=q^2-1$, while $\lambda \geq -r/2$ gives the displayed bound. Thus every vertex satisfies $x \leq 1, y \leq 1$, and only C can meet either equality line. Convexity makes both adjacent-arm intersections zero-length, contradicting the T3-like condition. Hence $M_0 \notin T_0$.

For the tradeoff, the translated T3-like trace has the exhaustive Type-II form

$$\begin{aligned}(1-t)x+ty &\geq 0,\\ \beta+tx-y &\geq 0, \qquad 0<t<1, \quad z=\sqrt{1-t+t^2}, \quad 0<\beta<z.\\ \beta+x-(1-t)y &\leq z,\end{aligned}$$

For the selected support on r_1, substitution in the Type-II half-planes gives

$$b=z-\beta, \qquad c=\frac{\beta}{1-t}, \qquad \ell=\frac{\beta+1-z}{1-t}.$$

The condition $M_1 \in T$ is $\beta \leq z-(1+t)/2$. Therefore

$$b+c \leq z+\frac{t}{1-t}\left(z-\frac{1+t}{2}\right)=1-\frac{(1-z)^2}{2(1-t)}<1,$$

whereas

$$b+\ell-1=\frac{t(1-z+\beta)}{1-t}>0.$$

For a crossed adjacent pair, coverage forces $c_i \geq \ell_{i+1}$ and $c_{i+1} \geq \ell_i$. Combining these two inequalities with (23) gives both $b_i<b_{i+1}$ and $b_{i+1}<b_i$, a contradiction. □

Lemma 4.9 (The support-isolated four-label inequality). *Assume T_C is CE1 or CE2, $N_+=0$, no role is Vd1 or Vd2, and, after normalization and reflection, T_0 is T3-like with selected adjacent support on r_1. Then the two endpoint rows in the isolated configuration satisfy*

$$F_{c_5^*}(a_5^*)+F_{c_1^*}(a_1^*)<1, \tag{24}$$

where a_1^*, a_5^* are the exact boundary inputs and c_1^*, c_5^* the exact radial demands defined below.

Proof. Use the CE1 side variables $0<\lambda<1$, $\rho=\sqrt{1-\lambda+\lambda^2}$ and center slacks

$$X=F_0(O), \qquad Y=F_2(O).$$

Then

$$a_C=\lambda-Y, \qquad \lambda s=X+Y+1-\rho,$$

and

$$\gamma_1=\min\left\{\frac{Y}{\lambda},\frac{\rho-X-Y}{1-\lambda}\right\}, \qquad \gamma_5=\min\left\{\frac{X}{1-\lambda},\frac{\rho-X-Y}{\lambda}\right\}.$$

For CE2 the second boundary interval is $[S,T]$, where

$$S=\frac{X+Y+1-\rho}{1-\lambda}, \qquad T=X+\lambda.$$

The T3-like normal form supplies $D>1$, $0<R<1$, and $\alpha, p \geq 0$ with

$$R^2-DR+D^2=1, \qquad \alpha+p=1/D, \qquad q=1-p=\alpha+(D-1)/D<1/2.$$

Its interval on r_1, measured from V_1, is

$$[c,u], \qquad c=Dq/R, \qquad u=q+(1-R)/D.$$

For an initial trace $[0,p]$ and a center interval $J=[L,U]$, define

$$\Theta(p,J)=\begin{cases}1-p, & J=\varnothing \text{ or } p<L,\\ 1-\max\{p,U\}, & p\ge L.\end{cases}$$

Then

$$a_1^*=\Theta(p,[s,t]),\qquad a_5^*=\Theta(\alpha,[S,T]),$$

with the second interval empty in CE1, and

$$c_5^*=1-\gamma_5,\qquad c_1^*=\begin{cases}c, & 1-u\le\gamma_1\le 1-c,\\ 1-\gamma_1, & \text{otherwise.}\end{cases}$$

If $a_1^*+a_5^*>1$, the two caps already imply $F_{c_5^*}(a_5^*)+F_{c_1^*}(a_1^*)<1$. Assume the hard region $a_1^*+a_5^*\le 1$. The following table is an exhaustive audit of the four possible first labels; the middle column records the exact decisive inequality.

first label	right labels	conclusion
L	Full	$e(\gamma_5)<a_1^*$
L	L,T_-	$F_{c_5^*}(a_5^*)+F_{c_1^*}(a_1^*)<1$
L	T_+^{hi}	$e(\gamma_5)<a_1^*$
T_-	L,T_-	$F_{c_5^*}(a_5^*)+F_{c_1^*}(a_1^*)<1$
T_-	Full, T_+^{hi}	$F_{c_5^*}(a_5^*)<a_1^*$
Full	all	impossible
T_+^{hi}	Full, T_+^{hi}	impossible
T_+^{hi}	L,T_-	$F_{c_5^*}(a_5^*)+F_{c_1^*}(a_1^*)<1.$

We verify the nonformal entries. Put $\kappa=(1-\rho)/(1-\lambda)$. From $X+(1-\lambda)Y<\rho(1-\rho)$ one obtains

$$\gamma_1\ge a_C\implies e(\gamma_5)<a_C, \tag{25}$$

$$S\le Y/\lambda\implies e(\gamma_5)<S. \tag{26}$$

For (25), one has $\gamma_5<a_C-\kappa$ and $a_C-\kappa<1/8$; then (31) applies, and the remaining comparison is

$$(a_C-\kappa)+5(a_C-\kappa)^2\le\kappa.$$

After squaring its positive sides, the difference is $\lambda(23\lambda^3+58\lambda^2+7\lambda+72)>0$. For (26), the premise forces $0<\lambda<3/8$ and

$$\gamma_5\le\frac{X}{1-\lambda}<\frac{1}{8},\qquad S\ge\frac{X+1-\rho}{1-2\lambda}.$$

The concave quadratic in X obtained from $2X/(1-\lambda)+5X^2/(1-\lambda)^2<S$ is positive at both endpoints; clearing radicals gives respectively the positive squares recorded in (25) and $\lambda^2(1-2\lambda)^2(1-\lambda)^2(-R_8)$, where every Bernstein coefficient of $-R_8$ on $[0,3/8]$ is positive.

The L separation follows from its exact polynomial: if its output is $z=e(\eta)$ at input $1-\beta$, then

$$\beta\ge z+\eta.$$

Indeed the selector polynomial $P(x,z)=x^4-4x^3+5x^2-(2+z)x+z-z^2$ is strictly decreasing on the low range and vanishes at $x=\eta$. Equations (25)–(4.4), with the three possibilities for Θ, prove every first-L entry. For a first T_- label the exact root test

$$T_-(a,c)<z\iff c^2(a+z)^2-(a^2-ac+c^2)>0$$

is increasing in z. Substitution of a_1^*, followed by the center inequality, proves $F_{c_5^*}(a_5^*) < a_1^*$ except in the low–low entries, which are immediate from $T_-(a,c) \le \min\{a, 1-a\}$ and $L < 1/2$. First-coordinate Full would force either $X \ge (1-\lambda)\alpha$ and $Y \ge \lambda - \alpha$, contradicting $s \le p$, or, in the CE2 overlap case, $S \ge 1/2$, contradicting $S \le \alpha < 1/2$.

For a first T_+^{hi} label put $r = 1-\lambda$, $y = Y/\lambda$ and choose β so that

$$m = \sqrt{\beta^2 - \beta + 1}, \quad 1 - \gamma_5 = \frac{m}{r+\beta}, \quad F_{c_5^*}(a_5^*) = \frac{\beta m}{r+\beta}.$$

The exact selector is

$$\beta \ge \max\left\{r, \frac{1}{2}, \frac{1-r^2}{1+2r}\right\}, \qquad r \ge (1-\beta)(r+\beta)^2. \tag{27}$$

The center inequalities give $y < 1/10$ and, with

$$y_* = \frac{r}{1-r}\left(\frac{m}{r+\beta} - \frac{1}{2} - \frac{1-r}{1+\sqrt{r^2-r+1}}\right),$$

give $y < y_*$ and

$$3y_* \le 1 - \frac{\beta m}{r+\beta}.$$

This proves the right-L entry by $e(y) < 3y$. It excludes right Full and right high by the exact high-input filter. For right T_- with input a_C, direct substitution gives a sum below one. For input q, put $c = 1-y$, $q = tc$, and let $\widehat{b}_1 = F_{c_1^*}(q)$. The selected T_- equations give

$$\widehat{b}_1 \le \kappa := \frac{\sqrt{13}-1}{6}, \qquad q \le \tau < \frac{93}{200},$$

where $\tau \in (0, 1/2)$ is the unique root of $\tau^4 - 3\tau^3 + 3\tau^2 - 3\tau + 1 = 0$. In this CE2 overlap subcase the support ordering gives $q > S$. The common left-high identities are

$$S = S_0 + \frac{1-r}{r}y, \qquad S_0 = 1 - \frac{m}{r+\beta} + \frac{1-r}{1+\sqrt{r^2-r+1}}.$$

Since $y > 0$, we have the complete chain

$$S_0 < S < q \le \tau < \frac{93}{200}.$$

Thus the S_0 hypothesis of Theorem B.2 holds; together with the exact selector domain (27), it gives

$$F_{c_5^*}(a_5^*) < \frac{5657}{10000} < \frac{7-\sqrt{13}}{6}.$$

Thus the final sum is also below one. The table is exhaustive, proving (24). □

Proposition 4.10 (The $N_+ = 0$ T3-like branch)**.** *Assume that T_C is CE1 or CE2, $N_+ = 0$, no vertex role is Vd1 or Vd2, and at least one vertex role is T3-like. Then the seven roles do not cover H.*

Proof. Normalize the unique center midpoint to M_0 and simultaneously apply the closed-trace domination of Proposition 2.6 to every T3-like role. If T_0 were not T3-like, a maximal consecutive block of T3-like indices among $1, \dots, 5$ would have to match its roles bijectively to the same block of uncovered midpoints. The matching uses only adjacent indices, because a T3-like role misses its own midpoint and Vd0 roles cannot cover adjacent midpoints. Its first two roles therefore form a crossed pair forbidden by Lemma 4.8. Hence T_0 is T3-like; reflect so that it supports r_1.

The diagonal trace bounds from Section 3 imply that there are at most two T3-like roles: if there were $k \geq 3$, the center, T3-like, and nonsupercritical Vd0 diagonal caps would give

$$6 < \frac{3}{2} + \frac{k}{2} + (6 - k) \leq 6.$$

Midpoint locality then forces T_2, T_4, T_5 to be Vd0. Consequently r_1 has positive support only from T_C, T_0, T_1, and r_5 only from T_C, T_5.

The actual outgoing reaches of T_1, T_5 are bounded by the two safe maps in Lemma 4.9; an extra adjacent-ray demand on a possible T3-like T_1 only shrinks its feasible set. Thus their sum is less than one. Rows T_2, T_3, T_4 must cover more than three units of the four intervening boundary edges, while their three nonsupercritical caps sum to at most three. This contradiction proves the proposition. □

4.5. **The exact capped map.** For $1/2 < c < 1$ put

$$\ell(c) = \frac{c}{2}\left(1 - \sqrt{4c^2 - 3}\right), \qquad r(c) = c - \ell(c)$$

when $c \geq \sqrt{3}/2$, and

$$h(c) = \frac{2c}{1 + \sqrt{16c^2 - 3}}.$$

Also define

$$\begin{aligned} T_-(a,c) &= \frac{\sqrt{a^2 - ac + c^2}}{c} - a, \\ T_+(a,c) &= \frac{2(1-a^2)c^2}{2ac^2 + c + \sqrt{(2ac^2 + c)^2 - 4(1-c^2)(1-a^2)c^2}}. \end{aligned}$$

Proposition 4.11 (Four-label capped map)**.** *For $1/2 < c \leq \sqrt{3}/2$,*

$$F_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ T_+(a,c), & 1 - c < a < h(c), \\ h(c), & a = h(c), \\ T_-(a,c), & h(c) < a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases} \tag{28}$$

For $\sqrt{3}/2 < c < 1$,

$$F_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ T_+(a,c), & 1 - c < a \leq \ell(c), \\ \ell(c), & \ell(c) < a < r(c), \\ T_-(a,c), & r(c) \leq a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases} \tag{29}$$

Thus there are exactly four labels

$$\text{Full}, \qquad L, \qquad T_-, \qquad T_+^{\text{hi}}.$$

At $a = \ell(c)$ in (29), $F_c(a) = r(c)$; immediately to its right the value is $\ell(c)$. This downward jump is genuine. The formal other root of the T_+ quadratic is not feasible because it violates $c \leq 2\max\{a, F_c(a)\}$.

The maps F_c are nonincreasing, the maps $G_c = 1 - F_c$ are nondecreasing and extensive, and

$$G_c(a) \le z \quad \Longleftrightarrow \quad G_c(1-z) \le 1-a. \tag{30}$$

Proof. The cap $a+b \le 1$ eliminates the S cell of Proposition 4.2. On the cap $b = 1-a$, the T inequality reduces to $M(c-M) \le 0$; hence the Full face is feasible exactly for $a \le 1-c$ or $a \ge c$. Off that face a maximal point lies on $F_L = 0$ or $F_T = 0$. The L equation has roots $\ell(c), r(c)$ and the selector $q \le 0$ keeps only the smaller coordinate. The ordered T halves give respectively the displayed lower quadratic root T_+ and the root T_-. Their order boundary is $a = b = h(c)$, while their $q = 0$ boundaries are $(\ell(c), r(c))$ and $(r(c), \ell(c))$. These observations give (28)–(29), including all equality assignments.

Down-closedness gives monotonicity. Finally,

$$G_c(a) \le z \iff F_c(a) \ge 1-z \iff (a, 1-z, c) \in \mathcal{A}, \quad a + 1 - z \le 1.$$

Reflection $a \leftrightarrow b$ gives (30). □

For $0 < X < 1 - \sqrt{3}/2$ write

$$e(X) = \ell(1-X).$$

The root equation gives the estimates

$$X < e(X) < \frac{2X}{1-2X}, \qquad e(X) \le 2X + 5X^2 \quad (0 < X \le 1/8), \tag{31}$$

and

$$a > e(X) \quad \Longrightarrow \quad G_{1-X}(a) \ge 1 - e(X). \tag{32}$$

Indeed, substitution into $z^2-(1-X)z+(1-X)^2-(1-X)^4$ gives the two comparisons, and (32) is read directly from (29). We shall also use $e(X) < 2X + 3X^2$ for $0 < X \le 1/10$, obtained from $X^2(8X^2 + 19X - 2) < 0$.

Lemma 4.12 (Free strict-supercritical envelope)**.** *For $0 \le c < 1/2$,*

$$\sup\{b : \exists a, (a,b,c) \in \mathcal{A}, a+b > 1\} = \mathcal{B}(c) := \frac{c + \sqrt{c^2 - 8c + 4}}{2}.$$

The supremum is not attained. For $c \ge 1/2$ the strict feasible set is empty. The function $\mathcal{B}$ is strictly decreasing on $[0, 1/2]$.

Proof. Lemma 2.9 excludes $c \ge 1/2$. On the boundary $a = 1-b$, a support rotation has strict slack precisely when

$$c < \frac{1-b^2}{2-b}.$$

The right side decreases from 1/2 to zero. Solving equality gives the displayed larger root. Equality has no strict support slack, which explains nonattainment. Differentiation gives

$$\mathcal{B}'(c) = \frac{1}{2}\left(1 + \frac{c-4}{\sqrt{c^2-8c+4}}\right) < 0,$$

since $(4-c)^2 - (c^2 - 8c + 4) = 12$. □

4.6. **Exact center traces and complementary demands.** The normalization in Proposition 2.10 lets us assume that the unique center midpoint is M_0.

Proposition 4.13 (Exact CE1 and CE2 formulas)**.** *In the CE1 case there are $0 < \lambda < 1$, $0 \le s < t \le 1$, and $\rho = \sqrt{1-\lambda+\lambda^2}$ for which*

$$T_C \cap e_{0,1} = [s,t]$$

and

$$C_0 = \rho + \lambda s - t - \lambda, \qquad C_1 = 1 - \lambda s, \qquad C_2 = t + \lambda - 1.$$

The exact domain is

$$\begin{gathered} C_0 > 0, \quad C_2 > 0, \quad \lambda s \le \tfrac{1}{2}, \quad t < 1 - \tfrac{\lambda}{2}, \\ t > \rho + \lambda s - \frac{1+\lambda}{2}, \qquad t \ge \rho - \frac{\lambda^2}{1-\lambda} s. \end{gathered}$$

The center exits on the six arms are

$$\begin{aligned} d_0 &= C_1, & d_1 &= C_2/\lambda, \\ d_2 &= C_2, & d_3 &= \min\{C_0/\lambda, C_2/(1-\lambda)\}, \\ d_4 &= C_0, & d_5 &= C_0/(1-\lambda). \end{aligned}$$

In the CE2 case write

$$T_C \cap e_{5,0} = [x,u], \qquad T_C \cap e_{0,1} = [y,v], \qquad S = x+y, \quad D = \sqrt{x^2+xy+y^2}.$$

Then

$$\begin{gathered} 0 < x < u < 1, \qquad 0 < y < v < 1, \\ (u+v)S - xy = D, \qquad uS \ge x, \quad vS \ge y, \\ uS < x + y/2, \qquad vS < y + x/2, \end{gathered}$$

where the two weak center inequalities are strict for closures of original open center roles. The six exits are

$$\begin{aligned} d_0 &= 1 - \frac{xy}{S}, & d_1 &= \frac{vS-y}{x}, \quad d_2 = \frac{vS-y}{S}, \\ d_3 &= \min\left\{\frac{vS-y}{y}, \frac{uS-x}{x}\right\}, & d_4 &= \frac{uS-x}{S}, \quad d_5 = \frac{uS-x}{y}. \end{aligned}$$

In either case the complementary demand on row i is $c_i = 1 - d_i$; $c_0 \le 1/2$ and $c_i > 1/2$ for $i \ne 0$.

Proof. In the affine coordinates $X = V_0 + b(V_1 - V_0) + a(V_5 - V_0)$, the CE1 triangle is cut out by

$$\begin{aligned} F_1 &= \lambda b + (1-\lambda)a - \lambda s, \\ F_2 &= -b + \lambda a + t, \\ F_0 &= (1-\lambda)b - a + \rho + \lambda s - t. \end{aligned}$$

Substitution of $O, M_0, \ldots, M_5$ and of the six radial parametrizations gives the displayed domain and exits. The last domain inequality is exactly the absence of a positive interval on $e_{5,0}$.

For CE2, the side through the initial endpoints has outward normal $(\sqrt{3}S, y-x)/(2D)$. Rotating it by $\pm 120°$, the three support constants are

$$\frac{\sqrt{3}(S-xy)}{2D}, \qquad \frac{\sqrt{3}(vS-y)}{2D}, \qquad \frac{\sqrt{3}(uS-x)}{2D}.$$

Their sum is $\sqrt{3}/2$ exactly when $(u+v)S-xy=D$. Substitution of the midpoints gives the four remaining inequalities, and substitution of the radial rays gives the six exits. The inequalities $d_0 \geq 1/2 > d_i$ for $i \neq 0$ are precisely the exact-midpoint conditions. □

4.7. **Two reusable capped-loss inequalities.**

Lemma 4.14 (One-side capped loss)**.** *Let* $0<R<1$, $S=\sqrt{1-R+R^2}$, *and let* $s,u,w>0$ *satisfy* $s+u+w=1$ *and* $0<u<R$. *Put*

$$\delta = \frac{R}{1+S}, \qquad \gamma = u-\delta-\frac{R}{1-R}w,$$

and assume $\gamma>0$. *Set*

$$c_1 = u/R, \qquad c_5 = 1-\gamma.$$

Then

$$F_{c_5}(s)+F_{c_1}(u)<1. \tag{33}$$

Proof. Put $\eta=(1-R)\delta=1-S$ and

$$\widehat{b}_5 = F_{c_5}(s), \qquad \widehat{b}_1 = F_{c_1}(u).$$

The center identity may be written

$$c_5 = \frac{1-u+\eta-Rs}{1-R}. \tag{34}$$

The hypotheses imply $1/2<c_1,c_5<1$: for example $u>\delta$ gives $c_1>1/(1+S)>1/2$, and $\gamma<u-\delta<R-\delta=RS/(1+S)<1/2$ gives $c_5>1/2$. Thus the four high-radial labels of Proposition 4.11 are exhaustive.

Right low labels. If the right label is T_-, its exact frontier equation gives $(u+\widehat{b}_1)^2 = 1-R+R^2=S^2$, so $\widehat{b}_1=S-u$. For a right L label write $\widehat{b}_1=kc_1$, $0\leq k\leq 1/2$, and $C=1-k+k^2=c_1^2$. With $X_R=R+k$, the Cell-L selector factors as

$$q=c_1^2(X_R-1)G_k(X_R), \qquad G_k(X)=CX^3+CX^2-k(1-k)X+k^2.$$

For $X\geq k$, $G_k(X)>0$: when $k>0$, $G_k'(X)\geq k(1+2k-k^2+3k^3)>0$ and $G_k(k)=k^2(1+k+k^3)>0$. Hence $q\leq 0$ gives $X_R\leq 1$. If $h_R=1-X_R$, then

$$S^2-(u+\widehat{b}_1)^2 = h_R(1+2k^2+k(1-k)h_R)\geq 0.$$

Consequently either right low label satisfies

$$\widehat{b}_1 \leq X := S-u. \tag{35}$$

A left low label against right high or Full. For left T_- put $z=s/c_5=1-p$ and $h=\sqrt{1-p+p^2}$. The component selector gives $0\leq p\leq 1/2$, the frontier gives $s+\widehat{b}_5=h$, and its selector gives $s\leq(1-p)h$. Equation (34) becomes $c_5(1-Rp)=1-u+\eta$, whence

$$u\geq \eta+(1-h)+Rph.$$

Put $q_0 = ph$. If $q_0 > R$, then $p(1-p) > R(1-R)$ and the preceding lower bound would give $u > R$. Thus $q_0 \le R$, and

$$\eta + 1 - h > \frac{R(1-R) + p(1-p)}{2} \ge (1-R)q_0.$$

Substitution in the exact formula

$$w = \frac{(1-R)(1-u)p - \eta(1-p)}{1 - Rp}$$

now gives $w < 1-h$. A right high output has the form $H - u$, where $H = \sqrt{1-\beta+\beta^2} \le 1$; for Full take $H = 1$. Hence

$$\widehat{b}_5 + \widehat{b}_1 = h + H - (s+u) = 1 + w - (1-h) - (1-H) < 1.$$

For a left L label write

$$c_5 = h(x) := \sqrt{1-x+x^2}, \qquad \widehat{b}_5 = xh(x), \qquad 0 \le x \le \frac{1}{2},$$

and, along the center line,

$$s(x) = \frac{R - u + \eta + (1-R)(1-h(x))}{R}.$$

If $y = s(x)/h(x)$, the exact selector factorization is

$$\frac{q}{h(x)^2} = (y - (1-x))P_x(y),$$

where

$$\begin{aligned} P_x(y) &= (1-x+x^2)y^3 + (1+2x-2x^2+3x^3)y^2 \\ &\quad + (x+2x^2-x^3+3x^4)y + (x^2+x^3+x^5). \end{aligned}$$

Every coefficient is nonnegative on $0 \le x \le 1/2$, and the polynomial is positive at every nondegenerate point. Thus a selected L point has $s(x) \le (1-x)h(x)$. The two curves meet once in $(0, 1/2)$: at $x = 0$ the left side is smaller, while at $x = 1/2$ it is enough to use

$$u \le \frac{R}{1+R} \quad \text{(Full)}, \qquad u \le \min\{RS, S/2\} \quad (T_+^{\mathrm{hi}}).$$

Writing $e_0 = 1 - \sqrt{3}/2$, the corresponding endpoint differences are

$$\frac{(1-R)(R^2 + 2Re_0 + 2e_0)}{2(1+R)} > 0$$

for Full, and respectively $[2e_0(1-R) - R^3]/2 \ge 0$ for $R \le 1/2$ and $(1-R)(3R + 4e_0 - 2)/4 \ge 0$ for $R \ge 1/2$ in the high case. At the unique transition x_0, the L frontier is the $q = 0$ endpoint of the preceding T_- branch. Since $xh(x)$ increases, every selected left L output is no larger than that transition output. The preceding strict loss therefore also covers (L, T_+^{hi}) and (L, Full).

Full and high exclusions. Left Full forces $s < 1/2$ and $\gamma \ge s$. Right Full or right high gives $u \le 1/2$, but

$$\gamma - s = 2u - 1 - \delta + \frac{1-2R}{1-R} w < 0:$$

for $R \ge 1/2$ this is immediate, while for $R < 1/2$ use $w < 1-u$ to bound it by $(u-R)/(1-R) - \delta < 0$. Thus left Full cannot pair with either label. If the left label is high and the right one Full, then $s < 1/2$ and $u \le R/(1+R)$, whereas $\gamma > 0$

and $w > 1/2 - u$ force $u > R/2 + \eta > R/(1+R)$; the final inequality is equivalent to $2(1+R) > 1+S$. This pair is also impossible.

A right low label. Use (35). If $s > X$, the cap gives $\widehat{b}_5 + \widehat{b}_1 < 1$. Suppose $s \le X$ and put

$$c_X = X + 2\delta, \qquad f(c,z) = c^4 - c^2 + zc - z^2.$$

At the limiting value $u = R$, with $\kappa = \delta$, one has

$$X_0 = \frac{1-2\kappa}{1+\kappa}, \qquad c_0 = \frac{1+2\kappa^2}{1+\kappa}.$$

Now $c_0 > \sqrt{3}/2$ because $16\kappa^4 + 13\kappa^2 - 6\kappa + 1 > 0$, and

$$f(c_0, X_0) = \frac{\kappa^2(1-2\kappa)^2(4\kappa^4 + 4\kappa^3 + 10\kappa^2 + 6\kappa + 5)}{(1+\kappa)^4} > 0.$$

Along the u-line, $\frac{d}{du} f(c_X, X) = -4c_X^3 + c_X + X < 0$, so the same two signs hold before the limiting endpoint. Therefore, for $\ell(c) = c(1 - \sqrt{4c^2 - 3})/2$,

$$\ell(c_X) < X < c_X - \ell(c_X), \qquad \ell(c_X) < 1 - X.$$

Write $c = h(z) = \sqrt{1 - z + z^2}$ and let z_X satisfy $h(z_X) = c_X$. The actual $c_5(s)$ corresponds to $0 < z \le z_X < 1/2$, and

$$s(z) = s_0 + \frac{1-R}{R}(1 - h(z)), \qquad s_0 = \frac{R - u + \eta}{R} > 0.$$

The transition input $(1-z)h(z)$ decreases, and the preceding root ordering gives $s(z_X) < (1 - z_X)h(z_X)$, hence the same strict inequality for all $z \le z_X$. To check the order of the two boundary coordinates put

$$\phi(z) = s(z) - zh(z), \qquad k_R = \frac{1-R}{R}.$$

Its endpoint values are $s_0 > 0$ and $X - \ell(c_X) > 0$. Moreover

$$\phi'(z) = \frac{(k_R + z)(1-2z)}{2h(z)} - h(z).$$

For $k_R \le 2$ this is nonpositive; for $k_R > 2$ its zero is described by the strictly increasing function $K_0(z) = (2 - 3z + 4z^2)/(1-2z)$, so the minimum is again at an endpoint. Thus $zh(z) < s(z)$. Applying the already displayed polynomial P_z gives the strict Cell-L selector, and $s + zh(z) \le X + \ell(c_X) < 1$. Since $\partial_b F_L = c(1-2z) > 0$ and each feasible fiber is an interval, the capped maximum is

$$\widehat{b}_5 \le zh(z) \le z_X h(z_X) = \ell(c_X) < 1 - X.$$

Together with $\widehat{b}_1 \le X$, this proves the loss for every right low label.

High–high. Let $c_L(z)$ be the unique root in $[\sqrt{3}/2, 1]$ of $f(c,z) = 0$. With $D_c = 4c^3 - 2c + z > 0$, implicit differentiation gives

$$c_L''(z) = \frac{2(c^2 - cz + z^2) + 16c^2 z(c-z)}{D_c^3} > 0.$$

A selected left high label requires $s < 1/2$ and $c_5(s) \le c_L(s)$. Define

$$s_0 = \frac{R - u + \eta}{R};$$

then $0 < s_0 < s$ and $c_5(s_0) = 1$. The explicitly named function $g(z) = c_5(z) - c_L(z)$ is concave, with $g(s_0) > 0$. The right high bound $u \le \min\{RS, S/2\}$ gives $c_5(1/2) >$

$7/8$. For $R \le 1/2$ this follows after squaring from $17-10R+9R^2-64R^3-64R^4 > 0$ (a decreasing polynomial with value $9/4$ at $1/2$); for $R \ge 1/2$ it follows from $(3+R)^2 - 16S^2 = (1-R)(15R-7) > 0$. Finally

$$f(7/8, 1/2) = 33/4096 > 0$$

and f increases in c on this range, so $c_L(1/2) < 7/8$ and $g(1/2) > 0$. Concavity contradicts $g(s) \le 0$. Thus high–high is impossible.

If both labels are low, the cap already gives $\widehat{b}_5 + \widehat{b}_1 \le s + u = 1 - w < 1$. The preceding five paragraphs cover every other ordered pair of the four labels, proving (33). □

Lemma 4.15 (CE2 two-endpoint capped loss)**.** *Let $[x, U]$ and $[y, v]$ be an exact CE2 pair as in Proposition 4.13. Put*

$$p = 1 - U, \qquad q = 1 - v, \qquad r = \frac{x}{x+y}, \qquad w = \frac{y}{x+y},$$

and reflect if necessary so that $0 < w \le 1/2$ and $r = 1 - w$. Then

$$F_{p/w}(p) + F_{q/r}(q) < 1. \tag{36}$$

Proof. Put

$$S_0 = x + y, \quad K = \sqrt{1 - wr}, \quad z = \frac{w}{1+K}, \quad \zeta = \frac{r}{1+K}, \quad \alpha = \frac{p}{w}, \quad \beta = \frac{q}{r}.$$

Then $D = S_0 K$, $1 - K = wr/(1+K)$, and the exact midpoint inequalities give $1/2 < \alpha, \beta < 1$. Dividing the coupling equation $(U+v)S_0 - xy = D$ by S_0, and using successively $x < U$ and $y < v$, gives the strict endpoint inequalities

$$\beta + p > 1 + z, \qquad \alpha + q > 1 + \zeta. \tag{37}$$

Define

$$\Phi_w(\alpha) = F_\alpha(w\alpha), \qquad \Phi_r(\beta) = F_\beta(r\beta), \qquad h(t) = \sqrt{1 - t + t^2}.$$

Thus the desired sum is $\Phi_w(\alpha) + \Phi_r(\beta)$.

Selected branch parametrizations. For a side of weight $\sigma \in \{w, r\}$, every L label has

$$c = h(k), \qquad F_c(\sigma c) = kh(k), \qquad 0 \le k \le w, \tag{38}$$

on either side. On the large side the range follows from

$$Q_{\mathrm{cell}}/h(k)^2 = (k - w)G_k(r + k),$$

where $G_k(X) = h(k)^2X^3 + h(k)^2X^2 - k(1-k)X + k^2 > 0$ for $X \ge k$. The remaining selected parametrizations are

$$\alpha = \frac{h(a)}{w+a}, \qquad \Phi_w(\alpha) = a\alpha, \qquad r \le a < 1, \tag{39}$$

$$\beta = \frac{h(b)}{r+b}, \qquad \Phi_r(\beta) = b\beta, \qquad r \le b < 1, \tag{40}$$

$$\beta = \frac{K}{r+t}, \qquad \Phi_r(\beta) = t\beta = K - r\beta, \qquad w \le t \le r. \tag{41}$$

For example, the small-high selector factors as

$$h(a)^2 a\left(\frac{w}{(w+a)^2} - (1-a)\right), \qquad w - (1-a)(w+a)^2 = (a-r)(a^2 + wa - w),$$

which gives $a \geq r$; the other two factorizations are $(b-w)(b^2+rb-r)$ and $(t-w)(r^2-wt)$, respectively. These formulas specify the selected geometric components, not discarded algebraic roots.

The small-side T_- label is absent for $w < 1/2$. Large-side Full is incompatible with the first inequality in (37). Small-side Full can pair only with L: either high or T_- on the large side has $\beta \leq K$, which would force $p > (1+r)z > w/(1+w)$, contrary to Full. Thus, up to the symmetric $w = r = 1/2$ tie, the exact list is

$$(\mathrm{Full}, L), (T_+^{\mathrm{hi}}, L), (L, L), (L, T_-), (L, T_+^{\mathrm{hi}}), (T_+^{\mathrm{hi}}, T_-), (T_+^{\mathrm{hi}}, T_+^{\mathrm{hi}}). \tag{42}$$

Low–low sums are at most $K < 1$: an L output is at most wK, while (41) is at most rK. For Full–L, put

$$b_{\mathrm{thr}} = 1 + z - \frac{w}{1+w}, \qquad \kappa = \frac{w}{1 + (1+w)z}.$$

Full and (37) give $\beta > b_{\mathrm{thr}}$ and $(1+\kappa)b_{\mathrm{thr}} = 1 + z$. Direct reduction yields

$$b_{\mathrm{thr}}^2 - h(\kappa)^2 = \frac{w^2(A(w) + KC(w))}{D_0(w,K)} > 0,$$

where

$$\begin{aligned} A(w) &= 8 - 5w + 22w^2 + 6w^3 + 5w^4 + 5w^5 - w^6, \\ C(w) &= 8 - w + 14w^2 + 4w^3 - 2w^4 + w^5, \\ D_0(w,K) &= ((1+w)(1+K))^2(1 + K + w + w^2)^2. \end{aligned}$$

Both numerator polynomials are positive on $[0, 1/2]$. Since h decreases and $(1+k)h(k)$ increases there, $\beta = h(k) > b_{\mathrm{thr}} > h(\kappa)$ implies $(1+k)\beta < 1 + z$; combined with (37), this gives $k\beta < p$ and hence the strict Full–L loss.

It remains to verify the four pairs containing a high label on the small side.

(L, T_+^{hi}). Use parameters $0 \leq k \leq w$ and $r \leq t < 1$:

$$\alpha = h(k), \quad \Phi_w = kh(k), \qquad \beta = \frac{h(t)}{r+t}, \quad \Phi_r = \frac{th(t)}{r+t}.$$

The first output and the last output increase in their parameters, while $h(t)/(r+t)$ decreases. If $w \leq 2/5$, then

$$\Phi_w + \Phi_r < wK + \frac{1}{2-w} < 1;$$

after using $K \leq 1 - wr/2$, the positive cleared gap is $P(w) = w^4 - 3w^3 + 4w^2 - 6w + 2$, which decreases on $[0, 2/5]$ and has $P(2/5) = 46/625$. If $w \geq 2/5$, the first endpoint inequality gives $\beta > r + z \geq 3/4$. Since $h(3/4)/(r + 3/4) = \sqrt{13}/(7 - 4w) < 3/4$, monotonicity forces $t < 3/4$. Therefore

$$\Phi_w + \Phi_r < \frac{\sqrt{3}}{4} + \frac{3\sqrt{13}}{20} < 1.$$

(T_+^{hi}, T_-). Use a, t from (39) and (41). Both demands are at most K. The first endpoint inequality gives

$$\alpha > \frac{1 + z - K}{w} = \frac{1+r}{1+K} =: a_0.$$

Thus $K > 1-w^2$, which forces $w > 1/3$, and comparison at $a = 2/3$ forces $a < 2/3$. The function

$$G(a) = \frac{(1+a)h(a)}{w+a}$$

decreases on $[r, 2/3]$, because its derivative numerator $2a^3+(4w-1)a^2+(1-w)a+w-2$ is at most its value $-7/54$ at $(2/3, 1/2)$. Hence $G(a) \le (1+r)K$. Using the second endpoint inequality and $\Phi_r = K - r\beta$ gives

$$\Phi_w + \Phi_r < (2+r)K - 1 - \zeta < 1.$$

After multiplication by $1+K$, the last positive gap is $r(3w-w^2-K)$; its squared comparison is

$$(3w-w^2)^2 - K^2 = w^4 - 6w^3 + 8w^2 + w - 1 > 0$$

for $w > 1/3$ (value $1/81$ at $1/3$, positive derivative thereafter).

$(T_+^{\mathrm{hi}}, T_+^{\mathrm{hi}})$. Use parameters a, b from (39)– (40). The endpoint inequalities first force $w > 2/5$: for $w \le 2/5$, the asserted contrary inequality $K(w+1/(2r)) \le 1+z$ reduces, after multiplication by $2r(1+K)$, to

$$K(2w^2-4w+1) + 2w^4 - 4w^3 + w^2 - w + 1 > 0,$$

which is immediate according as the coefficient of K is nonnegative or, using $K \le 1$, from the decreasing polynomial $2w^4 - 4w^3 + 3w^2 - 5w + 2 \ge 172/625$.

For $F_c(t) = (1+t)h(t)/(c+t)$, the derivative numerator $2t^3 + (4c-1)t^2 + (1-c)t + c - 2$ is increasing on the present rectangle, so F_c has at most one interior minimum. Endpoint comparison gives

$$\Phi_w + \alpha \le \frac{2}{1+w}, \qquad \Phi_r + \beta \le \frac{2}{1+r}.$$

The nontrivial endpoint signs are exactly

$$4 - (1+r)^2(1+w)^2K^2 = -w^2(w-1)^2(w^2-w-3) > 0$$

and

$$16r^2 - (1+r)^4h(r)^2 = (1-r)(r^5 + 4r^4 + 7r^3 + 9r^2 - 4r - 1) > 0;$$

the bracket is increasing on $[1/2, 1]$ and equals $13/32$ at $1/2$. Weighting the two strict endpoint inequalities by w/K^2 and r/K^2 gives

$$\alpha + \beta > \frac{2K+1}{K(1+K)}.$$

Consequently

$$\Phi_w + \Phi_r < \frac{2}{1+w} + \frac{2}{1+r} - \frac{2K+1}{K(1+K)} < 1.$$

The numerator of the last gap is $3Kwr - Q(w)$, where $Q(w) = w^4 - 2w^3 + 7w^2 - 6w + 2 > 0$. Its squared difference is

$$D(w) = -w^8 + 4w^7 - 9w^6 + 13w^5 - 41w^4 + 65w^3 - 55w^2 + 24w - 4.$$

Here $D'(w) = (1-2w)H(w)$ with

$$H(w) = 4w^6 - 12w^5 + 21w^4 - 22w^3 + 71w^2 - 62w + 24 > 0$$

on $[2/5, 1/2]$; use $71w^2 - 62w + 24 \ge 743/71$ and the remaining cubic block at least $-11/4$. Thus D increases and $D(2/5) = 4904/390625 > 0$, proving the claim.

(T_+^{hi}, L). Use the fully specified parameters

$$\alpha = \frac{h(t)}{w+t}, \quad p = w\alpha, \quad b = t\alpha, \quad \frac{1}{2} \le t < 1,$$

and

$$\beta = H = h(k), \quad \widehat{b}_L = kH, \quad 0 \le k \le w.$$

Since $p + b = h(t)$, the first endpoint inequality gives

$$1 - (b + kH) > G := 2 + z - h(t) - (1+k)H. \tag{43}$$

Put

$$p_0 = 1 + z - K = \frac{w(2-w)}{1+K},$$

let $t_0 \in (1/2, 1)$ be the unique point with $p(t_0) = wh(t_0)/(w+t_0) = p_0$, and put $b_0 = t_0 p_0/w$. The exact endpoint lemma is

$$b_0 < 1 - wK. \tag{44}$$

To verify it, set $b_* = 1 - wK$ and $T_* = wb_*/p_0$. One has $0 < T_* < 1$, and

$$p_0^2(w+T_*)^2 - w^2 h(T_*)^2 = -\frac{w^3(KE(w) + J(w))}{(1+K)^2(2-w)^2} > 0,$$

where

$$E(w) = 2w^5 + 3w^3 - 20w^2 + 23w - 6,$$

$$J(w) = w^6 + 8w^5 - 22w^4 + 12w^3 + 6w^2 + 2w - 6.$$

Here $J \le -111/64$; if $E > 0$, then $KE + J < E + J < 0$, because $E + J$ is increasing and equals $-139/64$ at $w = 1/2$. Thus $p(T_*) < p(t_0)$. Since $p(t)$ decreases, $t_0 < T_*$ and (44) follows.

Finally put $P = 1 + z - p$. If $P < K$, then $t < t_0$ and $k \le w$, so

$$G > 2 + z - h(t_0) - (1+w)K = 1 - b_0 - wK > 0.$$

If $P \ge K$, then $K \le P \le P_1 := 1 + z - w/(1+w) \le 1$ and there is $a \in [0, w]$ with $h(a) = P$. Since $P < H$, one has $k < a$ and $(1+k)H < (1+a)P = P + \ell(P)$, where $\ell(P) = P(1 - \sqrt{4P^2 - 3})/2$. Hence

$$G > \mathcal{D}(P) := 1 - b(P) - \ell(P).$$

This function is concave. Indeed $\ell''(P) > 0$, and the high output b is convex in $p = 1+z-P$: with $A_t = 2 + w - (1+2w)t > 0$ and $Q_t = 8wt^2 + 4t^2 - 8wt - 13t - w + 4 < 0$,

$$\frac{d^2 b}{dp^2} = -\frac{2h(t)(t+w)^3 Q_t}{w^2 A_t^3} > 0.$$

At $P = K$, (44) gives $\mathcal{D}(K) = 1 - b_0 - wK > 0$. At $P = P_1$, set $\kappa = w/(1 + (1+w)z)$. The Full–L comparison already proved gives $P_1 > h(\kappa)$, hence $\ell(P_1) < \kappa P_1 = w/(1+w)$, while $b(P_1) = 1/(1+w)$. Thus $\mathcal{D}(P_1) > 0$, and concavity gives $G > 0$ throughout. Equation (43) proves the final hard pair.

The exact list (42) is now exhausted, proving (36). $\square$

4.8. **The exactly-one-Vd1/Vd2 placements.** We now assume CE2, $N_+ = 1$, and exactly one row is Vd1 or Vd2. The corner normal form of Proposition 3.10 will be used in the following form: for some $t > 0$ and $d = \sqrt{t^2+t+1}$,

$$x - (t+1)y \le a, \qquad ty - (t+1)x \le tb, \qquad tx + y \le d - a - tb. \tag{45}$$

In particular $a + b < 1/2$ for a positive-support row.

Lemma 4.16 (Vd1 adjacent rescue)**.** *Assume an original open-role skeleton cover in the CE2 exact-M_0 branch, with $T_C \cap \{M_0, \dots, M_5\} = \{M_0\}$. Suppose T_0 is the unique Vd1 row, $M_1 \in T_0$, T_1 is uniquely supercritical, and $T_2, \dots, T_5$ are nonsupercritical Vd0 rows. Suppose also that these are the only positive-adjacent-support roles. Then the roles do not cover the perimeter together with r_1.*

Proof. From (45), the r_1 interval is

$$[c, u], \qquad c = \frac{t(1-b)}{t+1}, \qquad u = \frac{d - a - tb - 1}{t}.$$

The midpoint inequalities and absence of positive r_5 support force $t \ge 1$. They also give

$$a \le F(t,c) := d - 1 - \frac{3t}{2} + c(t+1) \le L(c) := \sqrt{3} - \frac{5}{2} + 2c.$$

With $A(c) = 1 - \mathcal{B}(c)$, squaring positive sides gives

$$2L(c) \le A(c),$$

because the difference is $4(20c^2 + (18\sqrt{3} - 52)c + 47 - 24\sqrt{3}) > 0$ on $[0, 1/2]$. Writing $\delta = 1 - u$, the same normal form yields

$$\frac{a}{a+\delta} \le \frac{L(c)}{L(c) + 1/2} \le 2L(c) \le A(c).$$

Thus $a \le A(c)$. If the CE2 interval on $e_{5,0}$ hides the gap after T_0, its side parameters X, Y, λ, ρ satisfy $Y \ge \lambda\delta$ and $S \le a$, whence

$$T = X + \lambda \le a + \lambda(u - a) \le \frac{a}{a+\delta} \le A(c).$$

In every other ordering its forced tail starts no later than a. Consequently the far-row reach is always $h \ge \mathcal{B}(c)$.

Coverage of r_1 gives the supercritical row a radial demand at least c. Therefore Lemma 4.12 gives $b_1 < \mathcal{B}(c) \le h$. The four remaining row caps would then have to satisfy

$$\sum_{i=2}^{5} (a_i + b_i) \ge (1 - b_1) + 1 + 1 + 1 + h > 4,$$

a contradiction. □

Lemma 4.17 (Adjacent positive-support placement)**.** *Assume an original open-role skeleton cover in the CE2 exact-M_0 branch, with $T_C \cap \{M_0, \dots, M_5\} = \{M_0\}$. Suppose T_0 is uniquely supercritical, T_1 is the unique Vd1/Vd2 row, and T_2, T_3, T_4, T_5 are nonsupercritical Vd0 rows; in particular, there is no additional positive-adjacent-support row. Then the perimeter together with r_1, r_2 is not covered. The reflected T_5 placement is also impossible.*

Proof. Use the CE2 variables $S = x + y$, $D = \sqrt{x^2 + xy + y^2}$ and $E = vS - y$. The center length on r_2, measured from O, is $d_2^C = E/S$. Let A, H be the boundary demands forced on T_1 from $e_{0,1}$ and the propagated $e_{5,0}$ tail. The corner cap gives

$$A + H < \frac{1}{2}. \tag{46}$$

The exact radial envelope of Proposition 4.2, including the Cell-T selector $c \le 2M$, gives

$$c_{\max}\left(\frac{1}{2} + A, H\right) \le 1 - \frac{H}{3}. \tag{47}$$

For completeness, at $c = 1 - H/3$ the Cell-L polynomial is

$$\frac{H}{81}(H^3 - 12H^2 - 63H + 27) > 0 \quad (0 < H \le 3/8),$$

while the Cell-T expression is increasing in the larger boundary input $p \in [1/2, 1 - H]$; at the order-change point it is nonnegative, and if the cell is already active at $p = 1/2$ its endpoint sign is $H^4 - 5H^3 + 9H^2/4 + 12H - 9/2 > 0$. Also $1 - H/3 \le 2p$, so the selected, rather than discarded, component is used. This proves (47).

The CE2 coupling and (46) give the strict outer ratio

$$d_2^C < \frac{H}{3}. \tag{48}$$

There are two endpoint sources for H. If $H = 1 - u$, put $K = x + 2y - xy - D$; then $d_2^C/H = 1 - K/(SH)$, and the diameter endpoint bounds give $K/(SH) > 2/3$. If $H = 1 - a_0$, the disk inequality gives $H \ge L(y) = (2 + y - \sqrt{4 - 3y^2})/2$; squaring the CE2 comparison yields $(D + xy - x - y)/S < L(y)/3 \le H/3$. These are precisely the two alternatives, proving (48).

Now r_2 can be covered only by T_2, the center, and a possible bridge from T_1. The normal form makes a bridge imply $H < d_2^C$, contrary to (48). Without a bridge, boundary propagation gives T_2 inputs strictly exceeding $1/2 + A$ and H; then (47) gives $c_2 \le 1 - H/3 < 1 - d_2^C$, so T_2 cannot meet the center interval either. □

Lemma 4.18 (Nonadjacent positive-support placements)**.** *Assume an original open-role skeleton cover in the CE2 exact-M_0 branch, with $T_C \cap \{M_0, \dots, M_5\} = \{M_0\}$. Suppose T_0 is uniquely supercritical, T_τ is the unique Vd1/Vd2 row for $\tau \in \{2, 3, 4\}$, and every other vertex row is nonsupercritical Vd0; in particular, there is no additional positive-adjacent-support row. Then the perimeter together with r_τ is not covered.*

Proof. Let A, H be the two boundary demands propagated to T_τ, and put $B = 1 - A$, $U = 1 - H$. The distance endpoint

$$\beta(z) = \frac{-z + \sqrt{4 - 3z^2}}{2}$$

gives $B \le \beta(x)$ and $U \le \beta(y)$, whether an extent comes from the supercritical row or from the corresponding CE2 endpoint. The exact CE2 comparison

$$\frac{D + xy}{S} + \beta(x) < 2 \tag{49}$$

follows by squaring: with $b = \beta(x)$, the required quadratic in y is concave, positive at $y = 0$, and at $y = 1$ its numerator reduces to $4x^2(x^4 + 4x^3 + 9x^2 + 11x + 8)$. Reflection gives the y version. Since $u + v = (D + xy)/S$, it follows that

$$u + v - 1 < \min(A, H) =: m. \tag{50}$$

Put $P = vS - y$ and $Q = uS - x$. Then $P + Q = (u + v - 1)S$, while the center lengths on r_2, r_3, r_4 are

$$\frac{P}{S}, \qquad \min\left\{\frac{P}{y}, \frac{Q}{x}\right\}, \qquad \frac{Q}{S}.$$

Equation (50) makes each strictly less than m. On the other hand (45) gives $c_\tau < 1 - m$. Hence the vertex-side demand $1 - d^C_\tau$ is strictly larger than c_τ, leaving r_τ uncovered. □

Lemma 4.19 (Vd1–supercritical replacement)**.**
Assume an original open-role perimeter-and-radial-skeleton cover in the CE2 exact-M_0 branch and the complementary no-additional-positive-support case. Let T_i, $i \neq 0$, be uniquely supercritical, and let the unique Vd1 row, distinct from T_0 and adjacent to T_i, contain M_i. Assume every other vertex row is nonsupercritical Vd0 and the center has no boundary trace on the shared edge of this pair. Then the pair can be replaced by two open nonsupercritical Vd0 roles preserving all boundary and radial demands used by Proposition 4.7.

Proof. Normalize the Vd1 row at V_0 and the supercritical row at V_1. By the shared-edge hypothesis, no center trace intervenes on $e_{0,1}$; by the no-additional-positive-support hypothesis, no third vertex role has positive boundary or radial support in the handoffs used below. In (45), midpoint rescue and one-sided support imply $t \geq 1$ and, with

$$c = \frac{d - a - tb}{t + 1}, \quad \lambda = \frac{t(1 - b)}{t + 1}, \quad \mu = \frac{d - a - tb - 1}{t},$$

give the strict consequences

$$a + c < 1, \qquad a < \lambda \leq \frac{1}{2}, \qquad b < \frac{1}{2}, \qquad \mu < 1 - a. \tag{51}$$

The shared edge forces the supercritical input $a_i \geq 1 - b > 1/2$, and radial coverage forces $c_i \geq \lambda$. A direct consequence of Proposition 4.2 is

$$(x, y, z) \in \mathcal{A}, \quad x \geq 1/2, \quad z \leq 1/2 \quad \Longrightarrow \quad y \leq 1 - z.$$

Indeed, substituting $x = 1/2, y = 1 - z$ in the selected supercritical polynomial gives $z^2(2z - 5)(2z - 1)/4 \geq 0$, and the polynomial is increasing in both relevant coordinates. Hence $b_i \leq 1 - c_i \leq 1 - \lambda$ and $a + b_i < 1$. Open radial handoff further gives

$$c_i^{\mathrm{req}} < \max\{c_i, \mu\} < \max\{1 - b_i, 1 - a\}.$$

Choose $a < p_1 < p_2 < 1 - b_i$ within these strict margins. In local coordinates use the unit triangles

$$\Delta_p^- = \mathrm{conv}\{(0, 1 - p), (1, 1 - p), (0, -p)\}, \quad p \leq 1/2,$$

and

$$\Delta_p^+ = \mathrm{conv}\{(p, 0), (p, 1), (p - 1, 0)\}, \quad p \geq 1/2.$$

Their incident reaches are $p, 1 - p$, their own-radial reach is respectively $1 - p, p$, and they have zero adjacent support. Small translations, chosen below every strict margin above, make the two replacements open Vd0 roles with boundary sum < 1 while preserving the shared boundary and both radial handoffs. Thus the datum reduces to Proposition 4.7. □

Proposition 4.20 (Complementary exact Vd1/Vd2 placements)**.** *Consider the CE2 exact-M_0, $N_+ = 1$ branch with exactly one Vd1 or Vd2 role. Assume that no other vertex row has positive adjacent support and that the unique Vd2 role, if present, covers neither neighboring midpoint. Then the configuration is impossible.*

Proof. By the vertex classification, the no-additional-positive-support hypothesis makes every vertex row other than the unique Vd1/Vd2 row Vd0; all such rows other than the unique supercritical row are nonsupercritical. This is precisely the complementary branch in which Lemmas 4.16–4.19 apply.

Let T_σ be uniquely supercritical and T_τ the unique Vd1/Vd2 row. They are distinct by the corner cap. If $\sigma = 0$, the adjacent placements $\tau = 1, 5$ are excluded by Lemma 4.17, and $\tau = 2, 3, 4$ by Lemma 4.18. If $\tau = 0$, midpoint forcing makes $\sigma = 1$ or 5 and T_0 cover that midpoint. Vd1 is excluded by Lemma 4.16; Vd2 would cover the neighboring midpoint M_σ, contrary to the proposition's hypothesis.

If neither index is zero, midpoint forcing makes the two rows adjacent and the positive-support row rescue the supercritical midpoint. Vd2 is again excluded by the neighboring-midpoint hypothesis; Vd1 reduces by Lemma 4.19 to the $N_+ = 0$ all-Vd0 obstruction. These alternatives exhaust the complementary exact-placement branch. □

4.9. **Strategy 2 terminal routing.**

Proposition 4.21 (Exact-demand terminal branches)**.** *The exact-demand method proves the following terminal exclusions:*

1. *CE1 or CE2, $N_+ = 0$, all vertex roles Vd0;*
2. *CE1 or CE2, $N_+ = 0$, with at least one T3-like role and no Vd1/Vd2;*
3. *CE1 or CE2, $N_+ = 1$, all vertex roles Vd0, with at least one boundary gap;*
4. *CE1 or CE2, $N_+ = 1$, exactly one T3-like role and no Vd1/Vd2;*
5. *the complementary exact-placement subcases of CE2, $N_+ = 1$, with exactly one Vd1 or Vd2: no additional positive-support row and no Vd2 neighboring-midpoint rescue.*

All gap alternatives include singleton gaps.

Proof. The five items are respectively Propositions 4.7, 4.10, 4.5, 4.6, and 4.20. The strictness at open handoffs appears in the weak endpoint bounds, the nonattained supercritical envelope, and the strict replacement margins; no positive-length assumption is made about a V-gap itself. □

5. Strategy 3: area loss

Let

$$A_\triangle = \frac{\sqrt{3}}{4}$$

be the area of a unit equilateral triangle. Throughout this section areas are divided by $A_\triangle$. Thus a unit triangle has normalized area one and H has normalized area six.

5.1. **The local square-loss inequalities.** Fix a vertex V_i. Let P_a and P_b be the points at distances a and b from V_i on its incoming and outgoing boundary edges. For a closed unit equilateral triangle T containing V_i, P_a, P_b, put

$$G(T) = \frac{\operatorname{area}(T \setminus H)}{A_\triangle}.$$

Equivalently, if

$$f(a,b) = \max_T \frac{\operatorname{area}(T \cap H)}{A_\triangle}, \qquad G(a,b) = 1 - f(a,b),$$

then $G(a,b)$ is the least possible normalized loss. Reflection in the radial axis interchanges a and b, and increasing either demand can only decrease f.

Lemma 5.1 (Local wedge reduction)**.** *After rotating indices to $i = 0$, write*

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0), \qquad W = \{(x,y) : x \geq 0,\ y \geq 0\}.$$

If a closed unit equilateral triangle T contains V_0, then

$$T \cap H = T \cap W.$$

Normalized Euclidean area is twice the corresponding (x,y)-coordinate area.

Proof. The two basis vectors have length one and angle $120°$, so

$$\|(x,y)\|^2 = x^2 + y^2 - xy, \qquad |\det(V_1 - V_0, V_5 - V_0)| = \frac{\sqrt{3}}{2}.$$

The hexagon is

$$H = \{0 \leq x, y \leq 2,\ |x-y| \leq 1\} \subset W.$$

Conversely, if $(x,y) \in T \cap W$, then the diameter-one condition gives $x^2 + y^2 - xy \leq 1$. Hence

$$|x-y|^2 \leq x^2 + y^2 - xy \leq 1, \qquad \frac{3}{4}x^2 \leq x^2 + y^2 - xy,$$

and the same estimate holds for y. Thus $|x - y| \leq 1$ and $0 \leq x, y \leq 2/\sqrt{3} < 2$, proving $(x,y) \in H$. The determinant identity proves the area assertion. $\square$

Theorem 5.2 (Unconditional local square loss)**.** *For every feasible pair $0 \leq a, b \leq 1$ and every realizing closed unit equilateral triangle T,*

$$G(T) \geq \min(a,b)^2. \tag{52}$$

If $a + b > 1$, then

$$G(T) \geq \max(a,b)^2. \tag{53}$$

Consequently the same inequalities hold for $G(a,b) = 1 - f(a,b)$.

Proof. In the coordinates of Lemma 5.1, physical rotation through $60°$ is $R(x,y) = (x - y, x)$. Put

$$0 \leq t \leq 1, \qquad D = 1 - t + t^2, \qquad z = \sqrt{D}.$$

After choosing one of the six directed sides, every orientation has one of the following two forms:

$$\begin{array}{c|cc} & d & Rd \\ \hline \mathrm{I} & z^{-1}(1,t) & z^{-1}(1-t,1) \\ \mathrm{II} & z^{-1}(t,-(1-t)) & z^{-1}(1,t). \end{array} \tag{5.1}$$

Indeed these forms cover, respectively, the physical direction sectors $[120°, 180°]$ and $[60°, 120°]$; their endpoints cover the common boundary orientations.

For Type I introduce

$$U = \alpha + y - tx, \qquad V = \beta + x - (1-t)y.$$

Then T is the simplex $U, V \geq 0$, $U + V \leq z$. Containment of $(0,0), (0,a), (b,0)$ gives

$$\alpha \geq tb, \quad \beta \geq (1-t)a, \quad \alpha + \beta + (1-t)b \leq z, \quad \alpha + \beta + ta \leq z. \tag{5.2}$$

The two inequalities defining W become

$$(1-t)U + V \geq (1-t)\alpha + \beta, \qquad U + tV \geq \alpha + t\beta.$$

For fixed t, lowering α or β therefore enlarges $T \cap W$. The outside loss is minimized at $\alpha = tb$, $\beta = (1-t)a$. At that placement the portion outside W has vertices, in the (U, V)-plane,

$$(0,0), \quad (a + tb, 0), \quad (tb, (1-t)a), \quad (0, b + (1-t)a).$$

Since the Jacobian of $(x, y) \mapsto (U, V)$ has absolute value D,

$$G(T) \geq G_{\mathrm{I}}(a, b; t) = \frac{(1-t)a^2 + tb^2 + 2t(1-t)ab}{D}. \tag{54}$$

If $a \leq b$, then

$$D(G_{\mathrm{I}} - a^2) = t\{b^2 - a^2 + (1-t)(a^2 + 2ab)\} \geq 0;$$

the reflected factorization proves $G_{\mathrm{I}} \geq b^2$ when $b \leq a$. This proves (52) in Type I.

Suppose now that $a + b > 1$ and $a \leq b$. Set

$$r = \frac{a}{b}, \qquad t_0 = \frac{1 - r^2}{1 + 2r}.$$

Since $b > 1/(1 + r)$, feasibility in (54) and (5.2) excludes $0 \leq t < t_0$: for $0 < t < t_0$,

$$\left(\frac{1 + r(1-t)}{1 + r}\right)^2 - D = \frac{t(1 + 2r)(t_0 - t)}{(1 + r)^2} > 0,$$

so $b + (1-t)a > z$, while $t = 0$ would give $a + b \leq 1$. Hence $t \geq t_0$, and

$$D(G_{\mathrm{I}} - b^2) = (1-t)b^2\{r^2 - 1 + t(2r + 1)\} \geq 0.$$

If $b \leq a$, put $r = b/a$ and $t_1 = (r^2 + 2r)/(1 + 2r)$. The identical reflected calculation excludes $t > t_1$ and gives

$$D(G_{\mathrm{I}} - a^2) = ta^2\{r^2 + 2r - t(1 + 2r)\} \geq 0.$$

Thus (53) holds in Type I.

For Type II put

$$U = \alpha + (1-t)x + ty, \qquad V = \beta + tx - y.$$

Again T is the simplex $U, V \geq 0$, $U + V \leq z$. Its exact reaches on the positive y- and x-axes are

$$A = \beta, \qquad B = z - \alpha - \beta, \qquad A + B \leq z \leq 1. \tag{5.3}$$

In particular Type II is impossible when $a + b > 1$. Its wedge intersection is the quadrilateral with vertices

$$(0,0), \quad (B, 0), \quad \left(\frac{B + (1-t)A}{D}, \frac{A + tB}{D}\right), \quad (0, A).$$

Hence

$$G(T) = G_{\mathrm{II}}(A, B; t) = 1 - \frac{(1-t)A^2 + tB^2 + 2AB}{D}. \tag{55}$$

Let $S = A + B$. If $A \le B$, write $A = rS$, $B = (1-r)S$ with $0 \le r \le 1/2$. Since $S^2 \le D$, and with

$$C = (1-t)r^2 + t(1-r)^2 + 2r(1-r),$$

we have $G_{\mathrm{II}} \ge 1 - C$. The exact factorization

$$1 - C - r^2 D = (1-t)(1 - 2r + tr^2) \ge 0$$

gives $G_{\mathrm{II}} \ge r^2 D \ge A^2$. Reflection gives $G_{\mathrm{II}} \ge B^2$ when $B \le A$. Since $A \ge a$ and $B \ge b$, this proves (52). The two orientation types are exhaustive, so the theorem follows. □

5.2. **The direct** T3-like **loss.** The next theorem uses the classification data (o, n) from Section 2: o counts triangle vertices outside H, and n counts adjacent radial arms met in positive length. Thus a T3-like role has $(o, n) = (2, 1)$.

Theorem 5.3 (Direct T3-like loss)**.** *Let T be a closed unit V_i-triangle of* T3-like *type, and let a, b be lower bounds for its adjacent boundary reaches. Then*

$$a + b \le 1. \tag{56}$$

Moreover, for $0 \le m \le 1/2$*,*

$$a, b \ge m \quad \Longrightarrow \quad G(T) \ge 2m - 4m^2. \tag{57}$$

Proof. Use the two orientation forms (5.1). In Type I the three simplex vertices are

$$\begin{aligned}
Q_0 &= D^{-1}(-(1-t)\alpha - \beta, -\alpha - t\beta),\\
Q_1 &= D^{-1}(z - (1-t)\alpha - \beta, tz - \alpha - t\beta),\\
Q_2 &= D^{-1}((1-t)z - (1-t)\alpha - \beta, z - \alpha - t\beta).
\end{aligned}$$

The first coordinate of Q_1 and the second coordinate of Q_2 are nonnegative. If two vertices lie outside the wedge, then Q_0 and exactly one of Q_1, Q_2 do. If Q_1 is outside, the sole wedge vertex Q_2 has both coordinates strictly below one; this follows from $(1-t)/z \le 1$, with strictness either from $t > 0$ or from $\alpha > 0$ at $t = 0$. If Q_2 is outside, the reflected estimate $t/z \le 1$ gives the same conclusion for Q_1. Axis intersection points are also at coordinate at most one by the diameter bound. Hence neither adjacent radial arm is met in positive length. A Type I triangle with $o = 2$ has $n = 0$, so it cannot be T3-like.

For Type II retain the exact reaches A, B in (5.3), put $S = A + B$, and let

$$Q = \left(\frac{B + (1-t)A}{D}, \frac{A + tB}{D}\right)$$

be the unique triangle vertex in the wedge. Equation (5.3) immediately gives $a + b \le A + B \le 1$, proving (56). The endpoint orientations $t = 0, 1$ cannot have $(o, n) = (2, 1)$; hence $0 < t < 1$. Reflecting if necessary, assume that $Q_x > 1$ and $Q_y \le 1$. The first inequality is

$$S - tA > D.$$

Since $S \le z$, it follows that

$$A < L := \frac{z-D}{t} = \frac{z(1-z)}{t}. \tag{5.4}$$

The normalized inside area is

$$I(S,A) = \frac{(1-t)A^2 + t(S-A)^2 + 2A(S-A)}{D}.$$

For fixed t, A, this increases with S, so the loss is bounded below by its value at $S = z$. There

$$G(z,A) = \frac{tA^2 + (1-t)(z-A)^2}{D},$$

and this decreases for $A < L < (1-t)z$. Thus

$$G(T) \ge G(z,L). \tag{5.5}$$

Set $r = (1-z)/t$. Then $0 < r < 1/2$, and eliminating t, z gives

$$t = \frac{1-2r}{1-r^2}, \qquad z = \frac{r^2-r+1}{1-r^2},$$

$$L = A_0(r) := \frac{r(r^2-r+1)}{1-r^2}, \qquad z - L = B_0(r) := \frac{r^2-r+1}{1+r}.$$

The limiting normalized inside area and loss are

$$F_0(r) = \frac{(r^2-r+1)^2}{1-r^2}, \qquad G_0(r) = 1 - F_0(r).$$

Since the exact reach A is at least the required demand $a \ge m$, (5.4) gives $A_0(r) > m$. Direct simplification yields

$$G_0(r) - \{2A_0(r) - 4A_0(r)^2\} = \frac{r^2(5r^4 - 8r^3 + 13r^2 - 8r + 2)}{(1-r^2)^2} \ge 0,$$

because the last polynomial is at least $9r^2 - 8r + 2 \ge 2/9$ on $[0, 1/2]$. Also $A_0(r) \le r$, and

$$G_0(r) - \frac{1}{4} = \frac{(1-2r)(2r^3 - 3r^2 + 6r - 1)}{4(1-r^2)} \ge 0 \quad \text{when } A_0(r) \ge \frac{1}{4};$$

the cubic is increasing and has value $11/32$ at $r = 1/4$.

Finally $\psi(u) = 2u - 4u^2$ is increasing on $[0, 1/4]$ and never exceeds $1/4$ on $[0, 1/2]$. If $A_0(r) \le 1/4$, then $G_0(r) \ge \psi(A_0(r)) \ge \psi(m)$; if $A_0(r) \ge 1/4$, then $G_0(r) \ge 1/4 \ge \psi(m)$. Together with (5.5), this proves (57). The reflected crossing is identical. □

5.3. **The cyclic area certificates.** The selected handoffs of Theorem 2.8 have the form

$$(a_i, b_i) = (1 - x_{i-1}, x_i), \qquad 0 < x_i < 1,$$

with cyclic indices. Row i is selected supercritical exactly when $x_i > x_{i-1}$.

Lemma 5.4 (At least two selected ascents)**.** *Suppose every selected pair is feasible and at least two rows are supercritical. Then*

$$\sum_{i=0}^{5} f(1 - x_{i-1}, x_i) < \frac{99}{20} < 5.$$

Proof. Put $m = \min_i x_i$ and $M = \max_i x_i$. The reflection $y_i = 1 - x_{-i-1}$ swaps each row's two coordinates and preserves the sum and the number of ascents. Hence assume $m \le 1 - M$. Every row coordinate is at least m, so (52) gives loss at least m^2.

Choose an ascent out of a minimum plateau, say $x_{p-1} = m < x_p$. Its row contains the coordinate $1 - m$ and (53) gives loss at least $(1 - m)^2$. At a second ascent $q \ne p$, one row coordinate is strictly larger than $1/2$, so its loss is strictly larger than $1/4$. The other four rows lose at least m^2 each. Therefore the total loss is strictly larger than

$$4m^2 + (1 - m)^2 + \frac{1}{4} = 5\left(m - \frac{1}{5}\right)^2 + \frac{21}{20} \ge \frac{21}{20}.$$

Subtracting from six proves the assertion. □

Lemma 5.5 (One ascent and a T3-like row)**.** *Suppose there is exactly one selected supercritical row, all roles are* Vd0 *or* T3-like*, and at least one is* T3-like*. Then the total normalized loss of the six actual vertex triangles is strictly greater than one.*

Proof. Rotate so that the unique ascent is row 0. Then

$$x_0 > x_5, \qquad x_0 \ge x_1 \ge \cdots \ge x_5.$$

Put $M = x_0$, $m = x_5$, and reflect as above so that $m \le 1-M$. Every row coordinate is at least m, and $0 < m \le 1/2$. The supercritical row has pair $(1 - m, M)$, so its loss is at least $(1 - m)^2$. By (56), a T3-like row is not row 0; by Theorem 5.3 it loses at least $2m - 4m^2$. Each of the remaining four rows loses at least m^2. The total is therefore at least

$$(1 - m)^2 + (2m - 4m^2) + 4m^2 = 1 + m^2 > 1.$$

□

Proposition 5.6 (Branches closed by area)**.** *The following branches are impossible:*

(1) T_C *is* CE0*,* $N_+ \ge 2$*, with arbitrary vertex-role types;*
(2) T_C *is* CE0*,* $N_+ = 1$*, no role is* Vd1 *or* Vd2*, and at least one role is* T3-like*.*

Proof. In the CE0 class, the center role has no positive-length boundary trace. If the six vertex roles missed a boundary point, openness of the center role would give such a trace. Hence the vertex roles cover ∂H, so Theorem 2.8 applies.

If $N_+ \ge 2$, its at-least-two clause selects handoffs with at least two supercritical rows. Lemma 5.4 bounds the six vertex triangles' total normalized inside area by less than $99/20$. The center triangle contributes at most one, so the total is less than

$$\frac{99}{20} + 1 = \frac{119}{20} < 6,$$

which is insufficient to cover H.

If $N_+ = 1$, the exact-one clause preserves the unique actual supercritical index for every strict selection. The exhaustive vertex-type hypothesis gives the assumptions of Lemma 5.5; the six vertex triangles contribute less than five normalized area, and the center triangle contributes at most one. Again the sum is strictly below the normalized area six of H. □

6. Strategy 4: the zero-gap AB-set residual core

This section proves a center-class-independent theorem. Its hypotheses are that all six vertex roles are Vd0, that they cover the full boundary, and that exactly one actual boundary row is supercritical. The theorem will therefore apply both to the CE0 branch and to the zero-gap subcases of CE1 and CE2.

6.1. **Corner-clipped AB-unions.**

Let e_A, e_B be unit vectors with angle $120°$, put

$$O = 0, \qquad A = ae_A, \qquad B = be_B, \qquad C = \{ue_A + ve_B : u, v \geq 0\},$$

and define

(58)
$$\mathcal{R}(a,b) = C \cap \bigcup \{T : T \text{ is a closed unit equilateral triangle and } O, A, B \in T\}.$$

The finite description in Theorem B.1 applies to the four-point set $\{O, A, B, x\}$ and makes membership in (58) an exact finite semialgebraic condition of degree at most four. We shall also use the immediate antitonicity

$$a' \geq a, \quad b' \geq b \quad \Longrightarrow \quad \mathcal{R}(a', b') \subseteq \mathcal{R}(a,b). \tag{59}$$

The strict supercritical branch admits a simpler closed form. In the next statement write $x = ue_A + ve_B$ and $\|x\|^2 = u^2 + v^2 - uv$.

Theorem 6.1 (Strict supercritical AB-union)**.** *Assume*

$$a + b > 1, \qquad \rho = a^2 + ab + b^2 < 1,$$

put $h = \sqrt{3}/2$, $D = \sqrt{4\rho - 3}$, and define

$$\alpha = \frac{h(a + 2b - aD)}{2\rho}, \qquad \beta = \frac{h(a - b + (a+b)D)}{2\rho},$$
$$\gamma = \frac{h(-a + b + (a+b)D)}{2\rho}, \qquad \delta = \frac{h(2a + b - bD)}{2\rho}.$$

Then all four coefficients are positive and

$$\mathcal{R}(a,b) = (C \cap \mathcal{D}_A \cap \mathcal{H}_A) \cup (C \cap \mathcal{D}_B \cap \mathcal{H}_B), \tag{60}$$

where

$$\mathcal{D}_A = \{(u,v) : (u-a)^2 + v^2 - (u-a)v \leq 1\},$$
$$\mathcal{D}_B = \{(u,v) : u^2 + (v-b)^2 - u(v-b) \leq 1\},$$
$$\mathcal{H}_A = \{(u,v) : \alpha(u-a) + \beta v \leq 0\},$$
$$\mathcal{H}_B = \{(u,v) : \gamma u + \delta(v-b) \leq 0\}.$$

Its non-axis frontier consists, in counterclockwise order, of the A-circle, the A-line, the B-line, and the B-circle.

Proof. The identities

$$4\rho - 3(a+b)^2 = (a-b)^2, \qquad (a+b)^2 D^2 - (a-b)^2 = 4\rho((a+b)^2 - 1)$$

give $0 < D < 1$ and positivity of the four coefficients.

We apply the caliper criterion of Theorem B.1. For a point $x = (u,v)$ in the relative interior of C but outside $\triangle OAB$, the four hull vertices occur in the order

O, B, x, A. The two cone-axis hull edges cannot certify a unit triangle: their support sums, divided by h, are respectively

$$\max\{a, u\} + \max\{b, v - u\}, \qquad \max\{b, v\} + \max\{a, u - v\},$$

and are at least $a + b > 1$. It remains to test the Ax and Bx calipers.

For the Ax caliper put $p = u - a$ and $r^2 = p^2 + v^2 - pv$. Exact evaluation of the three supports gives

(61)
$$\frac{G_A}{h} = \frac{\max\{r^2, -ap + (a+b)v\} + N_A}{r}, \quad N_A = \begin{cases} 0, & p \le 0, \\ ap, & 0 \le p \le v, \\ (a+b)p - bv, & p \ge v. \end{cases}$$

No support label is omitted in these three sectors. If $p \le 0$, $G_A \le h$ is equivalent to

$$r \le 1, \qquad -ap + (a+b)v \le r.$$

The second inequality is the half-plane condition in (60), because

$$r^2 - (-ap + (a+b)v)^2 = \frac{(cp + dv)(c'p + d'v)}{4\rho},$$

where

$$c = a + 2b - aD, \quad d = a - b + (a+b)D, \quad c' = a + 2b + aD, \quad d' = a - b - (a+b)D,$$

and $c, c', d > 0 > d'$. Thus the low sector is exactly $\mathcal{D}_A \cap \mathcal{H}_A$. In the middle sector $0 < p \le v$ one has $r \le v$, whereas the numerator in (61) is at least $(a+b)v > r$, so no fit is possible.

In the high sector $p \ge v$, the inequalities $G_A \le h$ imply

$$r^2 + (a+b)p - bv \le r, \qquad bp + av \le r. \tag{62}$$

Write $x - A = ry(\theta)$, $0 \le \theta \le 60°$, with cone coordinates

$$y(\theta) = te_A + we_B, \quad t = \frac{2}{\sqrt{3}}\cos(\theta - 30°), \quad w = \frac{2}{\sqrt{3}}\sin\theta.$$

Then (62) gives

$$r \le f(\theta) = 1 - (a+b)t + bw, \qquad S(\theta) = bt + aw \le 1.$$

Let n_B be the unit vector with dual coordinates (γ, δ). The identities

$$b\gamma + (a+b)\delta = h, \quad (a+b)\gamma + a\delta = \frac{h}{2}(1 + D), \quad b\delta - a\gamma = \frac{h}{2}(1 - D)$$

show that the unique angle θ_B of this normal satisfies $S(\theta_B) = 1$ and $f(\theta_B) = (1 - D)/2$. The function S has at most one critical point, a maximum, so $S(\theta) \le 1$ implies $\theta \le \theta_B$. On the feasible part $f, f' \ge 0$; hence $f(\theta)\cos(\theta_B - \theta)$ is increasing. Consequently

$$\langle n_B, x - B\rangle \le a\gamma - b\delta + \frac{h}{2}(1 - D) = 0.$$

The diameter bound also gives $\|x - B\| \le 1$. Thus every high sector fit belongs to $\mathcal{D}_B \cap \mathcal{H}_B$. Reflection exchanges a, u with b, v and supplies the corresponding result for the Bx caliper. This proves (60) in the cone interior; the axes follow by taking limits, since both sides are closed.

For completeness, the line–circle junctions are obtained by solving the two displayed line and circle equations. The line normals satisfy

$$\alpha^2+\alpha\beta+\beta^2=h^2, \qquad \gamma^2+\gamma\delta+\delta^2=h^2,$$

and the line determinant is

$$\alpha\delta-\gamma\beta=\frac{3(1-D)(D+3)}{16\rho}>0.$$

The successive cone slopes are

$$+\infty, \quad \frac{2}{1-D}, \quad \frac{a(1-D)+2b}{2a+b(1-D)}, \quad \frac{1-D}{2}, \quad 0;$$

the two middle comparisons reduce to $a(4-(1-D)^2)>0$ and $b(4-(1-D)^2)>0$. This proves the asserted four-piece order. ☐

For each hexagon vertex V_i, let $R_i(a,b)$ denote the rotated copy of $\mathcal{R}(a,b)$ in its corner cone, with the first coordinate incoming and the second outgoing. Formula (59) is valid for every R_i.

6.2. **From zero gaps to two model cores.**

Lemma 6.2 (The extreme jump)**.** *Assume the six vertex roles cover ∂H and exactly one actual row is supercritical. Rotate its index to* 4*. The strict handoffs supplied by Theorem 2.8 satisfy*

$$x_3<x_2<x_1<x_0<x_5<x_4.$$

With

$$a=1-x_3, \qquad b=x_4, \qquad p=1-b, \qquad q=1-a,$$

one has

$$0<a,b<1, \qquad a+b>1, \qquad a^2+ab+b^2<1, \tag{63}$$

and, for $i\neq 4$, the selected demands obey

$$a_i\geq p, \qquad b_i\geq q.$$

Proof. Every selected row other than 4 is strictly nonsupercritical, so $x_i<x_{i-1}$ for $i\neq 4$, which is precisely (6.2). In particular x_3 is the global minimum and x_4 the global maximum. The two anchors of row 4 lie in its open role; their mutual squared distance is a^2+ab+b^2, strictly below the diameter squared of the closed unit triangle. This proves (63). Finally every $x_j\in[x_3,x_4]$, so $1-x_{i-1}\geq 1-x_4=p$ and $x_i\geq x_3=q$. ☐

Define the actual residual and the asymmetric and symmetric model cores by

$$\mathcal{U}_6^\circ=\mathrm{int}(H)\setminus\bigcup_{i=0}^{5}\mathrm{int}R_i(a_i,b_i), \tag{64}$$

$$\mathcal{C}_{\mathrm{asym}}(a,b)=\mathrm{int}(H)\setminus\left(\mathrm{int}R_4(a,b)\cup\bigcup_{i\neq 4}\mathrm{int}R_i(p,q)\right), \tag{65}$$

$$\mathcal{C}_{\mathrm{sym}}(a,b)=\mathrm{int}(H)\setminus\bigcup_{i=0}^{5}\mathrm{int}R_i(p,q). \tag{66}$$

If $P \in \operatorname{int}(H)$ is covered by an open vertex role, then a small ball about P lies both in its corner cone and in its closed associated triangle; hence P lies in the plane interior of its row union. Thus a full cover forces

$$\mathcal{U}_6^\circ \subseteq U_C.$$

Lemma 6.2 and antitonicity give

$$\boxed{\mathcal{C}_{\mathrm{sym}}(a,b) \subseteq \mathcal{C}_{\mathrm{asym}}(a,b) \subseteq \mathcal{U}_6^\circ \subseteq U_C.} \tag{67}$$

6.3. **The forced disk and the three asymmetric witnesses.** Put

$$s = p+q, \qquad m = \min(p,q), \qquad M = \max(p,q), \qquad \chi = s^4 - s^2 + pq.$$

The strict domain (63) gives

$$pq > (1-s)(2-s), \qquad m > 1-s, \qquad s > 1-m. \tag{68}$$

Let $c_* = c_{\max}(p,q)$ be the exact radial envelope of Proposition 4.2. On the present subcritical domain, equation (16) becomes

$$c_* = \begin{cases} C_L(m), & \chi \le 0, \\ \dfrac{2M}{1+\sqrt{4s^2-3}}, & \chi > 0, \end{cases} \tag{69}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

At $\chi = 0$ the two expressions agree and equal s.

Lemma 6.3 (Symmetric radial witnesses)**.** *One has*

$$0 < c_* < 1, \qquad c_* > 1-m. \tag{70}$$

For every k,

$$P_k = (1-c_*)V_k \in \mathcal{C}_{\mathrm{sym}}(a,b).$$

Consequently every convex set containing $\mathcal{C}_{\mathrm{sym}}(a,b)$ contains the centered closed disk

$$\mathcal{D}_\eta = \{x : \|x\| \le \eta\}, \qquad \eta = h(1-c_*). \tag{71}$$

Proof. On the C_L branch, $C_L(m) < 1$ because the defining polynomial is positive at 1; if $s < \sqrt{3}/2$, then $c_* \ge \sqrt{3}/2 > s > 1-m$, while if $s \ge \sqrt{3}/2$ its value at s is $\chi \le 0$, so $c_* \ge s > 1-m$. On the other branch put $E = \sqrt{4s^2-3}$. The selector gives $sE > M-m$, whence $c_* < s < 1$. Moreover, with

$$A_0 = \frac{2M}{1-m} - 1,$$

one has

$$A_0^2 - E^2 = \frac{4(1-s)\{(1-m)(1-2m) + m(2-m)(1-s)\}}{(1-m)^2} > 0.$$

Thus $A_0 > E$ and $c_* > 1-m$. This proves (70).

We need only the following consequence of the neighboring-ray formula in Theorem 3.1. If $0 < m \le M$ and $m + M \le 1$, then

$$C_+(M,m) = 1-m, \qquad C_+(m,M) \le 1-m. \tag{72}$$

For reference, when $M \le 1/2$ this follows by evaluating the defining cubic

$$f_M(z) = z^3 - (M+2)z^2 + 2(M+1)z - 1$$

at $z = 1-m$: $f_M(1-m) \ge m^2(1-2m) \ge 0$. The same evaluation with m in place of M handles the first two pieces of $C_+(m, M)$. On its final piece,

$$C_+(m, M) = m + \frac{1}{2} - \sqrt{(m+M)^2 - \frac{3}{4}}.$$

If $m \le 1/4$, this is at most $m + 1/2 \le 1 - m$. If $m > 1/4$, let $\pi = \pi(m)$ be the root of f_m in $[m, 1]$ and put $d = \pi - m$. Then $0 \le d \le 1 - 2m$ and the root equation is

$$d\pi(2-\pi) = 2\pi - 1.$$

The final branch begins after $\tau(m) = 1 - m - d(1-\pi)$, and the preceding identity gives

$$(m + \tau(m))^2 - (4m^2 - 2m + 1) = (d - 2m)(d + 2m - 1) \ge 0.$$

Indeed $m > 1/4$ implies $d \le 1 - 2m < 2m$, so both factors are nonpositive. Thus $M > \tau(m)$ gives $(m+M)^2 > 4m^2 - 2m + 1$, and hence $C_+(m, M) < 1 - m$. Together with (70),

$$c_* > \max\{C_+(p, q), C_+(q, p)\}. \tag{73}$$

It suffices to prove the assertion for P_0. In the local cone coordinates at V_j, its coordinates and squared distance from V_j are

j	P_0	$\|P_0 - V_j\|^2$
0	(c_*, c_*)	c_*^2
1	$(c_*, 1)$	$c_*^2 - c_* + 1$
2	$(1, 2 - c_*)$	$c_*^2 - 3c_* + 3$
3	$(2 - c_*, 2 - c_*)$	$(2 - c_*)^2$
4	$(2 - c_*, 1)$	$c_*^2 - 3c_* + 3$
5	$(1, c_*)$	$c_*^2 - c_* + 1.$

By definition of the radial maximum, P_0 lies on, but not in the plane interior of, $R_0(p, q)$. The diameter bound excludes it from R_2, R_3, R_4. Membership in R_1 or R_5 would contradict (73). Hence $P_0 \in \mathcal{C}_{\mathrm{sym}}$, and rotation gives all six points. Their convex hull is a regular hexagon of circumradius $1 - c_*$, whose incircle is exactly (71). □

We next define the three asymmetric witnesses. At V_4 use local coordinates

$$P = V_4 + u(V_5 - V_4) + v(V_3 - V_4), \qquad \mathsf{q}(u, v) = u^2 + v^2 - uv.$$

The outgoing demand is b and the incoming demand is a. Accordingly, set

$$\alpha = \frac{h(b + 2a - bD)}{2\rho}, \qquad \beta = \frac{h(b - a + (a+b)D)}{2\rho},$$
$$\gamma = \frac{h(-b + a + (a+b)D)}{2\rho}, \qquad \delta = \frac{h(2b + a - aD)}{2\rho},$$

where now $\rho = a^2 + ab + b^2$ and $D = \sqrt{4\rho - 3}$. The two line pieces of Theorem 6.1 are

$$L_-(\lambda) = (b - \beta\lambda, \alpha\lambda), \qquad L_+(\mu) = (\delta\mu, a - \gamma\mu).$$

They meet at $\lambda_* = \mu_* = 8h\rho/(3(D+3))$ in

$$J = \left(\frac{2b + a - aD}{D+3}, \frac{b + 2a - bD}{D+3} \right). \tag{74}$$

Put

$$E_- = (1-b, 2-b), \qquad E_+ = (2-a, 1-a).$$

The restrictions of the two unit-circle equations are

$$\begin{aligned} g_-(\lambda) &= \frac{3}{4}\lambda^2 - 3(\alpha + b\beta)\lambda + 3b^2 - 3b + 2, \\ g_+(\mu) &= \frac{3}{4}\mu^2 - 3(\delta + a\gamma)\mu + 3a^2 - 3a + 2. \end{aligned} \tag{75}$$

Let λ_- and μ_+ be their first roots; explicitly,

$$\begin{aligned} \lambda_- &= 2(\alpha + b\beta) - \frac{2}{3}\sqrt{9(\alpha + b\beta)^2 - 9b^2 + 9b - 6}, \\ \mu_+ &= 2(\delta + a\gamma) - \frac{2}{3}\sqrt{9(\delta + a\gamma)^2 - 9a^2 + 9a - 6}. \end{aligned}$$

Lemma 6.4 (Fixed-line sign lemma). *Throughout* (63)*,*

$$\lambda_* < \lambda_- < h^{-1}, \qquad \mu_* < \mu_+ < h^{-1}. \tag{76}$$

Moreover $J_u, J_v < 1/2$*. The point* $L_+(\mu_+)$ *satisfies*

$$(L_+(\mu_+))_u + (L_+(\mu_+))_v < 3 - 2a,$$

and the reflected bound holds for $L_-(\lambda_-)$*. Each first-root point lies outside the opposite comparison circle, while* J *lies strictly outside both comparison circles.*

Proof. Put

$$U = a + b - 1, \qquad z = a - b, \qquad D^2 = z^2 + 6U + 3U^2.$$

Then $0 < U < 1/6$ and $z^2 < \Omega := 1 - 6U - 3U^2$. We give the fixed-line calculation for the circle centered at $E_+ = (2-a, 1-a)$; reflection gives the calculation for E_-. Put

$$F(P,t) = \mathsf{q}(P - (1+t,t)) - 1, \qquad t_0 = 1 - a,$$

so this circle is $F(P, t_0) = 0$. At the junction, exact expansion gives

$$(D+3)^2 F(J, t_0) = 3D(z^2 + Uz + 1 - 3U) + P_J,$$

where

$$P_J = z^4 + 5z^2 + 6Uz^2 + 3U^2z^2 + 9Uz + 3U + 15U^2 > 0;$$

indeed $z^2 + Uz + 1 - 3U \geq 1 - 3U - U^2/4 > 0$, and $5z^2 + 9Uz + 15U^2$ is positive definite. Thus $F(J, t_0) > 0$.

On the opposite line L_-, the derivative at the junction, after multiplication by a positive factor, has numerator

$$N_0 = D(9 + 3z^2 + 6Uz - 9U^2) + P_0,$$

where

$$\begin{aligned} P_0 = {} & 18U^3 + 6U^2z^2 + 63U^2 + 18Uz^2 + 18Uz + 54U \\ & + 2z^4 + 9z^2 + 9. \end{aligned}$$

The coefficient of D is at least $9 - 6U - 9U^2 > 0$, while $P_0 \geq 9 + 54U - 18U > 0$. Hence $N_0 > 0$. Since the restriction of F to L_- is a convex quadratic, it is positive from the junction to the line endpoint and has no intersection there.

On the correct line L_+, the value at its outer endpoint h^{-1} has, after multiplication by a positive factor, numerator

$$A_0 = P_2 - 4DU(1+U+z),$$

where

$$P_2 = 3U^4 + 6U^3z + 6U^3 + 4U^2z^2 + 12U^2z + 12U^2 + 2Uz^3 + 6Uz^2 + 2Uz + 6U + z^4 + 2z^2 - 3.$$

For fixed U, replace the odd powers of z by $x = |z|$. The resulting polynomial $P_2^+(U, x)$ satisfies

$$\partial_x P_2^+ = 2(3U^3 + 4U^2x + 6U^2 + 3Ux^2 + 6Ux + U + 2x^3 + 2x) > 0,$$

and therefore its supremum for $z^2 < \Omega$ occurs at $x = \sqrt{\Omega}$. There

$$P_2^+(U, \sqrt{\Omega}) = 4U(U + \sqrt{\Omega} - 3) < 0.$$

As $1 + U + z = 2a > 0$, this gives $A_0 < 0$. The positive junction value and negative endpoint value force the first root strictly between them, proving the L_+ half of (76); reflection proves the other half.

For the coordinate bounds, the exact identities

$$D^2 - (3U - z)^2 = 6U(1 + z - U), \qquad D^2 - (3U + z)^2 = 6U(1 - z - U)$$

give $J_u, J_v < 1/2$. It remains to control the coordinate sum along L_+. For $E = L_+(h^{-1})$, the sign of $3 - 2a - E_u - E_v$ is the sign of

$$D(6U + 2z + 6) + B(U, z),$$

where

$$B(U, z) = -9U^3 - 9U^2z - 9U^2 - 3Uz^2 - 18Uz + 3U - 3z^3 + 3z^2 - 3z + 3.$$

Here

$$B_z = -3(3z^2 + 2Uz + 6U - 2z + 3U^2 + 1) < 0,$$

because the quadratic in parentheses has discriminant $-32U^2 - 80U - 8 < 0$. Hence its minimum on $z^2 < \Omega$ is at $z = \sqrt{\Omega}$, where $B(U, \sqrt{\Omega}) = 6(1 - 3U - \sqrt{\Omega}) > 0$ because $(1 - 3U)^2 - \Omega = 12U^2$. Thus $E_u + E_v < 3 - 2a$. Also

$$J_u + J_v = \frac{(1+U)(3-D)}{3+D} < 1.$$

Indeed this inequality is $3U < D(2 + U)$, which follows from $D^2 \geq 3U(2 + U)$ and $(2+U)^3 > 3U$. Hence $J_u + J_v < 1 < 3 - 2a$, and the coordinate sum is affine on the line segment, so the same strict bound holds at the first root. Reflection gives the bound on L_-. The convexity and signs above also give all same- and opposite-circle assertions. □

Define the local and global witnesses by

$$Q_-^{\mathrm{loc}} = L_-(\lambda_-), \qquad Q_0^{\mathrm{loc}} = J, \qquad Q_+^{\mathrm{loc}} = L_+(\mu_+), \tag{77}$$

and by the displayed conversion from (u, v) to the plane.

Lemma 6.5 (Asymmetric witness membership). *The three points Q_-, Q_0, Q_+ lie in $\mathcal{C}_{\mathrm{asym}}(a, b)$.*

Proof. The root order and positivity of the line coordinates give $0 < u, v < 1$ for all three points. Each lies on the frontier of a unit triangle containing V_4, so $\mathsf{q}(u,v) \le 1$; hence $|u - v| < 1$ and all three points lie in $\operatorname{int}(H)$.

In these coordinates the mandatory points of the five comparison unions are

i	0	1	2	3	5
	$(2,1)$	$(2,2)$	$(1,2)$	E_-	E_+.

Every $R_i(p,q)$ is contained in the unit disk about its mandatory point. For $P = (u,v)$,

$$\|P-(2,1)\|^2 - 1 = \mathsf{q}(u,v) + 2 - 3u, \quad \|P-(1,2)\|^2 - 1 = \mathsf{q}(u,v) + 2 - 3v.$$

The inequalities $J_u, J_v < 1/2$ and the line order exclude the appropriate two witnesses directly. For the remaining pair, the first-root circle and the sum bound in Lemma 6.4 give

$$\|Q_+ - (2,1)\|^2 - 1 = a(3 - a - (Q_+)_u - (Q_+)_v) > a^2,$$

and the reflected inequality for Q_-. All three points are farther than one from $(2,2)$, because writing $u = 1-\xi$, $v = 1-\eta$ gives

$$\|P-(2,2)\|^2 = 1 + \xi + \eta + \xi^2 + \eta^2 - \xi\eta > 1.$$

The circle signs in Lemma 6.4 exclude the interiors of R_3 and R_5. Finally all three points lie on the exact line frontier of $R_4(a,b)$, and hence not in its interior. This proves the claim. $\square$

By (67), Lemmas 6.3 and 6.5 force the compact set

$$K_{\mathrm{wit}}(a,b) = \mathcal{D}_\eta \cup \{Q_-, Q_0, Q_+\} \subset U_C. \tag{78}$$

6.4. **The enclosure inequality.** For a compact set K, let $\Lambda(K)$ be the least side length of a closed equilateral triangle containing K. With R denoting rotation through $2\pi/3$,

$$\Lambda(K) = \frac{1}{h} \min_{\|n\|=1} \sum_{j=0}^{2} h_K(R^j n). \tag{79}$$

Theorem 6.6 (Witness enclosure)**.** *For every* (a,b) *in* (63)*,*

$$\Lambda(K_{\mathrm{wit}}(a,b)) \ge 1.$$

Proof. The disk alone gives $\Lambda(K_{\mathrm{wit}}) \ge 3\eta/h = 3(1-c_*)$, so the assertion is automatic when $c_* \le 2/3$. Assume henceforth $c_* > 2/3$.

We first remove the square radicals in the first-root witnesses. Normalize $\bar\alpha = \alpha/h$, and similarly for the other three coefficients, and put

$$\xi = \lambda/h, \quad \zeta = \mu/h, \quad \xi_* = \zeta_* = \frac{8\rho}{3(D+3)}.$$

Let g_-, g_+ be the rescaled forms of (75), and take one Newton step at the junction:

$$\widehat{\xi}_- = \xi_* - \frac{g_-(\xi_*)}{g_-'(\xi_*)}, \qquad \widehat{\zeta}_+ = \zeta_* - \frac{g_+(\zeta_*)}{g_+'(\zeta_*)}.$$

Since each quadratic is strictly convex, positive and decreasing at the junction, its tangent zero lies strictly between the junction and the first root. Define, in local coordinates,

$$\begin{aligned} A &= \left(b - \frac{3}{4}\bar{\beta}\widehat{\xi}_-, \frac{3}{4}\bar{\alpha}\widehat{\xi}_-\right), \\ B &= J, \\ C &= \left(\frac{3}{4}\bar{\delta}\widehat{\zeta}_+, a - \frac{3}{4}\bar{\gamma}\widehat{\zeta}_+\right), \end{aligned}$$

and convert them to global vectors. Then

$$A \in (Q_0, Q_-), \qquad C \in (Q_0, Q_+).$$

Consequently, for

$$\widehat{K} = \mathrm{conv}(\mathcal{D}_\eta \cup \{A, B, C\}),$$

we have $\widehat{K} \subseteq \mathrm{conv}(K_{\mathrm{wit}})$ and

$$\Lambda(K_{\mathrm{wit}}) \geq \Lambda(\widehat{K}). \tag{80}$$

Put $S = \widehat{K} + R\widehat{K} + R^2\widehat{K}$. Then

$$h_S(n) = \sum_{j=0}^{2} h_{\widehat{K}}(R^j n),$$

and S contains the full R-orbits of the four disks

$$\mathbb{D}(A, 2\eta), \quad \mathbb{D}(B, 2\eta), \quad \mathbb{D}(C, 2\eta), \quad \mathbb{D}(W, \eta), \qquad W = C + RA. \tag{81}$$

For a disk $\mathbb{D}(X, r)$ let

$$I(X, r) = \{n : \|n\| = 1,\ \langle X, n\rangle + r \geq h\}.$$

If the caps of the disks in (81) cover the unit circle, then $h_S(n) \geq h$ for every n and (79) proves $\Lambda(\widehat{K}) \geq 1$.

Put $\tau = h - 2\eta = h(2c_* - 1)$. The rays of

$$A,\ B,\ C,\ W,\ RA \tag{6.1}$$

occur in this cyclic order from A to RA. Indeed the two line distances give

$$\frac{\mathrm{cross}(A, B)}{\|B - A\|} = \alpha(1 - b) + \beta > 0, \quad \frac{\mathrm{cross}(B, C)}{\|C - B\|} = \gamma + \delta(1 - a) > 0.$$

If $A = (-X, -Y)$ and $C = (-P, -Q)$ in the centered cone basis, then $0 < X, Y, P, Q < 1$, $X, Q > 1/2$, and

$$\frac{\mathrm{cross}(C, RA)}{h} = PX + (Q - P)Y > 0.$$

The last sign is immediate unless $Q < P$ and $X < Y$, in which case it is larger than $Q - P/2 > 0$. The ray of $W = C + RA$ lies between those of C and RA. The same coordinates give

$$\|A\|, \|C\| > h/2 > \eta. \tag{82}$$

For the two equal-radius overlaps it is enough to prove

$$\frac{\mathrm{cross}(A, B)}{\|B - A\|} \geq \tau, \qquad \frac{\mathrm{cross}(B, C)}{\|C - B\|} \geq \tau. \tag{83}$$

We record the exact analytic verification. By reflection take $a \le b$ and set $m = 1-b$, $M = 1-a$, $U = a+b-1$, $s = m+M$, and $w = M-m$. The smaller of the two normalized left sides in (83) is

$$d = \frac{\mathcal{A} + U(2m-1) + D(\mathcal{A}+U)}{2\rho}, \qquad \mathcal{A} = 1 - m + m^2. \tag{84}$$

On the C_L sheet, write $F_L(z) = z^4 - z^2 + mz - m^2$. This defining polynomial evaluated at $r = 1 - m/2 + m^2/2$ is

$$F_L(r) = \frac{m^2(m-1)}{16}(m^5 - 3m^4 + 11m^3 - 17m^2 + 28m - 12) > 0.$$

Indeed the polynomial in parentheses is increasing on $(0, 1/2)$ and has value $-33/32$ at $1/2$; its derivative is $28 - 34m + m^2(5m^2 - 12m + 33) > 0$. Since $F_L(h) = -(m - h/2)^2 \le 0$ and the selected root is unique, $2c_* - 1 \le \mathcal{A}$. Put

$$X_0 = \mathcal{A} + U, \qquad Y_0 = 2\rho\mathcal{A} - \mathcal{A} - U(2m-1).$$

Then

$$d - \mathcal{A} = \frac{D(\mathcal{A}+U) - Y_0}{2\rho},$$

and exact expansion gives

$$\begin{aligned} Y_0 = {} & 2(m^2 - m + 1)U^2 + (2m^3 - 2m + 3)U \\ & + (m^2 - m + 1)(2m^2 - 2m + 1) > 0. \end{aligned}$$

Furthermore

$$D^2 X_0^2 - Y_0^2 = 4m\rho\Phi(U, m),$$

where

$$\begin{aligned} \Phi(U, m) = {} & (1-m)(m^2 - m + 2)U^2 \\ & + (2 - m^2)(m^2 - m + 1)U \\ & - m(1-m)^2(m^2 - m + 1). \end{aligned}$$

In these variables

$$\chi = U^4 - 4U^3 + 5U^2 - (m+2)U + m(1-m).$$

Since $U^2(U^2 - 4U + 5) > 0$, the selector $\chi \le 0$ implies $U > m(1-m)/(m+2)$; Φ is increasing in U and at that lower endpoint is

$$\frac{m^2(1-m)(m^4 - 2m^3 + 6m^2 - 6m + 4)}{(m+2)^2} > 0.$$

The last quartic equals $(m^2 - m)^2 + 5m^2 - 6m + 4 > 0$. Thus $d > \mathcal{A} \ge 2c_* - 1$.

On the other radial sheet let $E = \sqrt{4s^2 - 3}$. Then $0 \le w < sE$ and $c_* = (s+w)/(1+E)$. At fixed U, differentiation of (84) can be checked without an implicit estimate. Namely,

$$d = \frac{1+D}{2} - UB(D, w), \qquad B(D, w) = \frac{(1+U)(3+D) + w(1-D)}{D^2 + 3}.$$

Here $D^2 = w^2 + 6U + 3U^2$, so $0 \le D' = w/D \le 1$, and

$$B_w = \frac{1-D}{D^2 + 3} \ge 0.$$

Writing $N=(1+U)(3+D)+w(1-D)$, direct differentiation gives

$$B_D=\frac{(1+U-w)(D^2+3)-2DN}{(D^2+3)^2}>-\frac{34}{27}.$$

Indeed the first term in the numerator is positive, while $1+U-w=2a>0$, $N<(7/6)4+1=17/3$, $D<1$, and $D^2+3\geq 3$. If $B_D<0$, then $B_DD'\geq B_D$; otherwise $B_DD'\geq 0$. Hence $B'=B_w+B_DD'>-34/27$, and $U<1/6$ yields

$$d'<\frac{1}{2}+\frac{1}{6}\frac{34}{27}=\frac{115}{162}<1,$$

whereas $(2c_*-1)'=2/(1+E)\geq 1$. Hence their difference decreases with w. At the boundary $w=sE$, where $\chi=0$ and $c_*=s$, the preceding sheet applies: this is an admissible point because $h<s<1$ and $D^2=4s^4-12s+9<1$. Indeed $s^4-3s+2=(s-1)(s^3+s^2+s-2)<0$ on this interval; the second factor is increasing and already equals $(7h-5)/4>0$ at $s=h$. The preceding sheet gives $d\geq 2s-1$. Therefore (83) holds on both sheets, and reflection supplies the other line.

It remains to check the two mixed overlaps. Put

$$\Delta=\operatorname{cross}(C,RA),\qquad u=\langle C,RA\rangle,\qquad Z_X=\tau\|X\|^2-\eta u,$$

$$P_X=(\|X\|^2-\eta^2)\Delta^2-Z_X^2,\qquad X\in\{A,C\}. \tag{85}$$

To compare these quantities with the certificate, represent global vectors as (x,hy), put $C'=R^{-1}C$, and use the notation $\Delta_0,r_X^0,\ell_X^0,P_X^0$ of (104). Rotation invariance and $\eta=h(1-c_*)$, $\tau=h(2c_*-1)$ give the exact scaling

$$\Delta=-h\Delta_0,\qquad Z_X=-h\ell_X^0,\qquad P_X=h^2P_X^0, \tag{86}$$

and $r_X^0=\|X\|^2-\eta^2$. The outward-rounded certificate in Theorem B.3 proves, on the complete strict domain, that the two radicands are nonnegative, $\Delta_0\leq 0$, and $P_A,P_C\geq 0$; the ray-order calculation above makes $\Delta>0$. Consequently

$$\sqrt{\|X\|^2-\eta^2}\,\Delta\geq Z_X,\qquad X=A,C.$$

For $X=A$ this is exactly the common-support inequality for the caps of $\mathbb{D}(C,2\eta)$ and $\mathbb{D}(W,\eta)$; for $X=C$ it is the analogous inequality for $\mathbb{D}(W,\eta)$ and $\mathbb{D}(RA,2\eta)$.

Thus the four consecutive pairs of caps centered on the rays (6.1) overlap. Every cap is a single arc of half-width less than $\pi/2$, and the positive crosses put each overlap in the intervening short sector. They cover the sector from A to RA; rotation through $2\pi/3$ covers the circle. Hence $h_S\geq h$, so $\Lambda(\widehat{K})\geq 1$. Equation (80) completes the proof. $\square$

6.5. **The center-independent obstruction.**

Theorem 6.7 (Center-independent zero-gap obstruction)**.** *There is no cover of H by open unit equilateral triangles $U_C,U_0,\dots,U_5$ such that*

1. *every U_i is a* Vd0 *role at V_i;*
2. *$U_0,\dots,U_5$ cover ∂H; and*
3. *exactly one actual maximal boundary row is supercritical.*

No center class is assumed for the associated closed center role T_C.

Proof. The extreme-jump construction gives a parameter pair in (63), and (67) together with the two witness lemmas gives the compact containment $K_{\mathrm{wit}}(a,b) \subset U_C$. Theorem 6.6 says $\Lambda(K_{\mathrm{wit}}) \geq 1$.

On the other hand, a compact subset of an open unit equilateral triangle has positive clearance from its three sides. Moving all three sides inward by a sufficiently small equal amount gives a closed equilateral triangle of side strictly less than one still containing the compact set. Thus $K_{\mathrm{wit}} \subset U_C$ would imply $\Lambda(K_{\mathrm{wit}}) < 1$, a contradiction. □

Remark 6.8 (Why exact-one is essential). For a distinguished selected ascent $x_3 < x_4$, the five antitone comparisons used in (67) hold exactly when x_3 is a global minimum and x_4 a global maximum. The exact-one chain (6.2) guarantees this. With two or more ascents there need not be any minimum-to-maximum ascent, so replacing all other rows by $R_i(1 - x_4, x_3)$ is not justified. The $N_+ \geq 2$ branch is therefore handled by Strategy 3 or Strategy 1, not by this comparison.

Proposition 6.9 (Branches closed by the residual core)**.** *Theorem 6.7 closes precisely the following rows of the final routing table:*

(1) CE0*,* $N_+ = 1$*, all six vertex roles* Vd0*;*
(2) CE1 *or* CE2*,* $N_+ = 1$*, all six vertex roles* Vd0*, and zero boundary gaps.*

Proof. In the first row, a missed boundary point would have to lie in the open CE0 center role, giving a positive-length center trace on an incident edge. Hence the six vertex roles cover ∂H. In the second row, zero gaps means by definition that the two adjacent open vertex traces cover every boundary edge, including every endpoint. Thus in both rows all three hypotheses of Theorem 6.7 hold. □

7. Exhaustive assembly

We now assemble the four mechanisms. The table is included to make the exhaustiveness audit visible: its rows are mutually exclusive after the center type and the value of N_+ have been fixed, except for the one row explicitly marked as a joint application of Strategies 1 and 2.

center and row count	exhaustive refinement	route
CE0, $N_+ = 0$	all vertex types	1
CE0, $N_+ = 1$	all six roles Vd0	4
CE0, $N_+ = 1$	a Vd1 or Vd2 role	1
CE0, $N_+ = 1$	no Vd1/Vd2 role and a T3-like role	3
CE0, $N_+ \geq 2$	all vertex types	3
CE1/CE2, $N_+ = 0$	all six roles Vd0	2
CE1/CE2, $N_+ = 0$	a Vd1 or Vd2 role	1
CE1/CE2, $N_+ = 0$	no Vd1/Vd2 role and a T3-like role	2
CE1/CE2, $N_+ = 1$	all Vd0, zero gaps	4
CE1/CE2, $N_+ = 1$	all Vd0, exactly one gap	2
CE2, $N_+ = 1$	all Vd0, exactly two gaps	2
CE1, $N_+ = 1$	a Vd1 or Vd2 role	1
CE2, $N_+ = 1$	at least two Vd1/Vd2 roles	1
CE2, $N_+ = 1$	exactly one Vd1/Vd2 role	1+2
CE1/CE2, $N_+ = 1$	no Vd1/Vd2 role and exactly one T3-like role	2
CE1/CE2, $N_+ = 1$	no Vd1/Vd2 role and at least two T3-like roles	1
CE1/CE2, $N_+ \geq 2$	all vertex types	1

Lemma 7.1 (Exhaustiveness of the gap split)**.** *On each edge the complement of the two incident vertex-role traces is either empty or one closed interval; a singleton interval counts as a gap. Every edge having a gap has positive-length intersection with the open center role. Consequently a* CE0 *center permits no gaps, a* CE1 *center permits at most one, and a* CE2 *center permits at most two.*

Proof. In the coordinate $X_i(x)$ of Section 2, the two incident open traces are intervals issuing from 0 and 1. Their complement is therefore either empty or the single interval $[B_i, 1 - A_{i+1}]$, including the equality case.

If this complement is nonempty, each of its points must be covered by U_C: a nonincident vertex role cannot contain an interior point of the edge, because that point is more than unit distance from the role's distinguished vertex. Choose any point of the gap. Since U_C is open, its intersection with the edge contains a relative neighborhood of that point, and hence has positive length. Thus different gap edges give different edges counted by $N_E(T_C)$. Proposition 2.3 now gives the assertion, including singleton gaps. □

Proof of Theorem 1.1. Suppose that seven open unit equilateral triangles cover H. By Lemma 2.2, label them as one center role and six vertex roles. Proposition 2.3 places the center in exactly one of $\mathrm{CE0}, \mathrm{CE1}, \mathrm{CE2}$, and (2) places the actual vertex rows in exactly one of $N_+ = 0$, $N_+ = 1$, and $N_+ \geq 2$.

First suppose the center is CE0. If $N_+ = 0$, the perimeter obstruction in Proposition 3.18 applies. If $N_+ \geq 2$, the strict area certificate in Proposition 5.6 applies. Let $N_+ = 1$. Proposition 2.5 gives exactly three refinements: all roles are Vd0; at least one is Vd1 or Vd2; or no such role occurs and at least one role is T3-like. The first refinement has zero gaps by Lemma 7.1 and contradicts Theorem 6.7; the second is closed by Proposition 3.18; and the third is closed by Proposition 5.6.

Now suppose the center is CE1 or CE2. When $N_+ = 0$, the same vertex-type trichotomy is exhaustive. The all-Vd0 and the no-Vd1/Vd2 but T3-like refinements contradict Proposition 4.21, while the refinement containing a Vd1 or Vd2 role contradicts Proposition 3.18. When $N_+ \geq 2$, the conditional skeleton obstruction in Proposition 3.18 applies.

It remains that the center is CE1 or CE2 and $N_+ = 1$. If all six vertex roles are Vd0, Lemma 7.1 gives zero or one gap in class CE1 and zero, one, or two gaps in class CE2. The zero-gap state contradicts Theorem 6.7; the one-gap states and the CE2 two-gap state contradict Proposition 4.21.

Suppose instead that a Vd1 or Vd2 role occurs. In class CE1, or in class CE2 when at least two such roles occur, Proposition 3.18 applies. The remaining possibility is CE2 with exactly one Vd1/Vd2 role. The exceptional placement theorem incorporated in Proposition 4.21, together with the neighboring-midpoint and conditional-skeleton subcases in Proposition 3.18, exhausts that hybrid package.

Finally suppose there is no Vd1 or Vd2 role but not all roles are Vd0. Then Proposition 2.5 says that the remaining non-Vd0 roles are precisely T3-like. Exactly one such role is excluded by Proposition 4.21, and at least two are excluded by Proposition 3.18. Every row of the table has now produced a contradiction. Therefore the assumed open cover does not exist. □

Corollary 1.2 follows at once from Proposition 2.1.

Appendix A. Exact local and center formulas

This appendix records the formulas used by the propagation arguments in Section 4. Every radical below denotes its nonnegative value, and every algebraic root is restricted to the stated geometric contact interval. This last convention is important: an unrestricted root of a squared support equation need not describe an enclosing triangle.

A.1. The local admissible set.

Put

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \qquad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$$

and, for $0 \leq a, b, c \leq 1$, let

$$K(a, b, c) = \operatorname{conv}\{0, au, bv, c(u + v)\}.$$

Write $L_{\min}(a, b, c)$ for the least side length of an equilateral triangle containing this set, and let

$$\mathcal{A} = \{(a, b, c) \in [0, 1]^3 : L_{\min}(a, b, c) \leq 1\}.$$

Set $s = a + b$, $D = (a^2 + ab + b^2)^{1/2}$, and

$$R_a = (a^2 - ac + c^2)^{1/2}, \qquad R_b = (b^2 - bc + c^2)^{1/2}.$$

When $s > 0$ and $cs > ab$, define

$$L_{OA} = b + \max\{a, c\}, \qquad L_{BO} = a + \max\{b, c\},$$

$$L_{AC} = \begin{cases} (a^2 + bc)/R_a, & a \geq c, \\ c(a + b)/R_a, & a < c \leq a + b, \\ c^2/R_a, & c > a + b, \end{cases}$$

and let L_{CB} be the same formula with a and b interchanged. The support-function computation in the proof of Proposition 4.2 gives

$$L_{\min}(a,b,c)=\begin{cases} c, & a=b=0,\\ D, & s>0 \text{ and } cs\le ab,\\ \min\{L_{OA},L_{BO},L_{AC},L_{CB}\}, & s>0 \text{ and } cs>ab. \end{cases} \tag{87}$$

For the polynomial form, put

$$m=\min\{a,b\},\quad M=\max\{a,b\},\quad d=a^2+ab+b^2,\quad q=s^4-s^2+ab,$$
$$F_L=c^4-c^2+mc-m^2,$$
$$F_T=(s^2-1)c^2+Mc-M^2,$$
$$F_S=(m^2-1)c^2+(2mM^2+M)c+(M^4-M^2).$$

Then

$$\mathcal{A}=\mathcal{A}_L\cup\mathcal{A}_T\cup\mathcal{A}_S, \tag{88}$$

where

$$\begin{aligned}\mathcal{A}_L&=\{d\le 1, s\le 1, q\le 0, F_L\le 0\},\\ \mathcal{A}_T&=\{d\le 1, s\le 1, q\ge 0, F_T\le 0, c\le 2M\},\\ \mathcal{A}_S&=\{d\le 1, s>1, F_S\le 0, c\le \tfrac12\}.\end{aligned}$$

All three sets are understood to lie in $[0,1]^3$. The selector $c\le 2M$ removes the spurious upper sheet of $F_T\le 0$, and $c\le 1/2$ selects the lower component in the supercritical cell.

A.2. **The exact outgoing map.** For $a,c\in[0,1]$, define

$$B_c(a)=\max\{b\in[0,1]:(a,b,c)\in\mathcal{A}\},\qquad \beta(a)=\frac{-a+\sqrt{4-3a^2}}{2}.$$

The fiber is the whole interval $[0,B_c(a)]$. We now give the finite piecewise-algebraic formula used in the proof. The following catalog assumes $0<a,c<1$ and writes $\beta=\beta(a)$; an inapplicable entry is assigned $-\infty$.

The inner and axial candidates are

$$\beta_{\text{in}}=\begin{cases}\beta, & c(a+\beta)\le a\beta,\\ -\infty, & \text{otherwise},\end{cases}\qquad \beta_{OA}=\min\{\beta,1-\max\{a,c\}\},$$

$$\beta_{BO}=\begin{cases}\min\{\beta,1-a\}, & a+c\le 1,\\ -\infty, & a+c>1.\end{cases}$$

For the two AC contacts set

$$U_A=\begin{cases}+\infty, & a=c,\\ ac/(a-c), & a>c,\end{cases}$$

and

$$\beta_{AC}^{(1)}=\begin{cases}\min\{\beta,U_A,(R_a-a^2)/c\}, & a\ge c \text{ and this minimum is nonnegative},\\ -\infty, & \text{otherwise},\end{cases}$$

$$\beta_{AC}^{(2)}=\begin{cases}\min\{\beta,R_a/c-a\}, & a<c \text{ and } R_a\ge c^2,\\ -\infty, & \text{otherwise}.\end{cases}$$

There are three CB contacts. Define

$$P_{a,c}(b) = b^4 + (2ac-1)b^2 + cb + (a^2-1)c^2$$

and

$$U_H = \begin{cases} \beta, & a \le c, \\ \min\{\beta, ac/(a-c)\}, & a > c. \end{cases}$$

If $c \le U_H$, let ρ_H denote the largest root of $P_{a,c}$ in $[c, U_H]$, when such a root exists. Then

$$\beta_{CB}^{(1)} = \begin{cases} U_H, & c \le U_H \text{ and } P_{a,c}(U_H) \le 0, \\ \rho_H, & c \le U_H, P_{a,c}(U_H) > 0, \text{ and } \rho_H \text{ exists}, \\ -\infty, & \text{otherwise}. \end{cases}$$

Next put

$$L_M = \max\{0, c-a\}, \quad U_M = \min\{c, \beta\},$$

$$Q_{a,c}(b) = (c^2-1)b^2 + (2ac^2+c)b + (a^2-1)c^2,$$

$$\Delta_M = (2ac^2+c)^2 - 4(1-c^2)(1-a^2)c^2, \quad r_\pm = \frac{2ac^2 + c \pm \sqrt{\Delta_M}}{2(1-c^2)}$$

when $\Delta_M \ge 0$. The middle contact is

$$\beta_{CB}^{(2)} = \begin{cases} -\infty, & L_M > U_M, \\ U_M, & L_M \le U_M \text{ and } Q_{a,c}(U_M) \le 0, \\ r_-, & \Delta_M \ge 0 \text{ and } L_M \le r_- < U_M < r_+, \\ -\infty, & \text{otherwise}. \end{cases}$$

If $\Delta_M < 0$, the second line applies whenever $L_M \le U_M$. Finally set $U_L = \min\{\beta, c-a\}$ and, for $c \ge \sqrt{3}/2$,

$$\ell(c) = \frac{c}{2}\left(1 - \sqrt{4c^2-3}\right), \qquad r(c) = \frac{c}{2}\left(1 + \sqrt{4c^2-3}\right).$$

Then

$$\beta_{CB}^{(3)} = \begin{cases} -\infty, & U_L < 0, \\ U_L, & U_L \ge 0 \text{ and } c < \sqrt{3}/2, \\ U_L, & c \ge \sqrt{3}/2 \text{ and } (U_L \le \ell(c) \text{ or } U_L \ge r(c)), \\ \ell(c), & c \ge \sqrt{3}/2 \text{ and } \ell(c) < U_L < r(c). \end{cases}$$

With

$$\mathcal{B}(a,c) = \{\beta_{\text{in}}, \beta_{OA}, \beta_{BO}, \beta_{AC}^{(1)}, \beta_{AC}^{(2)}, \beta_{CB}^{(1)}, \beta_{CB}^{(2)}, \beta_{CB}^{(3)}\} \setminus \{-\infty\},$$

the final formula is

$$B_c(a) = \begin{cases} \beta(a), & c = 0, \\ 1, & a = 0, 0 < c \le 1, \\ 0, & a = 1, 0 < c \le 1, \\ 0, & 0 < a < 1, c = 1, \\ \max \mathcal{B}(a,c), & 0 < a, c < 1. \end{cases} \tag{89}$$

Every entry in $\mathcal{B}$ is obtained from one of the exhaustive support contacts in (87); hence the maximum introduces no unverified algebraic branch.

A.3. **The capped nonsupercritical map.** Define

$$F_c(a) = \min\{B_c(a), 1-a\}, \qquad G_c(a) = 1 - F_c(a).$$

For $1/2 < c < 1$, set

$$h(c) = \frac{2c}{\sqrt{16c^2-3}+1},$$
$$T_-(a,c) = \frac{\sqrt{a^2-ac+c^2}}{c} - a,$$

and

$$T_+(a,c) = \frac{2(1-a^2)c^2}{2ac^2+c+\sqrt{\Delta(a,c)}}, \quad \Delta(a,c) = (2ac^2+c)^2 - 4(1-c^2)(1-a^2)c^2.$$

For $1/2 < c \le \sqrt{3}/2$,

$$F_c(a) = \begin{cases} 1-a, & 0 \le a \le 1-c, \\ T_+(a,c), & 1-c < a < h(c), \\ h(c), & a = h(c), \\ T_-(a,c), & h(c) < a < c, \\ 1-a, & c \le a \le 1. \end{cases} \tag{90}$$

For $\sqrt{3}/2 < c < 1$,

$$F_c(a) = \begin{cases} 1-a, & 0 \le a \le 1-c, \\ T_+(a,c), & 1-c < a \le \ell(c), \\ \ell(c), & \ell(c) < a < r(c), \\ T_-(a,c), & r(c) \le a < c, \\ 1-a, & c \le a \le 1. \end{cases} \tag{91}$$

At $a = \ell(c)$ the value is $r(c)$; immediately to its right it drops to $\ell(c)$. Thus endpoints may not be reassigned by continuity. The four genuine labels are Full, L, T_+, and T_-. The map is nonincreasing, $G_c(a) \ge a$, and reflection of $\mathcal{A}$ gives the exact duality

$$G_c(a) \le z \iff G_c(1-z) \le 1-a. \tag{92}$$

A.4. **Exact center models.** We record the normalized forms after rotating and, when necessary, reflecting so that the unique center-covered midpoint is M_0.

For CE1, use affine coordinates

$$X = V_0 + b(V_1 - V_0) + a(V_5 - V_0)$$

and write $T_C \cap e_{0,1} = [s,t]$. Put

$$0 < \lambda < 1, \qquad \rho = \sqrt{1-\lambda+\lambda^2},$$

$$\begin{aligned} F_1 &= \lambda b + (1-\lambda)a - \lambda s, \\ F_2 &= -b + \lambda a + t, \\ F_0 &= (1-\lambda)b - a + \rho + \lambda s - t, \end{aligned} \qquad T_C = \{F_0, F_1, F_2 \ge 0\}.$$

At O, the three slacks are

$$C_0 = \rho + \lambda s - t - \lambda, \quad C_1 = 1 - \lambda s, \quad C_2 = t + \lambda - 1, \quad C_0 + C_1 + C_2 = \rho.$$

The exact closed domain is

$$0 < \lambda < 1, \quad 0 \le s < t \le 1, \quad t + \lambda(1-s) \le \rho, \quad t \ge 1-\lambda,$$
$$\lambda s \le \tfrac{1}{2}, \quad t < 1 - \tfrac{\lambda}{2}, \quad t > \rho + \lambda s - \tfrac{1+\lambda}{2}, \quad t \ge \rho - \frac{\lambda^2}{1-\lambda} s.$$

For the closure of an open center role the first two center slacks satisfy $C_0, C_2 > 0$. The radial exits d_i and complementary demands $c_i = 1 - d_i$ are

$$\begin{aligned}
&d_0 = 1 - \lambda s, && c_0 = \lambda s,\\
&d_1 = C_2/\lambda, && c_1 = 1 - C_2/\lambda,\\
&d_2 = C_2, && c_2 = 1 - C_2,\\
&d_3 = \min\{C_0/\lambda, C_2/(1-\lambda)\}, && c_3 = 1 - d_3,\\
&d_4 = C_0, && c_4 = 1 - C_0,\\
&d_5 = C_0/(1-\lambda), && c_5 = 1 - C_0/(1-\lambda).
\end{aligned} \tag{93}$$

For CE2, parameterize the two adjacent edge traces from V_0 by

$$T_C \cap e_{5,0} = [x, u], \qquad T_C \cap e_{0,1} = [y, v]$$

and put

$$S = x + y, \qquad D = \sqrt{x^2 + xy + y^2}.$$

The exact domain is

$$\begin{aligned}
&0 < x < u < 1, \qquad 0 < y < v < 1,\\
&(u+v)S - xy = D, \qquad uS \ge x, \qquad vS \ge y,\\
&uS < x + \tfrac{y}{2}, \qquad vS < y + \tfrac{x}{2}.
\end{aligned}$$

The two center inequalities become strict for the closure of an open center role: $uS > x$ and $vS > y$. In particular,

$$u + v = \frac{D + xy}{x + y}; \tag{94}$$

the two edge intervals cannot be varied independently. The exact exits are

$$\begin{aligned}
&d_0 = 1 - xy/S, && d_1 = (vS - y)/x, \quad d_2 = (vS - y)/S,\\
&d_3 = \min\{(vS - y)/y, (uS - x)/x\}, &&\\
&d_4 = (uS - x)/S, && d_5 = (uS - x)/y,
\end{aligned} \tag{95}$$

and again $c_i = 1 - d_i$. In both normalized center classes,

$$d_0 \ge \frac{1}{2}, \qquad d_i < \frac{1}{2}\ (i \ne 0), \qquad c_0 \le \frac{1}{2}, \qquad c_i > \frac{1}{2}\ (i \ne 0).$$

Appendix B. Exact and interval certificates

This appendix records the finite certificates used above. A certificate is part of the proof only through the exact statement and verification argument given here. The listed programs are reproducibility aids; neither a plot nor an unrecorded floating-point convention is used.

B.1. **The finite caliper certificate.** Let R and J denote counterclockwise rotations through $2\pi/3$ and $\pi/2$, respectively. For a nonempty finite set $S \subset \mathbb{R}^2$ and a vector N, put

$$W_S(N) = \sum_{k=0}^{2} \max_{z\in S}\langle z, R^k N\rangle. \tag{96}$$

Theorem B.1 (Finite caliper certificate)**.** *If S has at least two distinct points, then S is contained in a closed unit equilateral triangle if and only if there are distinct $p, q \in S$ and $\varepsilon \in \{-1, 1\}$ for which*

$$W_S\big(\varepsilon J(q-p)\big) \le \frac{\sqrt{3}}{2}\|p-q\|. \tag{97}$$

If S consists of one point, it is contained in equilateral triangles of arbitrarily small positive side length.

For $S_x = \{O, A, B, x\}$, every clause of (97) *is a finite polynomial clause of degree at most four in the coordinates of A, B, x; at most $12 \cdot 4^3$ clauses are required, and repeated point labels merely delete zero-length pairs.*

Proof. For a unit vector n, write

$$G_S(n) = \sum_{k=0}^{2} h_S(R^k n), \qquad h_S(n) = \max_{z\in S}\langle z, n\rangle.$$

The three supporting half-planes with normals $n, Rn, R^2 n$ form an equilateral triangle of side $2G_S(n)/\sqrt{3}$. Hence S fits in a closed unit equilateral triangle precisely when $\min_{\|n\|=1} G_S(n) \le \sqrt{3}/2$.

On an open normal-direction cell on which the three support points are fixed, $G_S(\theta) = c\cos\theta + d\sin\theta$ and $G_S'' = -G_S < 0$ unless all points coincide. Thus a minimum occurs at a cell boundary, where some support line contains two distinct points p, q. After a cyclic rotation of the three normals, its normal is $\varepsilon J(q-p)/\|p-q\|$. Homogeneity yields (97). A nondegenerate segment is checked directly with its two normals.

For the polynomial expansion fix (p, q, ε), put $N = \varepsilon J(q-p)$ and $N_k = R^k N$, and choose support labels $z_0, z_1, z_2 \in S_x$. The support cell is

$$\langle z_k, N_k\rangle \ge \langle z, N_k\rangle \quad (z \in S_x,\ k = 0, 1, 2), \tag{98}$$

with $\|p-q\|^2 > 0$. Since $W_S(N) \ge 0$, the remaining test is

$$4\left(\sum_{k=0}^{2}\langle z_k, N_k\rangle\right)^2 \le 3\|p-q\|^2. \tag{99}$$

The points are affine-linear in the parameters, so (98)–(99) have degree at most four. Six unordered pairs, two signs, and 4^3 support-label triples give the asserted finite exhaustive list. $\square$

B.2. **The hard four-label interval certificate.**

Theorem B.2 (Rational interval certificate for the hard label pair)**.** *Let $0 \le r \le 1$ and $1/2 \le \beta \le 1$ satisfy the realized T_+^{hi} left-label conditions*

$$\beta \ge r, \qquad \beta \ge \frac{1-r^2}{1+2r}, \qquad r - (1-\beta)(r+\beta)^2 \ge 0. \tag{100}$$

Put

$$M_\beta = \sqrt{\beta^2 - \beta + 1}, \qquad M_r = \sqrt{r^2 - r + 1},$$
$$B_5 = \frac{\beta M_\beta}{r + \beta}, \qquad S_0 = 1 - \frac{M_\beta}{r+\beta} + \frac{1-r}{1+M_r}. \tag{101}$$

Then

$$S_0 < \frac{93}{200} \implies B_5 < \frac{5657}{10000} < \frac{7-\sqrt{13}}{6}. \tag{102}$$

Certificate verification. Start with the rational rectangle $[0,1] \times [1/2,1]$. On a box $[r_-, r_+] \times [\beta_-, \beta_+]$, discard it when rational interval arithmetic proves that one of the three conditions in (100) is impossible. Enclose each square root from above by a rational with denominator 10^{18}, increasing the numerator until an exact integer comparison proves that its square dominates the radicand. The box is certified if these outward bounds prove either

$$B_5 < \frac{5657}{10000} \quad \text{or} \quad S_0 \geq \frac{93}{200}.$$

Otherwise bisect the longer rational side. Stop only after certification or at width 2^{-30}. The exhaustive run has 1174 certified terminal boxes, maximum depth 30, and no unresolved box. Their pruning counts are

$$(112, 290, 20, 752)$$

for, respectively, the B_5 bound, the S_0 bound, the elementary domain conditions, and the cell selector. Finally $(7 - 6 \cdot 5657/10000)^2 > 13$ is an exact rational comparison, proving the last inequality in (102).

The reproducibility command is

```
python3 proof/4XXX_CE1CE2/40XX_Nplus0/
  407X_T3_like_no_Vd1Vd2/407X_computation/
  407b_T_hi_Tminus_qright_threshold_certificate.py
```

with the line breaks removed. Its expected final line is `B0 < (7-sqrt(13))/6 verified`. □

B.3. **The outward-rounded mixed-overlap certificate.**

Theorem B.3 (Mixed cap-overlap certificate)**.** *In the strict residual-core domain*

$$0 < a, b < 1, \qquad a + b > 1, \qquad a^2 + ab + b^2 < 1,$$

let c_*, A, C, η, Δ_0, u, and the quantities r_X^0, ℓ_X^0, P_X^0 for $X \in \{A, C\}$ be those of Section 6 and equations (104) *below. Then*

$$r_A^0 \geq 0, \qquad r_C^0 \geq 0, \qquad \Delta_0 \leq 0, \qquad P_A^0 \geq 0, \qquad P_C^0 \geq 0. \tag{103}$$

Both P inequalities hold at every fixed asymmetric parameter; reflection is used only to restrict the domain to $a \geq b$.

Certificate verification. Represent a global vector as $(x, (\sqrt{3}/2)y)$ and put $C' = R^{-1}C$. For $A = (x_A, (\sqrt{3}/2)y_A)$ and $C' = (x', (\sqrt{3}/2)y')$, define

$$\begin{aligned} &\Delta_0 = x_A y' - y_A x', && u = \langle A, C' \rangle, \\ &r_X^0 = \|X\|^2 - \frac{3}{4}(1-c_*)^2, && \ell_X^0 = (1-c_*)u - (2c_*-1)\|X\|^2, \\ &P_X^0 = r_X^0 \Delta_0^2 - (\ell_X^0)^2. \end{aligned} \tag{104}$$

By reflection take $z = a - b \geq 0$. Introduce all variables used by the pruning test:

$$\begin{aligned} p &= 1-b, & q &= 1-a, & s &= p+q, \\ U &= 1-s = a+b-1, & m &= q, & z &= p-q, \end{aligned} \tag{105}$$

and

$$D^2 = z^2 + 6U + 3U^2 = 4(a^2+ab+b^2) - 3. \tag{106}$$

Thus $p = (s+z)/2$, $q = (s-z)/2 = m$, and the inverse conversion is

$$a = 1-m, \qquad b = 1-(s-m). \tag{107}$$

The selected radial value has two complete rational charts. On the L sheet put $x = 1+t$ and

$$c_* = \frac{x^2+3}{4x}, \qquad m = \frac{(x^2+3)(x-1)(x+3)}{16x^2}, \qquad s = c_* - v.$$

Equivalently,

$$z = s-2m, \qquad p = s-m, \qquad a = 1-m, \qquad b = 1-s+m. \tag{108}$$

The identity $x = 2c_* - \sqrt{4c_*^2-3}$ and the displayed formula for m are exactly the selected-root equation $c_*^4 - c_*^2 + mc_* - m^2 = 0$. The actual reflected L-sheet domain is the subset satisfying

$$z \geq 0, \quad 0 < m \leq p < 1, \quad U > 0, \quad 0 < D^2 < 1, \quad c_* \geq s, \quad \chi = s^4 - s^2 + mp \leq 0.$$

It is enclosed by the closed rational superrectangle

$$0 \leq t \leq \frac{3}{4}, \qquad 0 \leq v \leq \frac{4}{25}.$$

Indeed $0 \leq t \leq \sqrt{3}-1 < 3/4$ and $0 \leq v \leq U < 2/\sqrt{3} - 1 < 4/25$.

On the other sheet put $x = 1+t$ and

$$s = \frac{x^2+3}{4x}, \qquad E = \frac{3-x^2}{2x}, \qquad c_* = s-v, \qquad z = sE - v(1+E).$$

The exact inverse conversion is

$$m = \frac{s-z}{2}, \qquad p = \frac{s+z}{2}, \qquad a = 1 - \frac{s-z}{2}, \qquad b = 1 - \frac{s+z}{2}. \tag{109}$$

Here $x = 2s - \sqrt{4s^2-3}$ and $c_* = (s+z)/(1+E)$, so $v \geq 0$ is the selected side of the sheet, with $v = 0$ its selector face. Its actual reflected domain is the subset satisfying

$$z \geq 0, \quad 0 < m \leq p < 1, \quad U > 0, \quad 0 < D^2 < 1, \quad c_* > \frac{2}{3}, \quad \chi = s^4 - s^2 + mp \geq 0,$$

and it is enclosed by

$$0 \leq t \leq \frac{3}{4}, \qquad 0 \leq v \leq \frac{1}{3}.$$

The bounds follow from $x \le \sqrt{3} < 7/4$ and $0 \le v < 1/3$. These closed superrectangles include all selector and endpoint faces. They also contain points outside the original domain; proving the certificate there only strengthens the enclosure.

Every binary64 endpoint operation is widened by one next-representable number toward the relevant infinity. Rational constants are enclosed from both sides. Square-root endpoints are moved until exact integer-ratio comparisons prove that their squares enclose the radicand. Thus the square root step has no unverified library-rounding premise.

A box is pruned only when outward intervals prove that it cannot satisfy a necessary domain condition

$$x^2 \le 3, \quad z \ge 0, \quad m \ge 0, \quad U \ge 0, \quad 0 \le D^2 \le 1,$$

and, on the second sheet, $c_* > 2/3$. Before certification, the entire box must lie in $D^2 > 0$ and the two Newton denominators must have negative upper endpoints. For each target f, centered first-order interval automatic differentiation uses

$$f(B) \subseteq f(t_0, v_0) + \partial_t f(B)(t - t_0) + \partial_v f(B)(v - v_0).$$

Unresolved boxes are bisected along the larger relative chart width through depth 16.

On the L chart the run processes 21040 boxes, prunes 1258, certifies 9907, and leaves none unresolved; the least certified lower bound for a mixed P is

$$2.522678787629918 \cdot 10^{-7}.$$

On the other chart it processes 18574, prunes 222, certifies 9208, and leaves none unresolved; the corresponding least bound is

$$9.92316118814129 \cdot 10^{-7}.$$

The deterministic output reports `result: PASS` and `total_unresolved_boxes: 0`.

The reproduction command is

```
python3 proof/3XXX_CE0/31XX_Nplus1/310X_all_Vd0/
  3103X_residual_core/3103X_computation/
  verify_tangent_witness_mixed_overlap.py
```

with line breaks removed. Directed enclosure of every retained box and exhaustive bisection of both superrectangles prove (103) on the full parameter domain. □